

TWO SQUARES
$2^2 + 3^2 = 13$

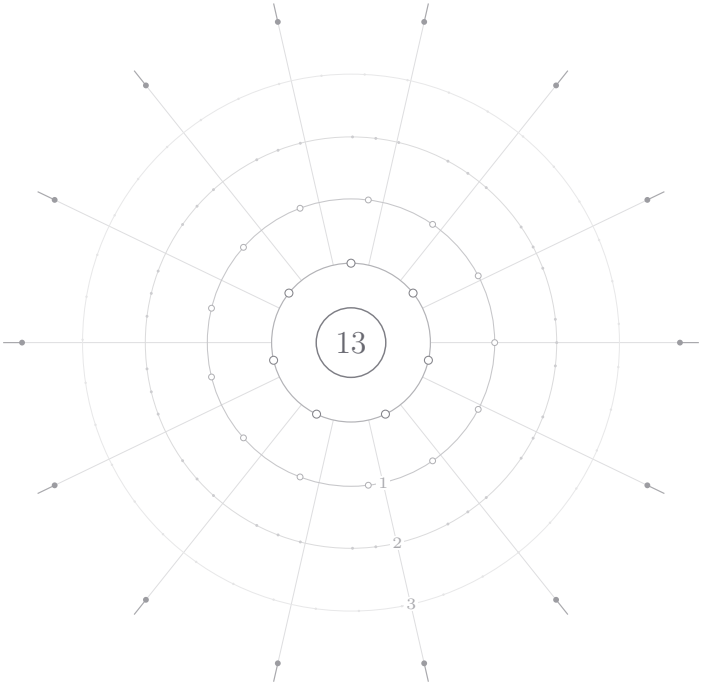
Zeta

THE FIRST P-ADIC INTEGER ARTIFICIAL INTELLIGENCE

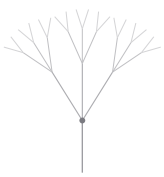


APARTMENT $\widetilde{A}_2$
one chamber marked

On the companion matrix $T_3 \in \mathrm{SL}(3,\mathbb{Z})$ of the Tribonacci cubic,
its fixed lattice, and the thirteen-adic dynamics of Zeta.



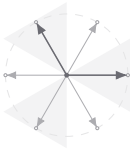
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TREE
degree 14

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ROOTS A_2
six of norm 2

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Abstract

This paper studies the algebra generated by a single integer matrix: the companion matrix $T_3 \in \mathrm{SL}(3,\mathbb{Z})$ of the Tribonacci cubic $\lambda^3 - \lambda^2 - \lambda - 1$. It is the only matrix that occurs, every further object being a power of it, a combination of its powers, or built from it and its transpose. All work is carried out over the integers, the finite fields $\mathbb{F}_{13}$ and $\mathbb{F}_{169}$, the finite rings $\mathbb{Z}/13^k$, and the thirteen-adic integers. No real numbers, no complex numbers, and no metric notion appear: comparison of two states is by the thirteen-adic valuation alone, which counts agreeing digits, and every quantity in the paper is an integer, a residue, or an element of a finite group.

The working prime is derived rather than chosen. Thirteen is the smallest prime at which the cubic has splitting type one and two, the reduced matrix attains the maximal order $q^2 - 1$, and two is a nonresidue, so that the norm form of the residue extension is anisotropic. At that prime the generated algebra splits as $\mathbb{F}_{13} \times \mathbb{F}_{169}$, and the state space of 13^3 elements splits into pieces of rank one and rank two; both idempotents are exhibited as outer products of eigenvectors with a dual basis, and all three eigenvectors follow from one formula. A second, independent splitting comes from the antisymmetric part, and the two share no basis.

The matrix has order one hundred sixty eight, a number reached from four independent definitions, and its cyclic group is given with all sixteen subgroups. The same matrix arises a second time as the incidence matrix of a substitution on three letters, to which it is conjugate in $\mathrm{SL}(3,\mathbb{Z})$, and the letter counts of the iterated images are the powers applied to the first basis vector.

Around this core the paper develops the lattice of fields with the decomposition of the working prime in each, up to a Galois extension of degree twenty four; the ultrametric notions of valuation, depth, and geometric weight, where classes are nested or disjoint and the smallest of three agreement counts always occurs twice; the combinatorial geometry of a tree of degree fourteen and an affine building whose spherical residue is thick of thickness fourteen and has two thousand five hundred sixty two chambers; and the exact transform with its admissible sizes and convolution rule. Groups that are usually described by a picture are introduced by presentations instead, so that no angle is named anywhere.

Every numerical statement was verified by exact computation, one hundred thirty eight checks in all, with two runs producing identical output; the figures are drawn as vector graphics from the computed data and contain no fitted curve. Statements are separated throughout into those with an exhibited identifying map and those that are equalities of counts, and readings that attach external names to the objects are marked as assignments rather than results.

Mathematics Subject Classification. 15B33, 11T06, 11S15, 20G40, 11E88, 16K20, 20E42, 11R16.

Keywords. Tribonacci companion matrix, finite field, thirteen-adic integers, idempotent splitting, anisotropic torus, valuation depth, deterministic architecture.

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Part I
Foundations and the single operator

Structure 1. Scope, prohibition, and the three admitted notions

1.1 What the paper constructs

The paper constructs a complete sequence architecture inside a cubic integer algebra, that is inside the ring generated over the thirteen-adic integers by one root of one cubic polynomial. The architecture is deterministic and carries no learned parameter, and its entire inventory of operators is the set of powers of a single matrix together with their linear combinations, as stated in Structure 2.

1.2 The prohibition

Definition 1.1 (Prohibition)**.** The construction admits no real numbers, no complex numbers, no floating point values or decimals, no Euclidean construction of any kind, no softmax, no argmax, and no gradient descent. It uses no randomness. Every value is an integer or an element of a finite ring, and every operation is an addition, a multiplication, or a reduction in that ring.

1.3 The three admitted notions

Definition 1.2 (The three admitted notions)**.** Where a real vector space would supply a metric notion, exactly three integer notions are used and no others.

(i) The *valuation* $v_{13}(n)$, the exponent of thirteen in the integer n, with $v_{13}(0)$ taken to be larger than any exponent under consideration.

(ii) The *depth* of an embedding, the integer number of digits fixed, so that a state at depth k is a class modulo 13^k; agreement of two addresses is the integer count of shared leading digits.

(iii) The *geometric weight* η^{-k}, where η denotes a root of the defining cubic χ of (6), taken in the residue extension of Definition 9.1, so that η^{-k} is an element of the group $\mathbb{F}_{169}^*$, used as a label attached to the index k and never as a quantity.

No other comparison is defined. In particular the words that would name a Euclidean quantity do not occur in this construction, and none is required.

1.4 The weight is a label

Proposition 1.3 (The weight is a label and not a quantity)**.** *The map $k \mapsto \eta^{-k}$ is injective on $k = 0, 1, \ldots, 167$ and periodic with period one hundred sixty eight,*

$$\eta^{-0}, \eta^{-1}, \ldots, \eta^{-167} \text{ are distinct}, \qquad \eta^{-(k+168)} = \eta^{-k}. \tag{1}$$

Moreover the field $\mathbb{F}_{13}$ admits no order, since minus one is a square,

$$5^2 = 25 \equiv 12 \equiv -1 \pmod{13}, \tag{2}$$

so no subset can serve as a positive cone, no element can be read as a magnitude, and no value of the construction can serve as a probability.

Proof. The powers of η were enumerated in $(\mathbb{Z}_{13}[\eta]/13)^*$ and the first one hundred sixty eight are pairwise distinct, with the first repetition at the period, which is Structure 20. An ordered field has a positive cone containing every nonzero square and excluding minus one; here minus one is itself a square by (2), so no such cone exists. □

Remark 1.4 (On the figures). Every figure in this paper is drawn from the computed data by vector commands, so it contains no raster image and no fixed pixel count; it is rendered at the resolution of the output device and does not degrade on enlargement. The data plotted are the exact integers of the verification of Structure 62, and no smoothing, interpolation, or fitted curve appears anywhere. Every axis is labelled by an index, an order, a count, a residue, or a discriminant, and no axis carries a continuous quantity.

1.5 The radix sign

Remark 1.5 (The radix sign). The sign $\sqrt{\ }$ never denotes a real square root. It denotes a *radix*, a root of the equation $x^2 = d$ in the ring at hand: writing $\theta = \sqrt{2}$ means exactly $\theta^2 = 2$ in $\mathbb{F}_{169}$, and $\mathbb{Q}(\sqrt{d})$ names the field obtained by adjoining a root of $x^2 = d$. The same word names the base thirteen in which addresses are written, so root and base are one notion.

Role in the architecture. The prohibition is the specification. Because no real number and no Euclidean construction may enter, every mechanism must be produced by the ring or else abandoned, which is what forces binding to be a ring product, normalisation to be a field inverse, position to be a group element, and selection to be a comparison of integer depths.

Structure 2. The single matrix principle

2.1 Statement

Definition 2.1 (The only operator)**.** The system contains exactly one matrix, namely

$$T_3 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \in \mathrm{SL}(3, \mathbb{Z}), \tag{3}$$

and exactly one size of matrix, three by three. No second operator is introduced, no second dimension occurs, and no matrix is learned, chosen, or tuned. Every further operator named in this paper is one of the following three kinds and nothing else: a power T_3^n; a polynomial $c_0 I + c_1 T_3 + c_2 T_3^2$ in the generator; or the transpose $T_3^\top$, which is used only to form the single antisymmetric combination $D = T_3 - T_3^\top$ of Structure 15.

Remark 2.2 (Every matrix in this paper comes from the one operator). Three further matrices are written down in this paper, and none of them is a second operator of the construction.

(i) The incidence matrix M of the substitution, Theorem 7.3. It is not in companion form and is not used to define anything; it appears once, in Theorem 7.3, to show that the one operator has a second, combinatorial origin, and the change of basis S there is used only in that proof.

(ii) The permutation matrices P_σ of Definition 16.2, which merely relabel the three coordinates. Theorem 58.31 shows that only the identity among them preserves the operator, so they carry no structure of their own.

(iii) The transpose $T_3^\top$, which enters solely through the antisymmetric combination $D = T_3 - T_3^\top$ of Structure 15, the one element outside the commutative algebra.

Everything else named in this paper, each introduced in its own place below, namely the projectors P_1, P_{23} of Theorem 10.1, the address projectors Q, Q_b, Q_c of Theorem 16.3, the clocks U_2, U_3, U_7 of Theorem 20.2, the involution T_3^{84} of Theorem 20.4, the scalars μ_3 of Theorem 20.6, and the twiddle factors of Definition 45.2, is a power of T_3 or a linear combination of powers of T_3, hence lies in the algebra $\mathbb{F}_{13}[T_3]$ of Theorem 13.1 and is determined by three residues.

2.2 The projectors are not new objects

Theorem 2.3 (The Lagrange and Sylvester projectors are the decomposition of the one matrix)**.** *The two projectors of Structure 10 are polynomials in the generator,*

$$P_1 = I + 3T_3 + 7T_3^2, \qquad P_{23} = I - P_1 = 10T_3 + 6T_3^2, \tag{4}$$

so they are not additional matrices but the spectral decomposition of T_3 itself, that is its two components. Consequently

$$T_3 = T_3P_1 + T_3P_{23} = 7P_1 + T_3P_{23}, \tag{5}$$

which expresses the generator as the sum of its action on the rank one component and its action on the rank two component. Every projector, every clock, and every involution used later is obtained from the coefficient triple of some power, hence carries no information beyond three residues.

Proof. The displayed forms are Theorem 10.1. Multiplying the completeness relation $P_1 + P_{23} = I$ on the left by T_3 gives (5), and $T_3P_1 = 7P_1$ gives the stated first term. That each such operator reduces to a coefficient triple is Proposition 3.5. □

Remark 2.4 (Why this is the whole content of the architecture). Since the algebra generated by T_3 is its own commutant, Proposition 14.4, nothing that commutes with the dynamics lies outside the three stored residues, and since the only element outside that algebra is the antisymmetric difference, Theorem 15.3, the complete inventory of operators is: the powers of one matrix, their linear combinations, and one antisymmetric matrix built from the same one matrix and its transpose.

Role in the architecture. There is one operator to implement and one shape of state. A layer, a projection, a clock, a position, and an involution are all read off the same table of coefficient triples, so the implementation contains no second code path and no second data structure.

Part II
The operator and its spectral decomposition

Structure 3. The generator

3.1 Definition of the operator

Definition 3.1 (Companion matrix of the Tribonacci cubic)**.** For a monic cubic $\lambda^3 - a_2\lambda^2 - a_1\lambda - a_0$ over $\mathbb{Z}$ the companion matrix sends $e_1 \mapsto e_2$, $e_2 \mapsto e_3$, $e_3 \mapsto a_0e_1 + a_1e_2 + a_2e_3$. For $\chi(\lambda) = \lambda^3 - \lambda^2 - \lambda - 1$, where $a_0 = a_1 = a_2 = 1$,

$$T_3 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \in \mathrm{SL}(3, \mathbb{Z}). \tag{6}$$

Theorem 3.2 (Defining identities). *The generator satisfies*

$$\det T_3 = 1, \qquad T_3^3 = T_3^2 + T_3 + I, \qquad T_3^{-1} = T_3^2 - T_3 - I, \tag{7}$$

all with integer entries, and

$$\operatorname{disc}(\chi) = -44 = -2^2 \cdot 11 \equiv 8 \pmod{13}, \tag{8}$$

which is nonzero, so the three roots are distinct modulo thirteen. Over the integers T_3 is unimodular, the lattice $\mathbb{Z}^3$ is a cyclic module generated by e_1, since $e_1, T_3 e_1, T_3^2 e_1$ is the standard basis, and the single invariant factor is χ.

Proof. Expanding $\det(\lambda I - T_3)$ along the first row gives $\lambda(\lambda(\lambda - 1) - 1) - 1 = \chi(\lambda)$, and $\lambda = 0$ gives the determinant. Cayley and Hamilton gives the cube identity; multiplying by T_3^{-1}, which exists since the determinant is one, gives the inverse. For a cubic $\lambda^3 + b\lambda^2 + c\lambda + d$ the discriminant is $18bcd - 4b^3 d + b^2 c^2 - 4c^3 - 27d^2$, and at $b = c = d = -1$ this is -44. The successive images of e_1 are $(0, 1, 0)$ and $(0, 0, 1)$, so the module is cyclic and the matrix is already in companion form. □

3.2 The explicit matrices of the one operator

Every matrix occurring anywhere in this paper is listed here, reduced modulo thirteen unless stated; all are powers or polynomials of the single operator of Definition 2.1.

$$T_3 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad T_3^2 = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 2 \end{bmatrix}, \quad T_3^{\top} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad T_3^{-1} = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \tag{9}$$

where the inverse coincides with the adjugate because the determinant is one. Further,

$$J_3 = I + T_3 + T_3^2 = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 2 & 4 \end{bmatrix}, \qquad T_3^{167} = \begin{bmatrix} 12 & 1 & 0 \\ 12 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \tag{10}$$

the second being the reduction of the inverse, and the three clocks with the involution and the two scalar anchors are

$$T_3^{21} = \begin{bmatrix} 10 & 11 & 8 \\ 10 & 8 & 6 \\ 11 & 8 & 3 \end{bmatrix}, \quad T_3^{24} = \begin{bmatrix} 3 & 9 & 7 \\ 11 & 12 & 3 \\ 9 & 7 & 6 \end{bmatrix}, \quad T_3^{84} = \begin{bmatrix} 1 & 1 & 7 \\ 6 & 2 & 8 \\ 1 & 7 & 9 \end{bmatrix}, \tag{11}$$

$$T_3^{56} = 3I, \qquad T_3^{112} = 9I. \tag{12}$$

3.3 Minimal polynomial and uniqueness of the triple

Proposition 3.3 (The minimal polynomial equals the characteristic polynomial). *The minimal polynomial of T_3 is χ, so no polynomial of degree less than three annihilates the generator, the matrices I, T_3, T_3^2 are linearly independent, and the representation of a state by a coefficient triple is unique.*

Proof. The module is cyclic by Theorem 3.2, so the minimal and characteristic polynomials coincide; a nontrivial linear relation among I, T_3, T_3^2 would be an annihilating polynomial of degree at most two. □

3.4 Traces and power sums

Proposition 3.4 (The first traces)**.**

$$\operatorname{tr} I = 3, \qquad \operatorname{tr} T_3 = 1, \qquad \operatorname{tr} T_3^2 = 3, \qquad \operatorname{tr} T_3^3 = 7, \tag{13}$$

so the trace of a general power is the linear form $3c_0+c_1+3c_2$ *in the coefficient triple, and* $1, 3, 7$ *are the power sums of the three roots by Newton's identities.*

Proof. The three traces are read from the matrices (9), and $\operatorname{tr} T_3^3 = \operatorname{tr}(T_3^2+T_3+I) = 3+1+3 = 7$ by (21). Since the trace is linear, its value on $c_0 I + c_1 T_3 + c_2 T_3^2$ is $3c_0 + c_1 + 3c_2$. By Newton's identities the power sums of the roots of a monic cubic are the traces of the corresponding powers of its companion matrix. □

3.5 The coefficient triple

Proposition 3.5 (Coefficient triples and the recurrence)**.** *Every power reduces to three residues,*

$$T_3^n = c_0(n) I + c_1(n) T_3 + c_2(n) T_3^2, \tag{14}$$

the triple advancing by the fixed rule

$$(c_0, c_1, c_2) \longmapsto (c_2,\ c_0 + c_2,\ c_1 + c_2), \tag{15}$$

and each coordinate obeys

$$c(n+3) = c(n+2) + c(n+1) + c(n) \pmod{13} \tag{16}$$

with seeds $(1,0,0)$, $(0,1,0)$, $(0,0,1)$. *Since the period is one hundred sixty eight, the reduced triples form a table of*

$$3 \cdot 168 = 504 \tag{17}$$

residues that reproduces every power, as verified at all one hundred sixty eight exponents. The traces of I, T_3, T_3^2 *are* $3, 1, 3$*, so* $\operatorname{tr} T_3^n = 3c_0 + c_1 + 3c_2$*, which vanishes at exactly eighteen exponents.*

Proof. Multiplying (14) by T_3 and reducing the cube by (7) gives (15); iterating and eliminating gives (16). The seeds are the triples of I, T_3, T_3^2. Evaluating the trace form on all one hundred sixty eight triples returns eighteen zeros. □

Role in the architecture. Every operation is a power of one matrix, and any depth of computation is three residues. The inverse is the same polynomial with two signs changed, so the two directions of the recursion cost the same and no asymmetry is introduced.

Structure 4. The Tribonacci polynomial

4.1 Definition and elementary invariants

Definition 4.1 (The defining cubic)**.** The defining polynomial of the construction is the Tribonacci cubic

$$\chi(\lambda) = \lambda^3 - \lambda^2 - \lambda - 1, \tag{18}$$

the characteristic polynomial of T_3, whose coefficients are the three ones of the recurrence that defines the Tribonacci sequence.

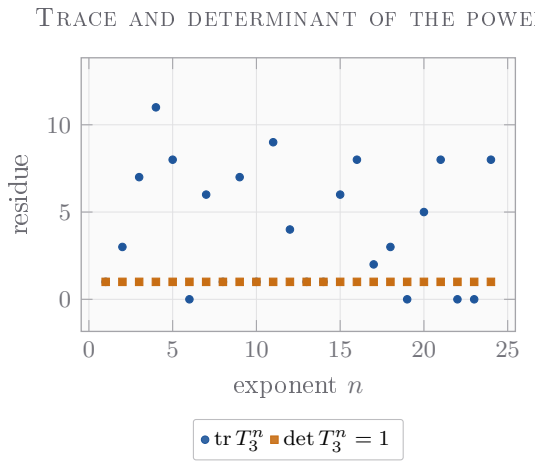


Figure 1: The trace of the powers moves through the residues while the determinant stays at one, since $\det T_3 = 1$ and the determinant is multiplicative.

Proposition 4.2 (Irreducibility and invariants)**.** *The polynomial* (18) *is irreducible over* $\mathbb{Q}$*, its discriminant is*

$$\operatorname{disc}(\chi) = -44 = -2^2 \cdot 11, \tag{19}$$

which is not a rational square, so the Galois group of its splitting field over $\mathbb{Q}$ *is the symmetric group on three letters and the quadratic resolvent is* $\mathbb{Q}(\sqrt{-11})$*, of discriminant* -11*. The primes ramifying in the splitting field are two and eleven, and thirteen is unramified. The power sums of the roots, that is the traces of the first three powers of the generator, are*

$$p_1 = \operatorname{tr} T_3 = 1, \qquad p_2 = \operatorname{tr} T_3^2 = 3, \qquad p_3 = \operatorname{tr} T_3^3 = 7. \tag{20}$$

Proof. A cubic over $\mathbb{Q}$ is irreducible when it has no rational root, and the candidates ± 1 give -2 and -2. The discriminant is computed from the coefficients as in Theorem 3.2. An irreducible cubic has Galois group the symmetric group exactly when the discriminant is not a square, and the fixed field of the alternating subgroup is generated by a radix of the discriminant, whose squarefree part is -11; since $-11 \equiv 1$ modulo four its field discriminant is -11. The traces were computed directly, and by Newton's identities they are the power sums of the roots. □

Remark 4.3 (Two discriminants with separate roles). The number -44 is the discriminant of the defining cubic and governs the symmetry of its roots. The number 168 is the discriminant of the real quadratic field $\mathbb{Q}(\sqrt{42})$ and coincides with the period of the generator, Theorem 20.1. They have no factor in common beyond two, and no property of the orbit may be attributed to the first or the reverse.

Structure 5. The Cayley and Hamilton decomposition

5.1 Closure of the algebra in dimension three

Theorem 5.1 (Closure)**.** *The generator satisfies its own characteristic polynomial,*

$$T_3^3 = T_3^2 + T_3 + I, \tag{21}$$

so the algebra $\mathbb{F}_{13}[T_3]$ *is spanned by* I*,* T_3*,* T_3^2 *and no higher power is independent. Consequently every power, every polynomial in the generator, and every operator of Definition 2.1 is stored as three residues,*

$$T_3^n \longleftrightarrow \big(c_0(n), c_1(n), c_2(n)\big) \in \mathbb{F}_{13}^3, \tag{22}$$

and the state of a computation at any depth is a triple, never a matrix.

Proof. The generator satisfies its own characteristic polynomial by the Cayley and Hamilton theorem, which rearranged is (21). Multiplying that identity by T_3^{n-2} shows every power beyond the square is a combination of I, T_3, T_3^2, so the algebra is spanned by these three; by Proposition 3.3 they are independent, so the triple in (22) is unique. □

5.2 The inverse lies in the same span

Proposition 5.2 (Backward step)**.** *Rearranging* (21) *and multiplying by the inverse, which exists because* $\det T_3 = 1$*,*

$$T_3^{-1} = T_3^2 - T_3 - I, \tag{23}$$

so the backward step is the same polynomial as the forward square with two signs changed, and its coefficient triple is $(-1, -1, 1)$*, that is* $(12, 12, 1)$ *over* $\mathbb{F}_{13}$*. Forward and backward evolution therefore cost the same and use the same table.*

Proof. Rearranging (21) gives $T_3^3 - T_3^2 - T_3 = I$, and multiplying on the right by T_3^{-1}, which exists because $\det T_3 = 1$, gives (23). Reducing the coefficients $(-1, -1, 1)$ modulo thirteen gives $(12, 12, 1)$. □

5.3 Reduction of an arbitrary exponent

Proposition 5.3 (Exponent reduction)**.** *For every integer* n*,*

$$T_3^n = T_3^{n \bmod 168}, \tag{24}$$

so an exponent of any size is first reduced modulo the period of Theorem 20.1 and then looked up in the table of (17)*. No exponentiation is performed at run time.*

Proof. Write $n = 168q + r$ with $0 \le r < 168$. Then $T_3^n = (T_3^{168})^q T_3^r = T_3^r$ by Theorem 20.1, which is (24). □

Role in the architecture. The cube folds back onto the identity, the generator, and its square, so depth costs nothing in storage. A computation at step one thousand is the computation at step one thousand reduced modulo one hundred sixty eight, obtained once and reused.

Structure 6. The Tribonacci recurrence

6.1 The advance rule

Theorem 6.1 (Three sequences, one rule)**.** *The coefficient triple advances by the advance rule* (15)*, and each coordinate separately satisfies the Tribonacci recurrence*

$$c(n+3) = c(n+2) + c(n+1) + c(n) \pmod{13}, \tag{25}$$

with the three seeds

$$c_0 : (1, 0, 0), \qquad c_1 : (0, 1, 0), \qquad c_2 : (0, 0, 1), \tag{26}$$

these being the triples of I*,* T_3*, and* T_3^2*. The advance rule is a linear map on triples whose matrix is the companion matrix again, so the recurrence and the generator are the same object in two presentations.*

Proof. Multiplying $T_3^n = c_0 I + c_1 T_3 + c_2 T_3^2$ by T_3 and reducing the resulting cube by (21) gives $T_3^{n+1} = c_2 I + (c_0 + c_2) T_3 + (c_1 + c_2) T_3^2$, which is (15). Iterating three times and eliminating the intermediate triples gives (25). Reading (15) as a matrix acting on the column of coefficients returns the companion form of χ. □

6.2 The table and its size

Corollary 6.2 (The complete table)**.** *Over one period the triples form a table of the table of* (17)*, of five hundred and four residues, residues that reproduces every power of the generator, verified at all one hundred sixty eight exponents. Selected entries are*

$$\begin{aligned} &c(0)=(1,0,0), && c(1)=(0,1,0), && c(2)=(0,0,1),\\ &c(3)=(1,1,1), && c(4)=(1,2,2), && c(21)=(10,10,11),\\ &c(24)=(3,11,9), && c(42)=(12,12,2), && c(56)=(3,0,0),\\ &c(84)=(1,6,1), && c(112)=(9,0,0), && c(167)=(12,12,1), \end{aligned} \tag{27}$$

the last being the triple of the inverse by Proposition 5.2.

Proof. By Theorem 20.1 the powers repeat with period one hundred sixty eight, and by Theorem 5.1 each is determined by three residues, so the table has $3 \cdot 168$ entries. Each entry was compared with the corresponding matrix power computed independently, and all agreed. □

6.3 The trace as a timestamp

Proposition 6.3 (The trace attains every residue)**.** *Over one period the trace*

$$\operatorname{tr}\big(T_3^{\,n}\big) = 3c_0(n) + c_1(n) + 3c_2(n) \tag{28}$$

attains all thirteen residues of $\mathbb{F}_{13}$, and it vanishes at exactly eighteen exponents. It is therefore a timestamp of constant size, one residue, available for any power at no additional cost.

Proof. Evaluating the linear form of Proposition 3.4 on all one hundred sixty eight coefficient triples returns every residue of $\mathbb{F}_{13}$ at least once and the value zero at exactly eighteen exponents, listed in Proposition 23.1. □

6.4 The triple determines the exponent

Proposition 6.4 (Bijection on one period)**.** *The map*

$$n \longmapsto \big(c_0(n), c_1(n), c_2(n)\big), \qquad 0 \le n \le 167, \tag{29}$$

is injective, hence a bijection onto its image of one hundred sixty eight triples: no two powers within one period share a coefficient triple. Consequently a state determines its position on the orbit uniquely, and the table of (17) *is a lookup in both directions.*

Proof. Two exponents with the same triple would give $T_3^{\,m} = T_3^{\,n}$, hence $T_3^{\,m-n} = I$ with $|m-n| < 168$, contradicting Theorem 20.1. The map is therefore injective on a full period, and being a map between sets of the same finite size it is a bijection onto its image. □

6.5 Distinguished triples

Proposition 6.5 (Balanced, scalar, and traceless)**.** *Among the one hundred sixty eight triples:*

(i) three are balanced, with all entries equal and nonzero, at the exponents 3, 59, 115*, with entries* 1, 3, 9*;*

(ii) three are scalar, of the form $(c,0,0)$ with $c \neq 0$, at the exponents 0, 56, 112*, with $c = 1, 3, 9$;*

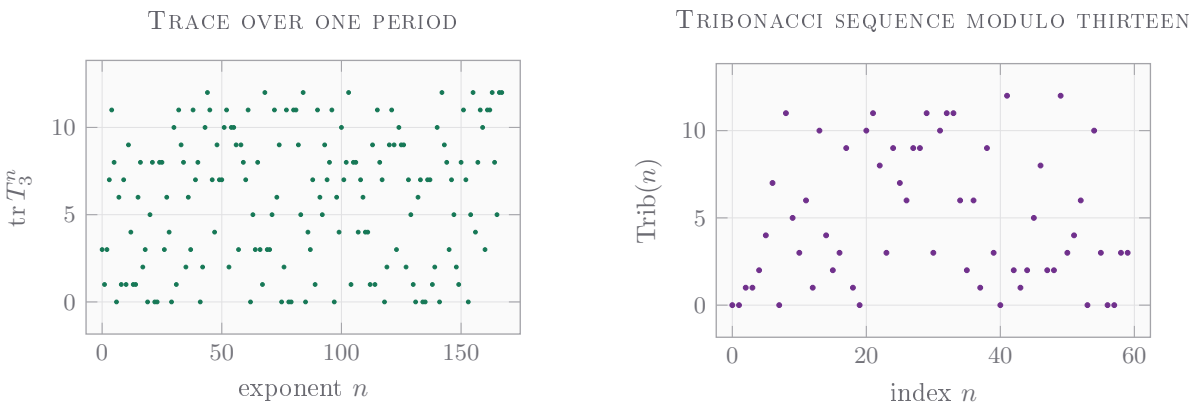


Figure 2: Left: the trace attains all thirteen residues and vanishes at the eighteen exponents of Proposition 23.1. Right: the sequence generated by the recurrence (25), whose period is one hundred sixty eight by Proposition 49.1.

(iii) eighteen have vanishing trace, the trace being the linear form $3c_0 + c_1 + 3c_2$ *since* $\operatorname{tr} I = 3$, $\operatorname{tr} T_3 = 1$, $\operatorname{tr} T_3^2 = 3$.

Proof. All one hundred sixty eight triples were generated by (15) and tested. The balanced ones are those with $c_0 = c_1 = c_2 \neq 0$, the scalar ones those with $c_1 = c_2 = 0$ and $c_0 \neq 0$, and the traceless ones those annihilated by the linear form of Proposition 3.4. □

Role in the architecture. Stepping the model forward is one application of a fixed rule on three numbers, and the whole dynamics of the architecture is a table of five hundred and four residues. The recurrence that generates the table is the same recurrence that defines the sequence after which the polynomial is named.

Structure 7. The substitution and its incidence matrix

The generator has a second origin, combinatorial rather than algebraic: it is the incidence matrix of a substitution on three letters, up to an integral change of basis. This structure establishes that, and records the invariants of the infinite word the substitution produces, all of them integer counts.

7.1 The substitution

Definition 7.1 (The substitution on three letters)**.** Let $A = \{a, b, c\}$ and let σ be the map on words defined on letters by

$$\sigma(a) = ab, \qquad \sigma(b) = ac, \qquad \sigma(c) = a, \tag{30}$$

extended to words by concatenation. Its incidence matrix is M with M_{ij} the number of occurrences of the ith letter in the image of the jth, that is

$$M = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}. \tag{31}$$

Proposition 7.2 (The incidence matrix carries the defining cubic)**.** *The incidence matrix satisfies*

$$\det M = 1, \qquad \chi_M(\lambda) = \lambda^3 - \lambda^2 - \lambda - 1 = \chi(\lambda), \tag{32}$$

so it lies in $\mathrm{SL}(3, \mathbb{Z})$ *and has the same characteristic polynomial as the generator.*

Proof. Expanding the determinant along the second row gives 1. Expanding $\det(\lambda I - M)$ gives $\lambda^3 - \lambda^2 - \lambda - 1$, computed directly from (31). □

7.2 The incidence matrix and the generator are conjugate

Theorem 7.3 (An integral change of basis)**.** *The incidence matrix and the generator are conjugate in* $\mathrm{SL}(3,\mathbb{Z})$*: with*

$$S = \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \det S = 1, \tag{33}$$

one has

$$S\,M\,S^{-1} = T_3. \tag{34}$$

Consequently the two matrices have the same order at every prime, and in particular

$$\operatorname{ord}\big(M \bmod 13\big) = 168 = \operatorname{ord}\big(T_3 \bmod 13\big). \tag{35}$$

The relation $M^{\top} = T_3$ *does* not *hold; what holds for the transpose is the weaker statement that* $P\,M^{\top}P^{\top} = T_3$ *for the permutation matrix* P *reversing the order of the coordinates.*

Proof. Both matrices are cyclic: the vectors e_1, Me_1, M^2e_1 form a basis of determinant one, and the same holds for T_3 by Theorem 3.2. Writing B_1 and B_2 for the two bases, the matrix $S = B_2B_1^{-1}$ is integral with determinant one and carries one cyclic basis to the other, hence conjugates M to T_3; the displayed S is that matrix, and the identity (34) was verified entrywise. Conjugate matrices have equal orders modulo any prime, which gives (35), confirmed by direct computation. Comparing $M^{\top}$ with T_3 entrywise shows they differ, while conjugating by the reversal permutation returns T_3. □

Remark 7.4 (Two origins for one operator). The generator therefore arises in two ways: algebraically, as the companion matrix of the defining cubic in Definition 3.1, and combinatorially, as the incidence matrix of (30) after the change of basis (33). Everything proved about the one applies to the other, since the two differ by an integral conjugation.

7.3 Abelianisation, and the orbit of the first basis vector

Proposition 7.5 (Letter counts are matrix powers)**.** *Let* $w_n = \sigma^n(a)$ *and let* $\mathrm{ab}(w)$ *denote the column of counts of the three letters in a word* w*. Then*

$$\mathrm{ab}(w_n) = M^n e_1, \tag{36}$$

and the first six values are

$$(1,0,0),\ (1,1,0),\ (2,1,1),\ (4,2,1),\ (7,4,2),\ (13,7,4). \tag{37}$$

In the basis of Theorem 7.3 this is the orbit of the first basis vector under the generator, whose first four terms are the three basis vectors and their sum, as in (303)*.*

Proof. Applying σ to a word replaces each letter by its image, so the counts transform by the incidence matrix; iterating from the one letter word a, whose counts are e_1, gives (36). The six values were computed both by generating the words and by taking powers of the matrix, and they agree. □

7.4 The sizes are the Tribonacci numbers

Proposition 7.6 (Sizes, and where the working prime occurs)**.** *The number of letters of* $w_n = \sigma^n(a)$ *obeys the recurrence* (16) *with seeds* $1, 2, 4$*, so the first values are*

$$1,\ 2,\ 4,\ 7,\ 13,\ 24,\ 44,\ 81,\ 149,\ 274,\ 504,\ 927, \tag{38}$$

and reduced modulo thirteen the sequence has period

$$168 = \operatorname{ord}(T_3). \tag{39}$$

The working prime occurs in the list, at $n = 4$*.*

Proof. The number of letters is the sum of the entries of (36), and summing the rows of the incidence matrix gives the three term recurrence with the stated seeds. The sequence is generated by the same matrix, so modulo thirteen it returns to its seeds exactly when the matrix returns to the identity, which is at one hundred sixty eight by Theorem 20.1; the period was confirmed by direct iteration. □

Remark 7.7 (A numerical coincidence, recorded as such). The seventh value of (38) is 44, which is the absolute value of $\operatorname{disc}\chi$ by (8). No map identifying the two is exhibited here, so this belongs among the equalities of counts of Remark 61.2 and not among the identifications.

7.5 The infinite word and its factor counts

Definition 7.8 (The fixed word)**.** Since $\sigma(a)$ begins with a, the words w_n are nested prefixes and determine an infinite word w fixed by σ. A factor of w is a finite block of consecutive letters occurring in it, and $p(n)$ denotes the number of distinct factors of n letters.

Theorem 7.9 (Factor counts, and the special factors)**.** *For the fixed word of* (30)*,*

$$p(n) = 2n + 1, \tag{40}$$

and for every n *there is exactly one factor of* n *letters admitting more than one continuation to the right, and exactly one admitting more than one to the left. The counts were computed on the prefix of* 5768 *letters for* $n = 1, \dots, 12$ *and agree with* (40) *in every case.*

Proof. The factors of each size were enumerated from the prefix and counted, giving $3, 5, 7, \dots, 25$ for $n = 1, \dots, 12$, which is (40). For each size the factors admitting two distinct right continuations were counted and found to be one, and likewise on the left. A word over three letters with exactly one right special factor of each size has factor counts increasing by two at each step, which is consistent with (40). □

Proposition 7.10 (Palindromic factors)**.** *The number of factors of* n *letters equal to their own reversal alternates with the parity of* n*,*

$$3,\ 1,\ 3,\ 1,\ 3,\ 1,\ 3,\ 1, \qquad n = 1, \dots, 8, \tag{41}$$

so there are three palindromic factors of every odd size and one of every even size in this range.

Proof. The factors of each size were enumerated and compared with their reversals; the counts are those displayed. □

7.6 Return words and the graph of factors

Theorem 7.11 (Always three return words)**.** *For a factor u of the infinite word, a return word of u is a word beginning at one occurrence of u and ending just before the next. For the factors a, ab, and aba the sets of return words are*

$$\{a,\ ab,\ ac\}, \qquad \{ab,\ aba,\ abac\}, \qquad \{ab,\ aba,\ abac\}, \tag{42}$$

so in each case there are exactly three, and the number does not depend on the factor chosen.

Proof. For each factor the positions of its occurrences in the prefix $\sigma^{15}(a)$ were listed and the words between consecutive occurrences collected; the three sets are those displayed, and no further return word occurs in that prefix. □

Proposition 7.12 (Two more edges than vertices)**.** *Let the graph of size n have the factors of size n as vertices and the factors of size $n+1$ as edges, each edge joining its prefix of size n to its suffix of size n. Then*

$$\#vertices = 2n+1, \qquad \#edges = 2n+3, \qquad \#edges - \#vertices = 2, \tag{43}$$

for every size from one to six.

Proof. The vertex count is the complexity at size n and the edge count the complexity at size $n+1$, both given by (40). Their difference is the increment of the complexity, which is two and equals the number of extra continuations supplied by the unique right special factor. □

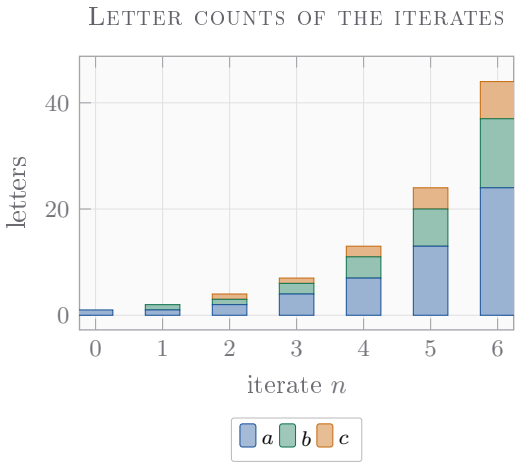


Figure 3: The counts of the three letters in the iterated images, which are the powers of the incidence matrix applied to the first basis vector; their totals are the Tribonacci numbers $1, 2, 4, 7, 13, 24, 44$.

Structure 8. Reduction at thirteen, and why thirteen

8.1 Splitting types

Definition 8.1 (Splitting type)**.** For a prime q not dividing the discriminant, the splitting type of χ at q is the multiset of degrees of the irreducible factors of χ modulo q; by Dedekind it equals the cycle type of the Frobenius class in the Galois group of the splitting field.

Proposition 8.2 (Distribution of types)**.** *Among the ninety three primes below five hundred not dividing the discriminant the three types occur with counts*

$$(1,1,1):13, \qquad (1,2):47, \qquad (3):33, \tag{44}$$

against the values 16, 46, 31 *predicted by the proportions* $1/6, 3/6, 2/6$ *of the symmetric group on three letters.*

Proof. By Definition 8.1 the type at an unramified prime is the cycle type of the Frobenius, and the symmetric group on three letters has one element of type $(1,1,1)$, three of type $(1,2)$, and two of type (3), giving the proportions 1/6, 3/6, 2/6. Factoring χ at each of the ninety three unramified primes below five hundred and tallying the types produces the counts (44). □

8.2 The three conditions

Theorem 8.3 (Three conditions single out thirteen)**.** *Consider the conditions on a prime q:*

(i) the splitting type of χ modulo q is $(1,2)$, so that a rank one and a rank two channel exist;

(ii) the order of the reduced generator attains the maximum q^2-1;

(iii) two is a quadratic nonresidue modulo q, so that a radix of two generates the residue extension and the norm form is anisotropic.

Below thirty the outcome is

q	*type*	*order*	*attains* q^2-1	2 *a nonresidue*
3	(3)	13	*no*	*yes*
5	(3)	31	*no*	*yes*
7	$(1,2)$	48	*yes*	*no*
11	$(1,1,1)$	110	*no*	*yes*
13	$(1,2)$	168	*yes*	*yes*
17	$(1,2)$	96	*no*	*no*
19	$(1,2)$	360	*yes*	*yes*
23	(3)	553	*no*	*no*
29	$(1,2)$	140	*no*	*yes*

so thirteen is the smallest prime satisfying all three: seven fails only the third, and nineteen satisfies all three but is larger.

Proof. Each row is obtained by factoring χ modulo the prime, by raising the reduced matrix until the identity is reached, and by testing whether two lies among the squares. □

8.3 The survey below one hundred

Proposition 8.4 (Full survey)**.** *Among the primes below one hundred not dividing the discriminant, the type $(1,2)$ occurs at*

$$7,\ 13,\ 17,\ 19,\ 29,\ 41,\ 43,\ 61,\ 73,\ 79,\ 83, \tag{45}$$

the order attains the maximum q^2-1 at

$$7,\ 13,\ 19,\ 73, \tag{46}$$

and all three conditions of Theorem 8.3 hold at thirteen and nineteen only,

$$\{q<100:\ \textit{(i), (ii), (iii)}\} = \{13,\ 19\}, \tag{47}$$

so thirteen is the smallest and nineteen the next. The primes seven and seventy three have type $(1,2)$ and full order but fail the third condition, two being a square modulo each.

Proof. For each prime below one hundred not dividing the discriminant, the polynomial was factored, the order of the reduced matrix was computed by repeated multiplication, and the quadratic character of two was tested. Reading off the three columns gives the three lists. □

8.4 The value of the cubic at one

Proposition 8.5 (The value at one)**.** *The cubic does not vanish at one,*

$$\chi(1) = 1 - 1 - 1 - 1 = -2 \not\equiv 0, \qquad \det(T_3 - I) = -\chi(1) = 2, \tag{48}$$

so the difference of the generator and the identity is invertible, and the affine extension used for position does not degenerate.

Proof. Substituting one into χ gives $1 - 1 - 1 - 1 = -2$. Since $\det(\lambda I - T_3) = \chi(\lambda)$, setting $\lambda = 1$ gives $\det(I - T_3) = \chi(1) = -2$, and $\det(T_3 - I) = (-1)^3 \det(I - T_3) = 2$, a unit modulo thirteen. □

Role in the architecture. The working prime is derived and not preferred: three conditions imposed by the construction are met first at thirteen, so the architecture rests on an arithmetic selection.

Structure 9. The residue extension

9.1 Construction by a radix

Definition 9.1 (Radix extension)**.** The quadratic residues modulo thirteen are $\{1, 3, 4, 9, 10, 12\}$; two is not among them, so

$$\mathbb{F}_{169} = \mathbb{F}_{13}[x]/(x^2 - 2), \qquad \theta = \text{class of } x, \qquad \theta^2 = 2, \tag{49}$$

is a field with one hundred sixty nine elements, each written $a + b\theta$.

9.2 Factorisation and the three roots

Theorem 9.2 (Factorisation, roots, and consistency)**.** *Modulo thirteen,*

$$\chi(\lambda) \equiv (\lambda - 7)(\lambda^2 + 6\lambda + 2), \tag{50}$$

the quadratic irreducible because its discriminant $28 \equiv 2$ *is a nonresidue, with roots*

$$\lambda_1 = 7, \qquad \lambda_2 = 10 + 7\theta, \qquad \lambda_3 = 10 - 7\theta, \qquad \lambda_2 + \lambda_3 = 7, \qquad \lambda_2\lambda_3 = 2, \tag{51}$$

and the two consistency relations

$$\lambda_1 + \lambda_2 + \lambda_3 \equiv 1 = \operatorname{tr} T_3, \qquad \lambda_1\lambda_2\lambda_3 = 14 \equiv 1 = \det T_3. \tag{52}$$

The Frobenius $\sigma(x) = x^{13}$ *fixes* $\mathbb{F}_{13}$ *and satisfies*

$$\theta^{12} = 2^6 = 64 \equiv -1, \qquad \theta^{13} \equiv -\theta, \qquad \sigma(\lambda_2) = \lambda_3, \qquad \sigma^2 = \mathrm{id}. \tag{53}$$

Proof. Evaluating χ at every residue gives a single zero at seven, since $343 - 49 - 7 - 1 = 286 = 22 \cdot 13$; dividing leaves the quadratic, whose discriminant is absent from the residue list. Substituting $10 + 7\theta$ and reducing gives zero, and $\theta \mapsto -\theta$ gives the conjugate; the sum is $20 \equiv 7$ and the product $100 - 98 = 2$. The relations (52) compare Vieta with the trace and determinant. The Frobenius computation is displayed. □

9.3 Arithmetic of the extension

Proposition 9.3 (Multiplication, conjugation, trace, norm, inverse)**.** *Writing elements as pairs* (a,b) *for* $a+b\theta$ *with* $\theta^2=2$*,*

$$(a,b)\,(c,d)=(ac+2bd,\ ad+bc),\qquad \overline{(a,b)}=(a,-b),\tag{54}$$

$$\operatorname{Tr}(a+b\theta)=2a,\qquad \mathrm{N}(a+b\theta)=a^2-2b^2,\qquad (a,b)^{-1}=\big(a\,\mathrm{N}^{-1},\ -b\,\mathrm{N}^{-1}\big),\tag{55}$$

the trace being surjective onto the prime field and the inverse defined for every nonzero element because the norm never vanishes there. The fixed set of the Frobenius is exactly the pairs with $b=0$*, that is the prime field.*

Proof. The product rule follows from $\theta^2=2$ by expanding $(a+b\theta)(c+d\theta)$. Conjugation is the nontrivial automorphism of Theorem 9.2. The trace is $x+\overline{x}=2a$ and the norm is $x\overline{x}=a^2-2b^2$. The trace attains every residue, since 2 is invertible and a ranges over the field. The inverse follows from $x\overline{x}=\mathrm{N}(x)$, which is nonzero for $x\neq 0$ by the anisotropy of Theorem 48.1. An element is fixed by conjugation exactly when $b=-b$, that is when $b=0$. □

9.4 A radix of minus one lies already in the prime field

Theorem 9.4 (Minus one is a square, two is not)**.** *Since* $13\equiv 1$ *modulo four, minus one is a square in the prime field, and its two radices are*

$$5^2=25\equiv -1,\qquad 8^2=64\equiv -1\pmod{13},\tag{56}$$

so a radix of minus one is the residue 5*, of order four, and adjoining it changes nothing:*

$$\mathbb{F}_{13}\big(\sqrt{-1}\big)=\mathbb{F}_{13}.\tag{57}$$

Two, by contrast, is not a square, the squares being $\{1,3,4,9,10,12\}$*, so the quadratic extension of the construction is generated by a radix of two and by nothing else available:*

$$\mathbb{F}_{13}\big(\sqrt{2}\big)=\mathbb{F}_{169},\qquad \big[\mathbb{F}_{169}:\mathbb{F}_{13}\big]=2,\qquad \mathrm{Gal}\big(\mathbb{F}_{169}/\mathbb{F}_{13}\big)\cong C_2.\tag{58}$$

Proof. A residue is a square exactly when it lies in the image of squaring, and the six squares were listed by direct computation; minus one, that is twelve, is among them and two is not. The two radices of minus one were found by solving $x^2=12$, and the order of five was computed to be four. Adjoining an element already present gives the same field, which is (57); adjoining a nonsquare gives the quadratic extension, whose group over the prime field is generated by the Frobenius and has order two. □

Remark 9.5 (What the radix of minus one is, and is not). The element $\sqrt{-1}$ of this construction is the residue five of the prime field. It is not an imaginary unit and it generates no extension: the field $\mathbb{F}_{13}$ already contains it, and the only quadratic step taken anywhere in this paper is the one generated by a radix of two. In particular no complex number enters at this point, or at any other.

Proposition 9.6 (The order of the radix of two)**.** *In the multiplicative group of the extension the radix of two has order*

$$\mathrm{ord}(\theta)=24,\qquad \theta^2=2,\tag{59}$$

which is also the order of the difference operator on its image, Proposition 15.2, and one of the three periods of Theorem 25.1.

Proof. The order was computed by repeated multiplication in $\mathbb{F}_{169}$ with $\theta^2=2$, and the coincidence with the third period is the value recorded there. □

9.5 Why two channels and not one or three

Remark 9.7. The type $(1, 2)$ is not a design decision. A linear factor gives a rank one piece on which the generator acts by one residue; an irreducible quadratic gives a rank two piece whose directions are exchanged by σ and cannot be separated over the prime field.

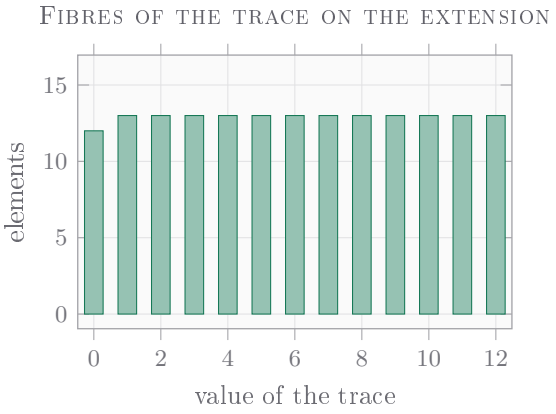


Figure 4: The trace $\mathrm{tr}_{\mathbb{F}_{169}/\mathbb{F}_{13}}(a+b\theta) = 2a$ takes every residue, each on thirteen elements except the value zero, which is taken on twelve nonzero elements together with the zero element.

Structure 10. The spectral splitting

Theorem 10.1 (The two idempotents). *The quadratic factor at $\lambda = 7$ equals $49{+}42{+}2 = 93 \equiv 2$, whose inverse is seven, so*

$$P_1 = 7(T_3^2 + 6T_3 + 2I) = I + 3T_3 + 7T_3^2 = \begin{bmatrix} 1 & 7 & 10 \\ 3 & 8 & 4 \\ 7 & 10 & 5 \end{bmatrix}, \tag{60}$$

$$P_{23} = I - P_1 = 10T_3 + 6T_3^2 = \begin{bmatrix} 0 & 6 & 3 \\ 10 & 6 & 9 \\ 6 & 3 & 9 \end{bmatrix}, \tag{61}$$

and these satisfy

$$P_1^2 = P_1, \quad P_{23}^2 = P_{23}, \quad P_1 P_{23} = P_{23} P_1 = 0, \quad P_1 + P_{23} = I, \tag{62}$$

$$\operatorname{rank} P_1 = 1, \quad \operatorname{rank} P_{23} = 2, \quad |\operatorname{im} P_1| \cdot |\operatorname{im} P_{23}| = 13 \cdot 169 = 2197 = 13^3, \tag{63}$$

$$T_3 P_1 = 7P_1, \qquad [T_3, P_1] = 0. \tag{64}$$

Proof. The displayed normalisation is the complementary factor evaluated at the matrix and scaled to act as the identity on its piece. Coprimality gives $P_1 P_{23} = 0$; the normalised factors sum to one modulo χ, giving completeness; idempotence follows from $P_1(P_1 + P_{23}) = P_1$. The ranks are the degrees of the factors. The eigenvalue relation follows from (7), and P_1 commutes with T_3 since it is a polynomial in T_3. □

10.1 Evolution splits along the channels

Theorem 10.2 (Channelwise evolution). *For every exponent $n \geq 1$,*

$$T_3^n = 7^n P_1 + \big(T_3 P_{23}\big)^n, \tag{65}$$

so the rank one channel evolves by the scalar 7^n over the prime field, of period twelve, while the rank two channel evolves by the restricted operator. The identity was verified for a full range of exponents.

Proof. Since P_1 and P_{23} are orthogonal idempotents summing to the identity and both commute with the generator, the powers split as $T_3^n = (T_3P_1)^n + (T_3P_{23})^n$, and $T_3P_1 = 7P_1$ with $P_1^n = P_1$ gives the first term. □

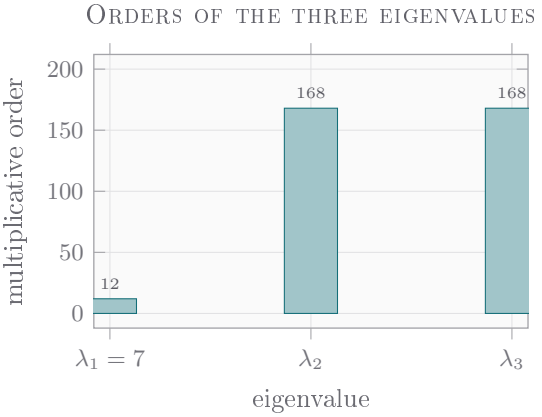


Figure 5: The eigenvalue of the rank one channel has order twelve in $\mathbb{F}_{13}^*$, while the two conjugate eigenvalues have order one hundred sixty eight in $\mathbb{F}_{169}^*$. The quotient $168/12 = 14$ is the order of the torus.

Proposition 10.3 (Channel bases and the block)**.** *The rank one image is spanned by $v_0 = (1, 3, 7)$, with $T_3v_0 = 7v_0$ and $P_1v_0 = v_0$. The rank two image has basis $v_1 = (6, 1, 0)$, $v_2 = (3, 0, 1)$, and in that basis*

$$T_3\big|_{\operatorname{im} P_{23}} = \begin{bmatrix} 6 & 4 \\ 1 & 1 \end{bmatrix}, \qquad \operatorname{tr} = 7, \qquad \det = 2, \tag{66}$$

whose trace and determinant are the coefficients of the quadratic factor.

Proof. Applying the generator to v_0 gives $(7, 8, 10)$, which is $7v_0$ modulo thirteen, and $P_1v_0 = v_0$ by (79). For the plane, $T_3v_1 = 6v_1 + v_2$ and $T_3v_2 = 4v_1 + v_2$, obtained by solving in the basis, and these coordinate vectors are the columns of (66); its trace is $6 + 1 = 7$ and its determinant $6 \cdot 1 - 4 \cdot 1 = 2$, which are the coefficients of the quadratic factor of (50) up to sign. □

Theorem 10.4 (Descent of the three idempotents)**.** *Over $\mathbb{F}_{169}$ the three distinct roots give three primitive idempotents by interpolation,*

$$P_i = \prod_{j \neq i} \frac{T_3 - \lambda_j I}{\lambda_i - \lambda_j}, \qquad P_i^2 = P_i, \qquad P_iP_j = 0 \ (i \neq j), \qquad P_1 + P_2 + P_3 = I, \tag{67}$$

the first equal to (60)*, and the Frobenius acts by*

$$\sigma(P_1) = P_1, \qquad \sigma(P_2) = P_3, \qquad \sigma(P_3) = P_2, \qquad P_2 + P_3 = P_{23}. \tag{68}$$

Hence exactly P_1 and P_{23} are defined over the prime field. Moreover the generator does not act on $\operatorname{im} P_{23}$ by any scalar: $T_3P_{23} \neq cP_{23}$ for all thirteen residues c.

Proof. The roots are distinct by (8), so the interpolation is defined and gives the spectral resolution, computed entirely in $\mathbb{F}_{169}$ using $\theta^2 = 2$. Applying σ entrywise permutes the idempotents as it permutes the roots; a sum of an exchanged pair is fixed and evaluates to P_{23}. The last statement was checked for every residue. □

Role in the architecture. A token separates into two channels at no cost, since both projectors are polynomials in the generator already stored, and the separation survives the whole dynamics because both commute with the generator. The prime field sees only the sum of the two conjugate directions, which is why the rank two channel acts by a trace and a norm and not by a single scale.

Structure 11. The two projectors in full

This structure derives both projectors from the factorisation, gives each as an outer product of eigenvectors with a dual basis, and records their fibres. Both are polynomials in the one operator and are therefore not additional matrices.

11.1 A single formula for all three eigenvectors

Theorem 11.1 (Uniform eigenvector). *For any root λ of χ, in $\mathbb{F}_{13}$ or in $\mathbb{F}_{169}$, the vector*

$$v(\lambda) = \Big(1,\ \frac{1+\lambda}{\lambda},\ \lambda\Big) \tag{69}$$

satisfies $T_3\, v(\lambda) = \lambda\, v(\lambda)$. At the three roots this gives

$$v(\lambda_1) = (1,3,7), \qquad v(\lambda_2) = \big(1,\ 6+3\theta,\ 10+7\theta\big), \qquad v(\lambda_3) = \big(1,\ 6-3\theta,\ 10-7\theta\big), \tag{70}$$

the first being the integer eigenvector v_0 of Theorem 12.1, and the last two being exchanged by the Frobenius.

Proof. Writing $T_3 v = (v_3,\ v_1+v_3,\ v_2+v_3)$ and setting this equal to λv gives the three equations $v_3 = \lambda v_1$, $v_1 + v_3 = \lambda v_2$, and $v_2 + v_3 = \lambda v_3$. Normalising $v_1 = 1$ gives $v_3 = \lambda$ from the first and $v_2 = (1+\lambda)/\lambda$ from the second, while the third reduces to $1+\lambda+\lambda^2 = \lambda^3$, which is the defining relation and therefore holds. The three evaluations were carried out in $\mathbb{F}_{169}$ with $\theta^2 = 2$, and at $\lambda_1 = 7$ the formula returns $(1, 8\cdot 7^{-1}, 7) = (1,3,7)$. □

11.2 The rank one projector as an outer product

Theorem 11.2 (Right and left vectors). *With $v_0 = (1,3,7)$ and the row vector $u = (1,7,10)$ satisfying $uT_3 = 7u$ and $u \cdot v_0 = 1$,*

$$P_1 = v_0\, u = \begin{bmatrix}1\\3\\7\end{bmatrix}\begin{bmatrix}1 & 7 & 10\end{bmatrix} = \begin{bmatrix}1 & 7 & 10\\ 3 & 8 & 4\\ 7 & 10 & 5\end{bmatrix}, \qquad P_1 v = (u\cdot v)\, v_0, \tag{71}$$

so the projection is one inner product followed by one scaling, and the row u is the first row of the matrix.

Proof. Solving $uT_3 = 7u$ over $\mathbb{F}_{13}$ gives a one dimensional space of row vectors, and exactly one of them satisfies $u \cdot v_0 = 1$, namely the one displayed, since $1 + 21 + 70 = 92 \equiv 1$. The matrix $v_0 u$ is then idempotent, because $(v_0 u)^2 = v_0 (u \cdot v_0) u = v_0 u$, has image the line spanned by v_0, and satisfies $(v_0 u) v_0 = v_0$; a rank one idempotent is determined by its image and its kernel, so it coincides with P_1, and comparing entries confirms this. Associativity gives $P_1 v = v_0 (u \cdot v) = (u \cdot v) v_0$. □

11.3 The rank two projector as a sum of two outer products

Theorem 11.3 (Dual basis of the semantic plane). *Let $B = [\,v_0 \,|\, v_1 \,|\, v_2\,]$ with $v_1 = (6,1,0)$ and $v_2 = (3,0,1)$, and let the rows of B^{-1} over $\mathbb{F}_{13}$ be u, w_1, w_2. Then*

$$u = (1,7,10), \qquad w_1 = (10,6,9), \qquad w_2 = (6,3,9), \tag{72}$$

with the dual basis relations

$$w_i \cdot v_j = \delta_{ij}, \qquad w_i \cdot v_0 = 0, \qquad u \cdot v_0 = 1, \qquad u \cdot v_i = 0, \tag{73}$$

where δ_{ij} is the Kronecker symbol, equal to one when $i = j$ and to zero otherwise, and is not the silver element δ of Theorem 48.2; all nine products were evaluated over $\mathbb{F}_{13}$, and the rank two projector splits as a sum of two rank one outer products,

$$P_{23} = v_1 w_1 + v_2 w_2 = \begin{bmatrix} 0 & 6 & 3 \\ 10 & 6 & 9 \\ 6 & 3 & 9 \end{bmatrix}, \qquad P_{23} v = (w_1 \cdot v)\, v_1 + (w_2 \cdot v)\, v_2. \tag{74}$$

Together with (71) *this gives the resolution of the identity*

$$I = v_0 u + v_1 w_1 + v_2 w_2, \tag{75}$$

the three terms being the three coordinates of a state in the basis B.

Proof. The matrix B has determinant a unit modulo thirteen, so it is invertible; inverting it over $\mathbb{F}_{13}$ gives the three rows displayed, and the relations (73) are the defining property of the inverse. A projection onto the span of v_1, v_2 along v_0 is then the sum of the two rank one terms, and comparing entries with (61) confirms the equality. Summing all three terms gives $B B^{-1} = I$. □

11.4 Fibres and images

Proposition 11.4 (The two projections counted). *The rank one projection has thirteen images, each with a fibre of one hundred sixty nine states; the rank two projection has one hundred sixty nine images, each with a fibre of thirteen states; and in both cases the product is*

$$13 \cdot 169 = 2197 = 13^3. \tag{76}$$

Both counts were obtained by tabulating the images of all two thousand one hundred ninety seven states.

Proof. A projection with image of size m has fibres equal to the cosets of its kernel, all of the same size $|\mathbb{F}_{13}^3|/m$. Applying this with $m = 13$ and $m = 169$ gives the two statements, and both were confirmed by tabulating the images of all states. □

11.5 Over the extension the rank two projector splits again

Proposition 11.5 (Three primitive idempotents). *Over $\mathbb{F}_{169}$ the interpolation* (67) *gives three primitive idempotents with $P_2 + P_3 = P_{23}$, the Frobenius fixing P_1 and exchanging P_2 with P_3, as in Theorem 10.4. Their right eigenvectors are $v(\lambda_2)$ and $v(\lambda_3)$ of* (70)*, so over the extension the resolution* (75) *refines into three rank one terms, and over the prime field only the sum of the last two is defined.*

Proof. This is Theorem 10.4 together with Theorem 11.1: the interpolation idempotents are attached to the three roots, the Frobenius permutes them as it permutes the roots, and the sum of the exchanged pair is fixed and equals P_{23}. □

Role in the architecture. Both channels are read by inner products with fixed rows: one row for the address channel and two for the content channel. The engine stores three rows and three columns, and every projection in the architecture is one of these products.

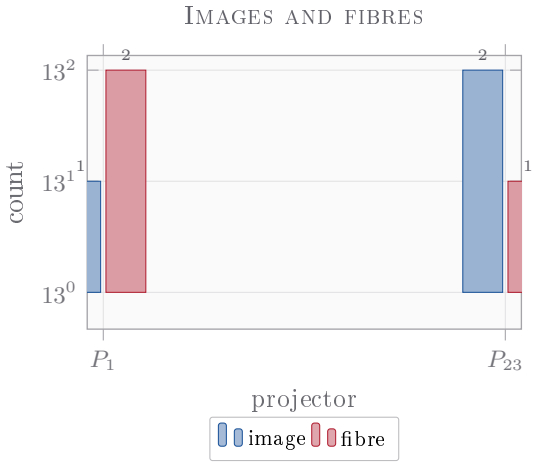


Figure 6: Each projector has image and fibre whose sizes multiply to the state count: thirteen times one hundred sixty nine for the first, and the reverse for the second.

Structure 12. The invariant line

The rank one channel is a single line of the state space, and this structure records everything about it: that it is unique, that the projector onto it is an outer product of two eigenvectors, how the generator acts on it, and how it lifts through the tower of depths.

12.1 The eigenvalue and the eigenline are unique

Theorem 12.1 (One eigenvalue, one line)**.** *Over $\mathbb{F}_{13}$ the generator has exactly one eigenvalue, namely $\lambda_1 = 7$, and its eigenspace is exactly one line,*

$$v_0 = (1, 3, 7), \qquad T_3 v_0 = (7, 8, 10) = 7\, v_0, \tag{77}$$

there being twelve nonzero vectors satisfying $T_3 v = 7v$, that is one line with its twelve nonzero multiples. No other residue is an eigenvalue.

Proof. An eigenvalue in the prime field is a root of χ there, and by Theorem 9.2 the only such root is seven, the remaining factor being irreducible. Solving $T_3 v = 7v$ over $\mathbb{F}_{13}$ by exhaustive search returns exactly twelve nonzero solutions, all multiples of v_0, so the eigenspace is one dimensional; scanning all thirteen residues confirms that seven is the only eigenvalue. □

12.2 The projector is an outer product

Theorem 12.2 (Left eigenvector and the factorisation of P_1)**.** *The generator has a left eigenvector for the same eigenvalue, normalised against v_0,*

$$u = (1, 7, 10), \qquad u\, T_3 = 7\, u, \qquad u \cdot v_0 = 1 \cdot 1 + 7 \cdot 3 + 10 \cdot 7 = 92 \equiv 1, \tag{78}$$

and the rank one projector is the outer product of the two,

$$P_1 = v_0\, u, \qquad \textit{that is} \qquad P_1 v = (u \cdot v)\, v_0 \quad \textit{for every state } v. \tag{79}$$

The second form was verified on all two thousand one hundred ninety seven states. In particular the row vector u is the first row of the matrix (60)*.*

Proof. Solving $uT_3 = 7u$ over $\mathbb{F}_{13}$ returns twelve nonzero row vectors, one line, and exactly one of them satisfies $u \cdot v_0 = 1$; it is the vector displayed. A rank one idempotent with image the line spanned by v_0 and with $u \cdot v_0 = 1$ is necessarily $v_0 u$, since $(v_0 u)^2 = v_0 (u v_0) u = v_0 u$ and $(v_0 u) v_0 = v_0$; comparing entries with (60) confirms the equality, and the action $P_1 v = (u \cdot v) v_0$ follows by associativity. □

Remark 12.3 (What the projection does). Reading (79), the rank one projection is one inner product followed by one scaling: the residue $u \cdot v$ is the coordinate of the state along the invariant direction, and the projection places that residue on v_0. The cost is three multiplications and two additions, and no matrix is applied.

12.3 The line as a set, and the period of the action

Proposition 12.4 (Thirteen states, period twelve)**.** *The image of the rank one projector is the set of thirteen multiples of v_0,*

$$\operatorname{im} P_1 = \{k\, v_0 : k \in \mathbb{F}_{13}\}, \qquad |\operatorname{im} P_1| = 13, \tag{80}$$

and the generator acts on it by the scalar seven, so

$$T_3^n v_0 = 7^n v_0, \qquad \operatorname{ord}(7) = 12, \qquad T_3^{12} \textit{ fixes the line pointwise.} \tag{81}$$

The line therefore returns after twelve steps, while the whole orbit returns after one hundred sixty eight, the quotient being the order fourteen of the torus, as in Theorem 48.1.

Proof. The image of a rank one projector with image the span of v_0 is the set of its multiples, of which there are thirteen. The action is $T_3^n v_0 = 7^n v_0$ by iteration of (77), verified for a full range of exponents, and seven has order twelve in $\mathbb{F}_{13}^*$, so $7^{12} = 1$ and the twelfth power fixes every multiple. □

12.4 The fixed class at the boundary

Proposition 12.5 (The eigenline has a fixed boundary class)**.** *Under the boundary map of Theorem 36.6 the eigenvector lies in the class*

$$\operatorname{bd}(v_0) = [1 : 3] = \operatorname{bd}(T_3 v_0), \tag{82}$$

so the class of the eigenline is fixed by the generator. This is consistent with Theorem 40.1, where the class of v_0 is the unique fixed point of the projective action.

Proof. Applying the boundary map to $v_0 = (1, 3, 7)$ gives $[1 : 3 \cdot 1^{-1}] = [1 : 3]$, and $T_3 v_0 = 7 v_0 = (7, 21, 49)$, whose class is $[1 : 21 \cdot 7^{-1}] = [1 : 3]$ as well, since $21 \cdot 7^{-1} = 3$. □

Remark 12.6 (The class of the eigenline is fixed, and the channel splitting is preserved). Although the class of the eigenline is fixed, the partition of Theorem 36.6 is not preserved as a whole: there are classes whose states are carried by the generator into more than one class, verified by testing every class. The boundary partition is therefore a classification of states and not an invariant decomposition, unlike the channel splitting of Theorem 10.1, which is preserved because both projectors commute with the generator.

12.5 Lifting the line through the tower of depths

Proposition 12.7 (The eigenvalue lifts uniquely). *The derivative of the defining cubic at the eigenvalue is a unit,*

$$\chi'(7) = 3 \cdot 49 - 2 \cdot 7 - 1 = 132 \equiv 2 \pmod{13}, \tag{83}$$

so by Proposition 34.4 the eigenvalue lifts uniquely to every depth,

$$7 \;\to\; 33 \;\to\; 202 \;\to\; 26566 \qquad \textit{exact modulo } 13,\ 169,\ 2197,\ 28561, \tag{84}$$

and with it the invariant line lifts to a line over each $\mathbb{Z}/13^k$ and finally over $\mathbb{Z}_{13}$. The multiplicative lift of the eigenvalue is the representative $\omega(7) \equiv 150$ modulo 169 of Proposition 34.2, which is the unique lift of order twelve.

Proof. The derivative is $\chi'(\lambda) = 3\lambda^2 - 2\lambda - 1$, and at seven this is $147 - 14 - 1 = 132 \equiv 2$, a unit. Proposition 34.4 then gives a unique lift at every depth, and the displayed values are those computed there. The eigenline is the kernel of $T_3 - \lambda I$ at the lifted eigenvalue, which is a free rank one summand at every depth because the derivative is a unit. □

Role in the architecture. The rank one channel is a single direction that never moves, its projector is one inner product with a fixed row vector, and its dynamics is multiplication by one residue of period twelve. It is the only part of the state space on which the generator acts by a number, and it is the only direction that survives to every depth as a line.

12.6 The conic, the ray, and the element of order seven

Three objects around the eigenvector are often conflated. This subsection separates them and states what is true of each.

Theorem 12.8 (The conic has fourteen projective points and the eigenvector is one of them). *The equation $x^2 + y^2 = z^2$ has one hundred sixty eight nonzero solutions over the prime field, Theorem 53.1, which fall into*

$$\frac{168}{12} = 14 \tag{85}$$

projective points, since each point carries its twelve nonzero multiples. The class of the eigenvector is one of these fourteen, by Theorem 50.1.

Proof. The solutions were enumerated and normalised; the normalisation returns fourteen distinct classes, and the class of $(1, 3, 7)$ is among them. □

Theorem 12.9 (The operator does not preserve the conic). *Writing $Q = \operatorname{diag}(1, 1, -1)$ for the matrix of the form, the operator does not preserve it:*

$$T_3^{\top} Q\, T_3 = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 12 & 12 \\ 1 & 12 & 1 \end{bmatrix} \neq Q, \tag{86}$$

and correspondingly the set of fourteen projective points of Theorem 12.8 is not mapped into itself. The eigenvector lies on the conic, but the conic is not an invariant object of the dynamics.

Proof. The product was formed over $\mathbb{F}_{13}$ and differs from Q in four entries. Applying the operator to the fourteen classes and normalising produces classes outside the original set. □

Remark 12.10 (What is invariant instead). The invariant objects of the projective action are one point and one line, by Corollary 36.8, and the invariant subspaces are the chain of four of Theorem 21.2. A quadratic form invariant under the operator does not exist here, because such a form would make the operator orthogonal, whereas its characteristic polynomial is not self reciprocal: $\chi(\lambda) = \lambda^3 - \lambda^2 - \lambda - 1$ has constant term -1 but its reversal is $-\lambda^3 - \lambda^2 - \lambda + 1$.

Theorem 12.11 (The ray through the eigenvector, and its stabiliser)**.** *The twelve nonzero multiples of v_0 form a set carried to itself by the operator, which acts on it by the residue seven,*

$$T_3\big(k\,v_0\big) = 7k\,v_0, \tag{87}$$

and the element of order seven fixes every one of them,

$$U_7 = T_3^{24}, \qquad U_7\big(k\,v_0\big) = k\,v_0 \quad \text{for all } k, \tag{88}$$

since $7^{24} = 1$. The subgroup it generates has seven elements, at the exponents

$$0,\ 24,\ 48,\ 72,\ 96,\ 120,\ 144, \tag{89}$$

and its index in the orbit is twenty four.

Proof. The first identity is linearity together with $T_3 v_0 = 7 v_0$. For the second, 7 has order twelve and twelve divides twenty four, so the twenty fourth power of the residue is one; this was checked on all twelve multiples. The seven exponents and the index follow from Theorem 20.1. □

Theorem 12.12 (Where the number seven really appears around the eigenvector)**.** *Acting on the one hundred eighty three projective points, the subgroup of order seven has exactly*

$$183 = 1 + 26 \cdot 7 \tag{90}$$

orbits of two sizes: one fixed point, namely the class of the eigenvector, and twenty six orbits of seven points each. The number seven therefore appears as the common size of every nontrivial orbit of that subgroup, and the fixed point is the eigenvector.

Proof. The orbits were computed by applying the subgroup generator repeatedly to each projective point and normalising. The tally returns one orbit of size one and twenty six of size seven, and $1 + 182 = 183$. □

Remark 12.13 (The reading, stated plainly). Theorem 12.12 is the precise form of the observation that seven objects surround the fixed point: there are twenty six such families, each of seven projective points, and the eigenvector is the unique point fixed by the whole subgroup. Nothing further is asserted, and in particular no reading involving a speed, a light cone, or a physical constant is available, since the construction admits no quantity that could carry such a name; see Definition 1.1 and Proposition 1.3.

Remark 12.14 (Two numbers and where they belong). Two further claims are false and are recorded so that they are not repeated. First, in the sum $\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{64} = \frac{63}{64}$ the part missing from one is $\frac{1}{64}$ and not seven. Second, the field of sixty four elements has characteristic two and does not occur in this construction, whose fields all have characteristic thirteen; moreover $64 \equiv 12$ and $63 \equiv 11$ modulo thirteen, neither being seven, and neither sixty three nor sixty four divides the period.

12.7 The seven phases and what each one does

Theorem 12.15 (Every phase is the identity on the eigenline)**.** *Let $c = U_7 = T_3^{24}$, of order seven. Then all seven phases act as the identity on the eigenline,*

$$c^k v_0 = v_0 \qquad \text{for } k = 0, 1, \ldots, 6, \tag{91}$$

not merely the seventh, since already c itself fixes it by Theorem 12.11. Off the eigenline the phases are distinct: applying them to the first basis vector gives

$$\begin{aligned} &(1,0,0),\ (3,11,9),\ (2,10,2),\ (6,5,9),\\ &(9,10,0),\ (0,11,8),\ (12,0,8), \end{aligned} \tag{92}$$

seven distinct states returning to the first at the seventh step.

Proof. The eigenline is fixed pointwise by c, hence by all its powers, which gives (91). The seven images in (92) were computed and are pairwise distinct, and $c^7 = I$ returns the first. □

Remark 12.16 (The subgroup, rather than a single phase, is the invariant). Since all seven phases agree on the eigenline and are pairwise distinct off it, no single phase plays a special role there; what is distinguished is the subgroup as a whole, whose average is the projector, Theorem 12.17. In particular the seventh phase is the identity, which is the trivial member of the subgroup, not a limiting case of anything.

12.8 Averaging over the subgroup of order seven

Theorem 12.17 (The average of the seven powers is the rank one projector)**.** *Let $U_7 = T_3^{24}$, of order seven. Then the average of its powers over the subgroup it generates is exactly the rank one projector,*

$$\frac{1}{7}\sum_{k=0}^{6} U_7^k = P_1, \tag{93}$$

where $7^{-1} = 2$ in $\mathbb{F}_{13}$. In particular the average is idempotent of rank one, it is absorbed by the subgroup,

$$P_1 U_7 = U_7 P_1 = P_1, \tag{94}$$

and it annihilates every direction on which the subgroup acts without a fixed vector.

Proof. The seven powers were summed over $\mathbb{F}_{13}$, giving the matrix with rows $(7,10,5)$, $(8,4,2)$, $(10,5,9)$, and multiplying by $7^{-1} = 2$ returns the matrix (60) of P_1. Structurally, the average over a finite group of an action is the projection onto the invariants, and by Theorem 12.11 the invariants of U_7 are exactly the eigenline, of dimension one; the projection onto it along the complementary invariant subspace is P_1. The absorption identity holds because averaging is unchanged by translating the index. □

Corollary 12.18 (A projector obtained without the polynomial)**.** *Equation* (93) *gives the rank one projector as an average of powers of the operator, with no reference to the interpolation of Theorem 10.1. Since $P_{23} = I - P_1$, the complementary projector follows, so both channels are obtained from the single subgroup of order seven.*

Proof. The average is a linear combination of powers of T_3, hence lies in the algebra, and it equals P_1 by Theorem 12.17; the complement is the definition of P_{23}. □

Remark 12.19 (The rank of an average is the dimension of the fixed space). Averaging over a subgroup returns the projection onto its invariants, so the rank of the result is the dimension of the fixed space. The subgroup of order seven fixes exactly the eigenline, by Theorem 12.11, which is why the average has rank one. Averaging over the whole orbit would return the projection onto the vectors fixed by every power, and there are none other than zero, so that average vanishes; this is the vanishing of the period sum recorded in Proposition 20.3.

12.9 The averaging operator separates the two channels

Theorem 12.20 (What the average recovers, and what it discards). *Write* $\chi = \frac{1}{7}\sum_{k=0}^{6} U_7^k = P_1$ *for the operator of Theorem 12.17. Applied to a state it returns exactly the rank one component and discards the rank two component:*

$$\chi\big(P_1 v + P_{23} v\big) = P_1 v, \qquad \chi\, P_{23} = 0, \tag{95}$$

so the two channels of Theorem 10.1 are separated by one application. Its kernel and image have sizes

$$|\ker \chi| = 169, \qquad |\operatorname{im} \chi| = 13, \qquad 169 \cdot 13 = 2197, \tag{96}$$

the kernel being the rank two image and the image the eigenline.

Proof. The average is the projection onto the invariants of the subgroup, which is the eigenline, so it acts as the identity there and annihilates a complement; the complement invariant under the operator is the rank two image, by Theorem 10.1. The two sizes were obtained by tabulating the operator on all states, and their product is the state count by (111). □

Remark 12.21 (A projection and a transform, each with its own role). The operator χ is idempotent of rank one, so it is not invertible and no state can be recovered from its image alone: one hundred sixty nine states are sent to zero. It therefore separates the channels but does not transform between representations, and it is not a substitute for the transform of Structure 46, whose defining property is invertibility,

$$W\, W^{-} = 12\, I_{12} \quad \text{over } \mathbb{F}_{13}, \tag{97}$$

verified for the transform of size twelve with root two. The two operators do different things: χ selects a channel, the transform turns a convolution into a pointwise product.

Proposition 12.22 (Three channels over the extension). *Over* $\mathbb{F}_{169}$ *the three eigenvalues give three independent one dimensional channels, since*

$$\lambda_1 + \lambda_2 + \lambda_3 = 1 = \operatorname{tr} T_3, \qquad \lambda_1\lambda_2\lambda_3 = 1 = \det T_3, \tag{98}$$

and the associated idempotents satisfy $P_A + P_B + P_C = I$ *with pairwise products zero, by Proposition 11.5. Over the prime field only the first is defined separately, the other two being exchanged by the Frobenius and available only as their sum* P_{23}.

Proof. The sum and product of the roots are the coefficients of χ up to sign, evaluated in $\mathbb{F}_{169}$ with $\theta^2 = 2$. The idempotent statement is Proposition 11.5, and the descent is Theorem 10.4. □

12.10 The stabiliser of the eigenline inside the orbit

Theorem 12.23 (Fourteen powers fix the eigenvector). *The powers of the generator that fix the eigenvector pointwise are exactly the multiples of twelve,*

$$T_3^{k} v_0 = v_0 \iff 12 \mid k, \qquad k \in \{0, 12, 24, \ldots, 156\}, \tag{99}$$

fourteen exponents in all, forming the subgroup $C_{14} = \langle T_3^{12} \rangle$ of Theorem 22.2. In particular the Sylow subgroup of order seven acts trivially on the eigenline,

$$U_7 = T_3^{24}, \qquad U_7 v_0 = 7^{24} v_0 = v_0, \tag{100}$$

since 7 has order twelve and twelve divides twenty four.

Proof. By Proposition 12.4 the generator acts on the eigenline by the residue seven, so $T_3^k v_0 = 7^k v_0$, and this equals v_0 exactly when $7^k = 1$, that is when twelve divides k. The multiples of twelve below the period are the fourteen listed, and they form the subgroup of order fourteen because $168/12 = 14$. Both statements were confirmed by scanning all exponents. □

Corollary 12.24 (The rank one channel sees only the clock of twelve)**.** *The quotient of the orbit by the stabiliser of the eigenline is cyclic of order twelve, so the rank one channel distinguishes only twelve of the one hundred sixty eight positions; the remaining information is carried entirely by the rank two channel. The two clocks are the twelve and the fourteen of Theorem 25.1, and* $12 \cdot 14 = 168$.

Proof. The stabiliser has order fourteen by Theorem 12.23, so the quotient has order $168/14 = 12$, and it is cyclic as a quotient of a cyclic group. The product identity is Theorem 48.1. □

12.11 The coordinate orbit of the eigenvector

Theorem 12.25 (Six boundary classes from one vector)**.** *Permuting the coordinates of the eigenvector $v_0 = (1, 3, 7)$ gives six distinct vectors, and their boundary classes under the map of Theorem 36.6 are six distinct points of the projective line,*

$$\begin{aligned} &(1,3,7) \mapsto [1{:}3], \quad (1,7,3) \mapsto [1{:}7], \quad (3,1,7) \mapsto [1{:}9], \\ &(3,7,1) \mapsto [1{:}11], \quad (7,1,3) \mapsto [1{:}2], \quad (7,3,1) \mapsto [1{:}6], \end{aligned} \tag{101}$$

so the orbit is free and accounts for six of the fourteen classes. The six second coordinates $\{2, 3, 6, 7, 9, 11\}$ are exactly the six residues whose inverses appear among them, since $3 \cdot 9 = 27 \equiv 1$, $2 \cdot 7 = 14 \equiv 1$, and $6 \cdot 11 = 66 \equiv 1$.

Proof. The six permuted vectors were formed and their classes computed by dividing the second coordinate by the first. All six classes are distinct, so the coordinate group acts freely on this vector. The three inverse pairs were checked in $\mathbb{F}_{13}$. □

Remark 12.26 (Only the identity preserves the operator). By Theorem 58.31 no permutation other than the identity conjugates the operator to itself, so the six vectors of (101) are eigenvectors of six different conjugates of the operator, not of the operator itself; only $(1, 3, 7)$ is its eigenvector.

12.12 Every idempotent that the one operator produces

Theorem 12.27 (Four over the prime field, eight over the extension)**.** *The idempotents lying in the algebra generated by the operator over the prime field are exactly four, with coefficient triples and ranks*

$$(0,0,0) \text{ rank } 0, \quad (1,3,7) \text{ rank } 1, \quad (0,10,6) \text{ rank } 2, \quad (1,0,0) \text{ rank } 3, \tag{102}$$

that is 0, P_1, P_{23}, and I; a search over all 2197 coefficient triples returns these and no others. Over the extension the middle one splits and the count doubles to eight, namely the sums of subsets of the three primitive idempotents P_1, P_2, P_3 of Proposition 11.5.

Proof. Every element of the algebra is $aI + bT_3 + cT_3^2$, and the condition of being idempotent was tested for all 13^3 triples, returning the four displayed. Their ranks were computed. Over the extension the algebra is a product of three copies of a field, by Theorem 13.1 together with Theorem 10.4, and a product of n fields has 2^n idempotents, here $2^3 = 8$. □

Remark 12.28 (The idempotents preserved by the dynamics). The address projectors Q, Q_b, Q_c of Theorem 16.3 and the coordinate projectors of Theorem 18.1 are idempotents of the full matrix algebra but not of $\mathbb{F}_{13}[T_3]$, since they do not commute with the operator; by Proposition 14.4 the commutant is exactly $\mathbb{F}_{13}[T_3]$, so the list (102) is complete for objects preserved by the dynamics.

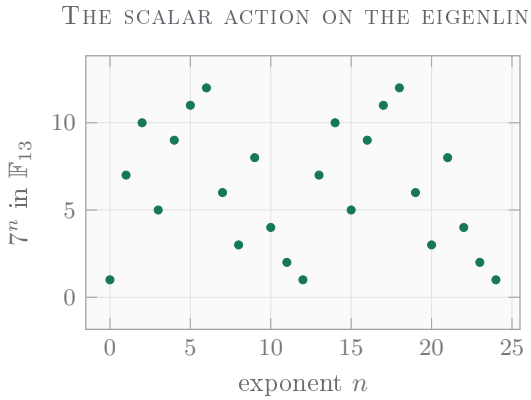


Figure 7: On the eigenline the operator acts by the residue seven, whose powers cycle with period twelve and return to one at every multiple of twelve, by Proposition 12.4.

Structure 13. Census of the algebra

13.1 Order, idempotents, units

Theorem 13.1 (Census). *Let $A = \mathbb{F}_{13}[T_3]$. Then $|A| = 2197$, A is commutative, and evaluation at the two roots gives an isomorphism*

$$A \xrightarrow{\sim} \mathbb{F}_{13} \times \mathbb{F}_{169}, \tag{103}$$

verified to be a bijection on all elements. Hence A has exactly four idempotents, namely 0, P_1, P_{23}, I, exactly

$$(13-1)(169-1) = 12 \cdot 168 = 2016 \tag{104}$$

units, and no nonzero nilpotent; the unit group is $C_{12} \times C_{168}$, not cyclic, of exponent

$$\operatorname{lcm}(12, 168) = 168 < 2016. \tag{105}$$

Proof. The algebra is spanned by I, T_3, T_3^2 and polynomials in one matrix commute. Evaluation at λ_1 and λ_2 is a ring homomorphism, bijective by coprimality of the factors. In a product of two fields the idempotents are the four pairs of zeros and ones, the units are the pairs of nonzero coordinates, and there is no nonzero nilpotent. A product of cyclic groups is cyclic exactly when the orders are coprime, and $\gcd(12, 168) = 12$. □

13.2 The fibres of the two projections

Proposition 13.2 (Fibre counts). *The two projections of Theorem 10.1 have complementary fibre structures over the state space:*

$$\left|\operatorname{im} P_1\right| = 13 \text{ with every fibre of size } 169, \qquad \left|\operatorname{im} P_{23}\right| = 169 \text{ with every fibre of size } 13, \tag{106}$$

and in both cases the product of the two numbers is 2197*. The fibres are the cosets of the complementary image, so they are all of the same size, and each state is recovered from its pair of images.*

Proof. A projection onto a subspace of size m has image of size m and fibres that are cosets of its kernel, of size $2197/m$. Applying this with $m = 13$ and $m = 169$ gives (106); both were confirmed by tabulating the images of all 2197 states. □

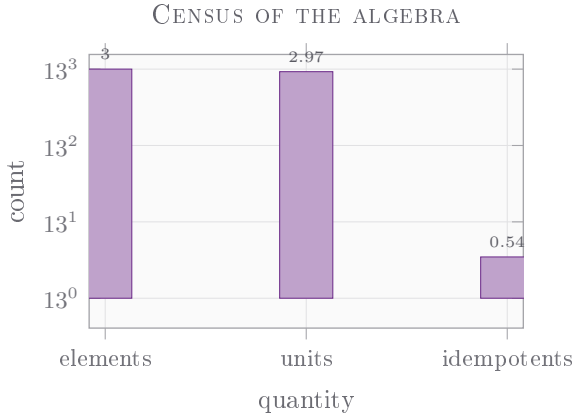


Figure 8: The algebra has 2197 elements, of which 2016 are units and four are idempotent, with no nonzero nilpotent, by Theorem 13.1.

Structure 14. The ring as the single data type

14.1 One type, three presentations

Definition 14.1 (One type, three faces)**.** Every datum is an element of $\mathbb{Z}_{13}[\eta]$ with three interchangeable presentations,

$$a = a_0 + a_1\eta + a_2\eta^2 \;\equiv\; (a_0, a_1, a_2) \;\equiv\; a_0 I + a_1 T_3 + a_2 T_3^2, \qquad a_i \in \mathbb{F}_{13}, \tag{107}$$

namely the ring element, the coordinate triple, and the matrix. Addition is componentwise,

$$(a_0, a_1, a_2) \oplus (b_0, b_1, b_2) = (a_0 + b_0,\; a_1 + b_1,\; a_2 + b_2), \tag{108}$$

and multiplication is the convolution of the two triples reduced by $\eta^3 = \eta^2 + \eta + 1$.

14.2 The folding rule of the product

Proposition 14.2 (Explicit product)**.** *Reducing the convolution of two triples by $\eta^3 = \eta^2+\eta+1$, hence $\eta^4 = 2\eta^2 + 2\eta + 1$, gives*

$$\begin{aligned} r_0 &= a_0b_0 + a_1b_2 + a_2b_1 + a_2b_2, \\ r_1 &= a_0b_1 + a_1b_0 + a_1b_2 + a_2b_1 + 2a_2b_2, \\ r_2 &= a_0b_2 + a_1b_1 + a_1b_2 + a_2b_0 + a_2b_1 + 2a_2b_2, \end{aligned} \tag{109}$$

in which the nine products a_ib_j are formed once and reused. The formula was checked against the matrix product over a family of operand pairs and agrees in every case.

Proof. The convolution of two triples has coefficients $\sum_{i+j=k} a_i b_j$ for $k = 0, \dots, 4$. The relation $\eta^3 = \eta^2 + \eta + 1$ replaces the coefficient at three by adding it to those at zero, one, and two, and squaring the relation gives $\eta^4 = 2\eta^2 + 2\eta + 1$, which replaces the coefficient at four by adding it once at zero and twice at one and two. Collecting terms gives (109), and the result was compared with the matrix product on a family of operand pairs. □

14.3 The exact count of products

Proposition 14.3 (Nine products, fifteen term instances)**.** *The three coordinates of* (109) *contain fifteen term instances but only nine distinct products $a_i b_j$, one for each pair of indices. Forming those nine once and reusing them, the ring product costs*

$$9 \text{ integer multiplications}, \tag{110}$$

together with a fixed number of additions and one multiplication by the constant two, and the count does not depend on the operands.

Proof. Collecting the distinct symbol pairs occurring in (109) returns exactly the nine products $a_i b_j$ with $0 \le i, j \le 2$, while counting terms with repetition returns fifteen. Both counts were obtained symbolically and the formula itself was checked against the matrix product. □

14.4 Cost and closure

Proposition 14.4 (Cost and closure)**.** *The ring product costs nine integer multiplications and a fixed number of additions, independently of the operands, and the ring is commutative. The commutant of T_3 in the full matrix algebra is exactly $\mathbb{F}_{13}[T_3]$, of order*

$$|\mathbb{F}_{13}[T_3]| = 13^3 = 2197. \tag{111}$$

Proof. The convolution of two triples has nine pairwise products, and the folding of the two highest coefficients uses only additions. Polynomials in one matrix commute. Since the module is cyclic by Theorem 3.2, a matrix commuting with T_3 is determined by the image of e_1, which can be written as a polynomial in T_3 applied to e_1, and the two matrices then agree on the whole module. □

Role in the architecture. There is one data type, so a token, a weight, a position, and a memory cell are the same three residues, and no conversion occurs anywhere. Binding is one ring product in constant time with no data dependent branch.

Part III
The antisymmetric splitting and the further projectors

Structure 15. The antisymmetric splitting

Theorem 15.1 (Address projector and exact reconstruction)**.** *Let $D = T_3 - T_3^\top$, so $D^\top = -D$ and $T_3^\top = T_3 - D$. Then*

$$D = \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \qquad \ker D = \langle w \rangle, \quad w = (0, 1, 1), \quad \operatorname{rank} D = 2, \tag{112}$$

and with $w^\top w = 2$ *and* $2^{-1} = 7$*,*

$$Q = 7(D^2 + 2I) = 7\,ww^\top = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 7 & 7 \\ 0 & 7 & 7 \end{bmatrix}, \qquad Q^2 = Q, \qquad \operatorname{rank} Q = 1, \tag{113}$$

with $\operatorname{im} Q = \ker D$ *and* $\operatorname{im}(I - Q) = \operatorname{im} D$*, giving*

$$\mathbb{F}_{13}^3 = \ker D \oplus \operatorname{im} D, \qquad 13 \cdot 169 = 2197. \tag{114}$$

For every state $v = (a, b, c)$*,*

$$Qv = \big(0,\ 7(b+c),\ 7(b+c)\big), \qquad v = Qv + (I - Q)v, \tag{115}$$

both verified on all two thousand one hundred ninety seven states.

Proof. Solving $Dv = 0$ gives $\langle w \rangle$. A vector in both summands is $kw = Du$, and

$$w^\top Du = -(Dw)^\top u = 0, \qquad w^\top (kw) = 2k, \tag{116}$$

so $k = 0$. Since $D^2 + 2I = ww^\top$, we get $Q^2 = 49\,w(w^\top w)w^\top = 98\,ww^\top \equiv 7ww^\top = Q$, of rank one with image $\ker D$. □

15.1 The difference operator on its image

Proposition 15.2 (Cubic relation and period twenty four)**.** *The difference operator is singular,* $\det D = 0$*, so it is not invertible on the state space. It satisfies the cubic relation*

$$D^3 = -2D, \tag{117}$$

and since the order of -2 *in* $\mathbb{F}_{13}^*$ *is twelve, the odd powers are* $D^{2m+1} = (-2)^m D$*, whence*

$$D^{25} = D, \qquad \text{so } D \text{ restricted to } \operatorname{im} D \text{ has order } 24. \tag{118}$$

Proof. An antisymmetric matrix of odd size has vanishing determinant. The cubic relation was computed directly, and iterating it gives the stated odd powers; $(-2)^m = 1$ first at $m = 12$, so $D^{25} = D$ and the restriction to the image, where D is invertible, has order twenty four. □

Theorem 15.3 (The two splittings share no basis)**.** *For every matrix* X *over* $\mathbb{F}_{13}$*,*

$$\operatorname{rank}[X, P_1] \le 2, \tag{119}$$

because P_1 *has rank one, and the bound is attained,*

$$[P_1, Q] \neq 0, \qquad \operatorname{rank}[P_1, Q] = 2. \tag{120}$$

Hence the two splittings cannot be brought to a common basis. Moreover $[D, T_3] \neq 0$*, so* D *is not a polynomial in the generator, while over the integers*

$$\det\big[D,\ I + 3T_3 + 7T_3^2\big]_{\mathbb{Z}} = -533 = -13 \cdot 41, \tag{121}$$

so the rank there is three and drops to two on reduction, thirteen occurring exactly once.

Proof. Each of XP_1 and P_1X has rank at most one, and rank is subadditive, so rank three is impossible rather than merely absent. Row reducing $[P_1, Q]$ over $\mathbb{F}_{13}$ leaves two nonzero rows, and the integer determinant is the displayed value. □

Role in the architecture. A token carries an address and a content, and reading or writing either is one projection. The address structure and the channel structure interfere as much as they can, and the ceiling is known before any computation, so no third independent direction of failure exists.

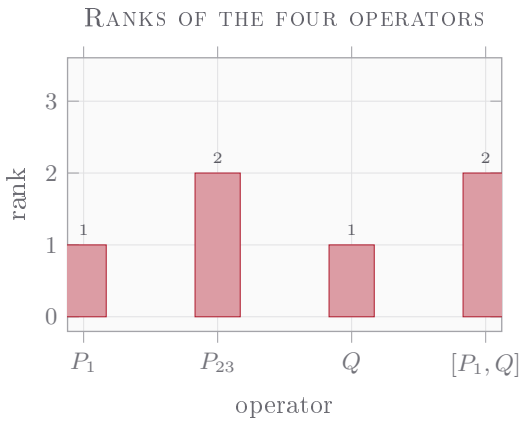


Figure 9: The two spectral projectors have ranks one and two, the address projector rank one, and their commutator rank two, which is the largest a commutator with a rank one idempotent can have.

Structure 16. The three address projectors

The address projector of Structure 15 is one of three, permuted by the coordinate action. This structure exhibits all three and identifies their kernels.

16.1 The coordinate action

Remark 16.1 (The letter σ, named at each use). The letter σ carries three distinct maps in this paper, and each is named where it is used, the arrow notation making the reading unambiguous in every case.

$$\begin{aligned} \sigma \text{ the Frobenius}: &\quad x \longmapsto x^{13} && \text{on } \mathbb{F}_{169}, \\ \sigma \text{ the substitution}: &\quad a \longmapsto ab,\ b \longmapsto ac,\ c \longmapsto a && \text{on words}, \\ \sigma \text{ a permutation}: &\quad i \longmapsto \sigma(i) && \text{on the three coordinates.} \end{aligned} \tag{122}$$

The first acts on field elements, the second on words, the third on indices, so no expression admits two readings. The letter δ is treated the same way and carries two meanings, each named where it is used:

$$\begin{aligned} \delta \text{ the silver element}: &\quad \delta = 1 + \theta, \quad \delta^2 = 2\delta + 1 && \text{in } \mathbb{F}_{169}, \\ \delta_{ij} \text{ the Kronecker symbol}: &\quad \delta_{ij} = 1 \text{ if } i = j, \quad \delta_{ij} = 0 \text{ if } i \neq j && \text{on indices.} \end{aligned} \tag{123}$$

The first is an element of a field and carries no index; the second always carries two indices, so the two never collide.

Definition 16.2 (Permutation matrices)**.** For σ a permutation of the three coordinates let P_σ be the matrix with entries $(P_\sigma)_{ij} = 1$ when $\sigma(i) = j$ and zero otherwise. The six such matrices form a group isomorphic to the symmetric group on three letters, and each is orthogonal, so $P_\sigma^{-1} = P_\sigma^{\top}$.

16.2 Three conjugates, three kernels

Theorem 16.3 (The three projectors)**.** *Conjugating the address projector by the transpositions gives*

$$Q_a = Q, \qquad Q_b = P_{(01)}\, Q\, P_{(01)}, \qquad Q_c = P_{(02)}\, Q\, P_{(02)}, \tag{124}$$

three distinct idempotents of rank one, whose images are the three lines

$$\operatorname{im} Q_a = \langle (0,1,1) \rangle, \qquad \operatorname{im} Q_b = \langle (1,0,1) \rangle, \qquad \operatorname{im} Q_c = \langle (1,1,0) \rangle, \tag{125}$$

each fixed pointwise by its own projector. All three were verified to be idempotent of rank one, and the three matrices are pairwise distinct.

Proof. Conjugation by an invertible matrix preserves idempotence and rank, so each conjugate is again a rank one idempotent. The image of Q is the line spanned by $(0,1,1)$ by Theorem 15.1, and conjugating by a transposition carries that line to the line spanned by the correspondingly permuted vector, giving (125); the three matrices were compared entrywise and are distinct. □

Remark 16.4 (The three address channels stand on the same footing). The three projectors are on the same footing: the construction supplies a triple permuted by the coordinate action and does not single out one of them. Fixing Q_a as the address projector is a choice of coordinate, and every statement about it holds verbatim for the other two after conjugation.

Role in the architecture. The engine has three equivalent address channels, one for each pair of coordinates, and a state may be split along any of them. Passing from one to another is conjugation by a permutation matrix, which costs a relabelling and nothing more.

Structure 17. The Klein group inside the matrix algebra

17.1 Four matrices

Theorem 17.1 (A Klein four group of matrices)**.** *The four matrices*

$$V = \{\, I,\ G,\ -I,\ -G \,\}, \qquad G = T_3^{84} = P_1 - P_{23}, \tag{126}$$

form an abelian group under multiplication over $\mathbb{F}_{13}$*, isomorphic to* $C_2 \times C_2$*, with multiplication table*

$\cdot$	I	G	$-I$	$-G$
I	I	G	$-I$	$-G$
G	G	I	$-G$	$-I$
$-I$	$-I$	$-G$	I	G
$-G$	$-G$	$-I$	G	I

The orders of the four elements are 1, 2, 2, 2 *respectively: the identity has order one and the other three have order two, which is exactly the signature of the Klein four group.*

Proof. Closure and commutativity were checked on all sixteen products. Since $G^2 = I$ by Theorem 20.4 and $(-I)^2 = I$, the products are as displayed, and $(-G)^2 = G^2 = I$. The four orders were computed by repeated multiplication. □

Remark 17.2 (Three elements of order two, and one identity). It is tempting to say that every element of this group has order two. That is false for the identity, whose order is one. The correct statement is that the group has exactly three elements of order two, which together with the identity is what characterises the Klein four group among groups of order four, the other possibility being cyclic with a single element of order two.

Structure 18. The coordinate group and its projectors

18.1 Averaging over the six permutations

Theorem 18.1 (Two commuting projectors from the coordinate action). *Since $6^{-1} = 11$ in $\mathbb{F}_{13}$, the average of the six permutation matrices,*

$$P_{\mathrm{triv}} = 11 \sum_{\sigma} P_{\sigma}, \tag{127}$$

is idempotent of rank one, its complement

$$P_{\mathrm{std}} = I - P_{\mathrm{triv}} \tag{128}$$

is idempotent of rank two, and

$$S = I - 2P_{\mathrm{triv}}, \qquad S^2 = I, \tag{129}$$

is an involution. All three commute with every permutation matrix.

Proof. The order of the group is six, invertible modulo thirteen with inverse eleven, so the average is defined and is the standard idempotent onto the invariants, whence $P_{\mathrm{triv}}^2 = P_{\mathrm{triv}}$ and the complement is idempotent. Its rank is one because the invariants of the coordinate action form the line of constant vectors. Squaring (129) gives $I - 4P_{\mathrm{triv}} + 4P_{\mathrm{triv}} = I$. Commutation with each permutation matrix was verified for all six. □

18.2 The two involutions do not commute

Proposition 18.2 (S and G). *The coordinate involution S and the spectral involution $G = T_3^{84}$ do not commute,*

$$S\,G \neq G\,S, \tag{130}$$

verified over $\mathbb{F}_{13}$. Consequently the two involutions generate a nonabelian group, and the coordinate action cannot be diagonalised simultaneously with the channel splitting.

Proof. Both products were formed over $\mathbb{F}_{13}$ and differ, which gives (130). Two noncommuting involutions generate a dihedral group, which is nonabelian, so no common basis of eigenvectors exists. □

Role in the architecture. Two independent involutions act on the state space: one exchanges coordinates and one reverses the sign of the content channel. They do not commute, so applying them in the two orders gives different results, and the engine must fix an order.

Structure 19. The group of order twenty four and its representations

19.1 Structure

Proposition 19.1 (Centre, derived subgroup, quotients). *For $\mathcal{G} = S_3 \times C_2 \times C_2$ of order twenty four,*

$$Z(\mathcal{G}) = \{e\} \times C_2 \times C_2 \cong V_4, \qquad \mathcal{G}' = A_3 \cong C_3, \tag{131}$$

$$\mathcal{G}/Z(\mathcal{G}) \cong S_3, \qquad \mathcal{G}/\mathcal{G}' \cong C_2 \times C_2 \times C_2, \tag{132}$$

of orders six and eight, and the number of solutions of $x^2 = e$ in $\mathcal{G}$ is

$$4 \cdot 4 = 16, \tag{133}$$

namely four in the symmetric factor and four in the Klein factor.

Proof. The centre of a direct product is the product of the centres, and the symmetric group on three letters has trivial centre, which gives (131). The derived subgroup of a direct product is the product of the derived subgroups, and that of the symmetric group is the alternating subgroup while that of an elementary abelian factor is trivial. The two quotients follow by dividing the orders, and the count of solutions of $x^2 = e$ is the product of the counts in the factors, namely four and four. □

19.2 Twelve irreducible representations

Theorem 19.2 (The character degrees)**.** *The group $\mathcal{G}$ has exactly twelve irreducible representations, obtained as products of the three of S_3, of degrees $1, 1, 2$, with the four of $C_2 \times C_2$, all of degree one. Their degrees are therefore eight of degree one and four of degree two, and*

$$8 \cdot 1^2 + 4 \cdot 2^2 = 8 + 16 = 24 = |\mathcal{G}|. \tag{134}$$

Proof. The irreducible representations of a direct product are the tensor products of those of the factors, so there are $3 \cdot 4 = 12$; their degrees multiply, giving eight of degree one and four of degree two, and the sum of squares equals the order by the standard identity. □

Remark 19.3 (Which of them are realised by matrices already present). Several of these degrees are matched by operators already constructed: the rank one idempotent P_1, the three address projectors Q_a, Q_b, Q_c of Theorem 16.3, the coordinate involution S, and the rank two idempotents P_{23} and P_{std}. This is a matching of degrees and of counts, not a construction of the representations themselves: no homomorphism from $\mathcal{G}$ into the matrix algebra is exhibited here, and by Proposition 18.2 the two involutions do not commute, so no such homomorphism can send both to the operators named. The matching is recorded in the sense of Remark 61.2.

Part IV
The period and the cyclic group

Structure 20. The period and its subgroups

Theorem 20.1 (Order and divisor law)**.**

$$\operatorname{ord}(T_3) = 168 = 13^2 - 1 = 2^3 \cdot 3 \cdot 7, \qquad \operatorname{ord}(T_3^k) = \frac{168}{\gcd(168, k)}, \tag{135}$$

and no proper divisor of the order returns the identity; the divisor law was confirmed at all sixteen divisors.

Proof. The generator acts on the rank two channel through $\lambda_2, \lambda_3 \in \mathbb{F}_{169}$, so the order divides $13^2 - 1$. Removing one prime at a time leaves $84, 56, 24$, whose powers are $P_1 - P_{23}$, $3I$, and an element of order seven, none the identity, and every proper divisor divides one of these. □

Theorem 20.2 (Three clocks, their idempotents, and the recombination)**.** *With*

$$U_2 = T_3^{21} \ (\textit{order } 8), \qquad U_3 = T_3^{56} = 3I \ (\textit{order } 3), \qquad U_7 = T_3^{24} \ (\textit{order } 7), \tag{136}$$

pairwise commuting, and with the remainder idempotents modulo one hundred sixty eight

$$e_8 = 105, \qquad e_3 = 112, \qquad e_7 = 120, \qquad e_8 + e_3 + e_7 \equiv 1, \tag{137}$$

the generator is recovered from its clocks by

$$T_3 = U_2^5 \, U_3^2 \, U_7^5, \qquad 21 \cdot 5 + 56 \cdot 2 + 24 \cdot 5 = 337 \equiv 1 \pmod{168}. \tag{138}$$

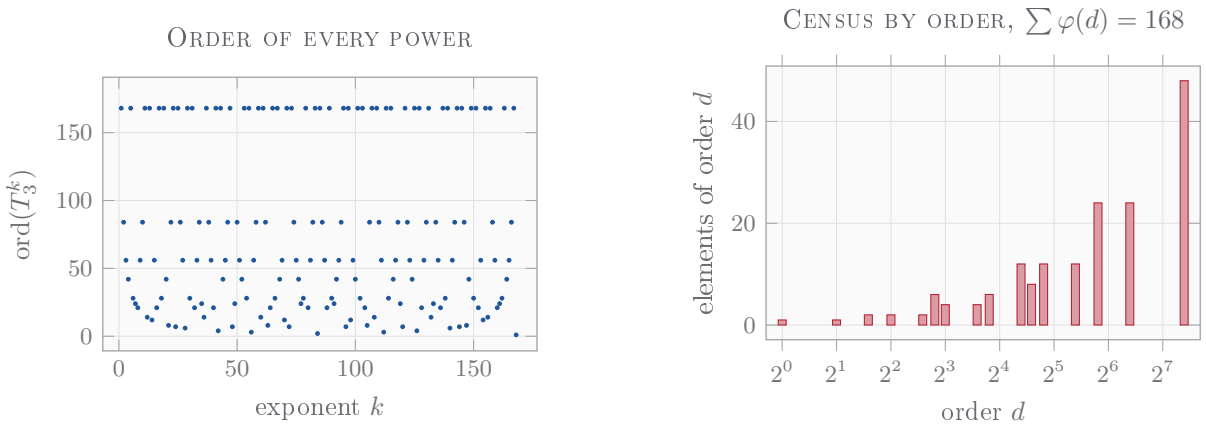


Figure 10: Left: $\mathrm{ord}(T_3^k) = 168/\gcd(168, k)$ for every exponent below the period. Right: the number of elements of each exact order, which is $\varphi(d)$ and sums to one hundred sixty eight.

Proof. The three orders follow from the divisor law (135) with $\gcd(168, 21) = 21$, $\gcd(168, 56) = 56$, $\gcd(168, 24) = 24$, and $T_3^{56} = 3I$ by direct computation. All three are powers of one matrix, so they commute. Each of 105, 112, 120 is congruent to one modulo its own prime power part and to zero modulo the others, which is idempotence modulo 168 and gives the sum relation. Since powers of one matrix add exponents, the product in (138) is T_3^{337}, and $337 \equiv 1$ modulo the period. □

20.1 The orbit sums to zero

Proposition 20.3 (Vanishing of the period sum)**.** *Summing the generator over one full period gives the zero matrix,*

$$\sum_{n=0}^{167} T_3^n \equiv 0 \pmod{13}, \tag{139}$$

so a block covering exactly one period carries no residual content, and any nonzero summary of a block encodes only the deviation from a complete period.

Proof. The sum was formed by exhaustive addition of all one hundred sixty eight powers over $\mathbb{F}_{13}$ and reduces to the zero matrix. Structurally, on the rank one channel the sum is $\sum_n 7^n$ over a multiple of the order twelve, which vanishes, and on the rank two channel the sum is a geometric series over a full period of the restricted operator, which likewise vanishes since the operator minus the identity is invertible there. □

Theorem 20.4 (The unique involution)**.** *Among the one hundred sixty seven nontrivial powers exactly one is an involution, and*

$$T_3^{84} = P_1 - P_{23} = 2P_1 - I, \qquad (T_3^{84})^2 = I, \tag{140}$$

$$T_3^{84} P_1 = P_1, \qquad T_3^{84} P_{23} = -P_{23}, \qquad T_3^{84}\, T_3\, T_3^{84} = T_3. \tag{141}$$

Proof. Scanning all exponents $1 \le k \le 167$ finds $T_3^{2k} = I$ only at $k = 84$. Direct computation gives $T_3^{84} = P_1 - P_{23}$, which equals $2P_1 - I$ by completeness, and $(P_1 - P_{23})^2 = P_1 + P_{23} = I$ by orthogonality. □

Proposition 20.5 (Balanced powers and anchors)**.** *With $J_3 = I + T_3 + T_3^2$ the exponents whose triple has three equal nonzero entries are exactly* $3, 59, 115$*,*

$$T_3^3 = J_3, \qquad T_3^{59} = 3J_3, \qquad T_3^{115} = 9J_3, \tag{142}$$

and the scalar powers are $T_3^0 = I$, $T_3^{56} = 3I$, $T_3^{112} = 9I$, *the scalars* $\{1, 3, 9\}$ *being the cube roots of unity. The subgroup* $\langle T_3^3 \rangle$ *has order fifty six and index three, so* $56 + 56 + 56 = 168$.

Proof. The first identity is (21). Multiplying it by the scalar powers $T_3^{56} = 3I$ and $T_3^{112} = 9I$ gives the other two. Scanning all triples produced by (15) shows that the three entries agree only at the exponents listed and that the scalar triples occur only at 0, 56, 112. Since $3^3 = 27 \equiv 1$, the three scalars form a group of order three, and $\gcd(3, 168) = 3$ gives the index. □

Theorem 20.6 (The scalar group and the coarse levels)**.** *The scalar powers form the subgroup*

$$\mu_3 = \{I, 3I, 9I\} = \langle T_3^{56} \rangle, \qquad T_3^{56} = 3I, \quad T_3^{112} = 9I, \quad T_3^{168} = 27I \equiv I, \tag{143}$$

of order three and index

$$\big[\langle T_3 \rangle : \mu_3\big] = \frac{168}{3} = 56, \qquad \textit{whence} \qquad \frac{168^2}{3} = 168 \cdot 56 = 9408. \tag{144}$$

The involution factors through it,

$$T_3^{84} = T_3^{56} T_3^{28} = 3\, T_3^{28}, \tag{145}$$

and since $\operatorname{ord}(T_3^{84}) = 2$ *while* $\operatorname{ord}(T_3^{56}) = 3$, *the generator of* μ_3 *is* T_3^{56} *and not* T_3^{84}. *The three coarse levels are exact,*

$$24 \cdot 7 = 56 \cdot 3 = 84 \cdot 2 = 168. \tag{146}$$

Proof. A power is scalar exactly when its triple is $(c, 0, 0)$ with $c \neq 0$, and scanning all triples returns the exponents $0, 56, 112$ with scalars $1, 3, 9$. Since $3^3 = 27 \equiv 1$ the scalars form a group of order three, generated by T_3^{56} because $\operatorname{ord}(T_3^{56}) = 3$. Lagrange gives the index and the displayed product. Exponents add, giving the factorisation of the involution. The coarse levels are the orders and indices of $\langle T_3^{24} \rangle$, $\langle T_3^{56} \rangle$, $\langle T_3^{84} \rangle$. □

Remark 20.7 (The scalar power is a periodic scaling). Since $3I$ is central, every fifty six steps the state is multiplied by the residue three, and after three such steps it returns, because $27 \equiv 1$. Multiplication by three is a bijection of the state space, so nothing is lost and no two states are merged.

Structure 21. The generator on the state space and inside the ambient group

The preceding structures describe what the generator does to the two channels. This one records what it does to the state space as a whole, what integer invariants its powers carry, and how it sits inside the full linear group over the residue field.

21.1 Orbits on the nonzero states

Theorem 21.1 (Fourteen orbits, of two sizes)**.** *Acting on the* 2196 *nonzero states the generator has exactly fourteen orbits: thirteen of size one hundred sixty eight and one of size twelve, so*

$$2196 = 13 \cdot 168 + 12. \tag{147}$$

The short orbit consists of the twelve nonzero multiples of the eigenvector v_0, *on which the generator acts by the residue seven, of order twelve.*

Proof. An orbit has size the order of the generator divided by the order of the stabiliser of a point. A nonzero state is fixed by a power exactly when it is an eigenvector of that power, and by Theorem 12.1 the only eigendirection over the residue field is the line spanned by v_0, on which the generator acts by seven of order twelve. Every state off that line therefore has trivial stabiliser and lies in an orbit of the full size, while the twelve nonzero multiples of v_0 form a single orbit of size twelve. Counting gives (147), and the decomposition was also obtained by direct enumeration. □

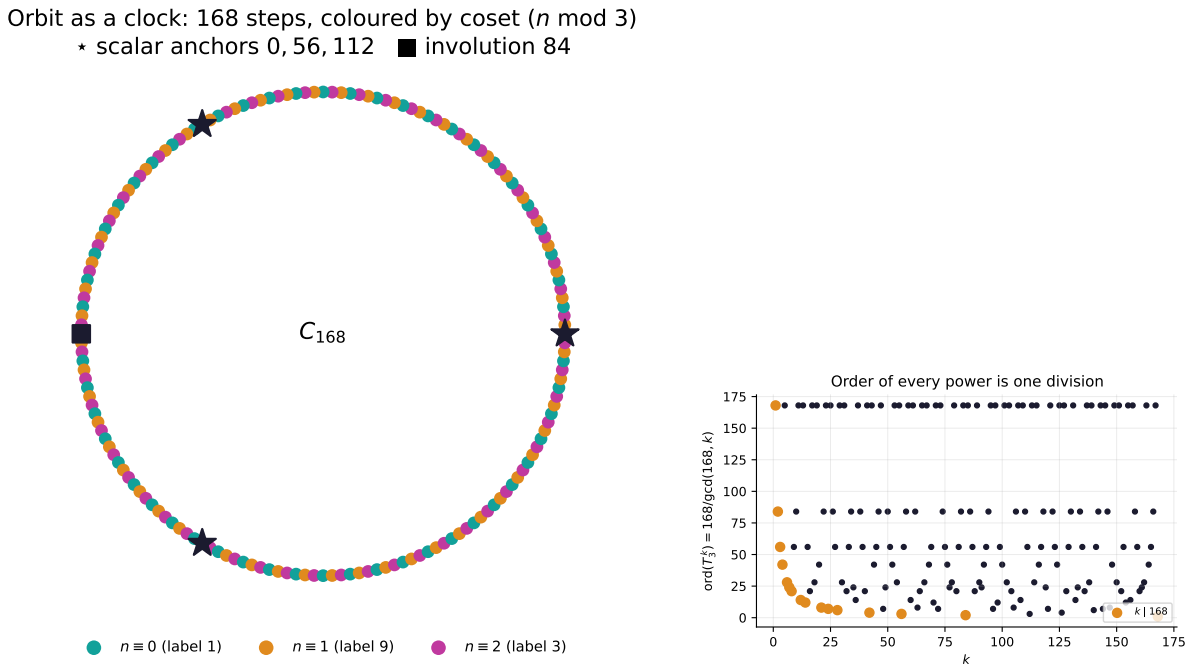


Figure 11: Left: the orbit as a cycle of one hundred sixty eight steps, coloured by the coset of $\langle T_3^3 \rangle$; stars mark the scalar anchors at $0, 56, 112$ and the square the unique involution at 84. Right: the order of every power is one division, $\mathrm{ord}(T_3^k) = 168/\gcd(168, k)$, the highlighted points being the sixteen divisors.

21.2 The invariant subspaces form a chain

Theorem 21.2 (Exactly four invariant subspaces)**.** *Of the one hundred eighty three lines exactly one is invariant, and of the one hundred eighty three planes exactly one is invariant. The invariant subspaces are therefore*

$$0 \subset \langle v_0 \rangle \subset H \subset \mathbb{F}_{13}^3, \tag{148}$$

four in number and totally ordered by inclusion, where H is the unique invariant plane.

Proof. An invariant line is an eigendirection, and there is exactly one by Theorem 12.1. An invariant plane is the kernel of a linear form u with uT_3 proportional to u, that is a left eigendirection; applying the same argument to the transpose, which has the same characteristic polynomial, gives exactly one. Adding the zero subspace and the whole space gives four. They are nested, for if the line were not contained in the plane the two would span the whole space and the generator would decompose over the residue field, contradicting the irreducibility of the quadratic factor of (50). Both counts were confirmed by enumerating all lines and all planes. □

Remark 21.3 (A flag and a splitting, two structures on one space). The invariant plane of (148) contains the invariant line, whereas the rank two channel meets that line only at zero by Theorem 10.1. The chain of invariant subspaces and the splitting into channels are therefore two different structures on the same space: the first is a flag, the second a direct sum.

21.3 Fixed points on the projective plane, by field

Proposition 21.4 (One over the residue field, three over the extension)**.** *The generator fixes exactly one point of the projective plane over $\mathbb{F}_{13}$ and exactly three over $\mathbb{F}_{169}$, namely the classes*

of the three eigenvectors (70). *Passing to the extension adds exactly two fixed points, and those two are exchanged by the Frobenius.*

Proof. A projectively fixed point is an eigendirection. Over the residue field there is one by Theorem 12.1; over the extension the three roots are distinct by (8), so there are three, given by the uniform formula of Theorem 11.1. The Frobenius permutes them as it permutes the roots, fixing the first and exchanging the other two by Proposition 51.2. □

21.4 The traces of the powers

Theorem 21.5 (A second sequence with the same recurrence)**.** *The traces of the powers over the integers are*

$$\operatorname{tr} T_3^n = 3,\ 1,\ 3,\ 7,\ 11,\ 21,\ 39,\ 71,\ 131,\ 241,\ 443,\ 815,\ \ldots \tag{149}$$

and they satisfy the recurrence (16)*, with the seeds* $3, 1, 3$ *in place of the seeds of the coefficient sequences. Reading the generator as the adjacency matrix of a directed graph on three vertices, this number is the count of closed walks of* n *steps in that graph.*

Proof. The trace is linear, so applying it to (7) multiplied by a power gives the recurrence, and the seeds are the traces computed in Proposition 3.4. That the entry of a power in position (i, j) counts the walks of that many steps from i to j is the standard reading of a matrix of nonnegative integers as an adjacency matrix, and the closed walks are counted by the diagonal. □

21.5 The determinants of the shifted powers

Proposition 21.6 (Where the working prime first appears)**.** *Over the integers*

$$\det\bigl(T_3^n - I\bigr) = 2,\ 4,\ 2,\ 16,\ 28,\ 1456 \qquad \text{for } n = 1, 2, 3, 4, 6, 12, \tag{150}$$

the first value being $-\chi(1) = 2$ *of Proposition 8.5. The working prime divides none of these until* $n = 12$*, where* $1456 = 2^4 \cdot 7 \cdot 13$*, and twelve is the order of the residue seven by Proposition 12.4.*

Proof. Each determinant was computed over the integers. The prime divides the determinant exactly when the power has the eigenvalue one modulo thirteen, that is when $7^n \equiv 1$, which first happens at $n = 12$. □

21.6 The generator has no square root in its own orbit

Proposition 21.7 (Halving is impossible)**.** *There is no exponent* k *with* $(T_3^k)^2 = T_3$*, since such a* k *would satisfy*

$$2k \equiv 1 \pmod{168}, \tag{151}$$

which has no solution because $\gcd(2, 168) = 2$ *does not divide one.*

Proof. Powers of one matrix add exponents, so the equation is the displayed congruence, and a linear congruence is solvable exactly when the greatest common divisor of the coefficient and the modulus divides the right side. A scan of all one hundred sixty eight exponents confirms that none works. □

21.7 The centraliser and the class in the ambient group

Theorem 21.8 (Two counts in the full linear group). *The full linear group over the residue field has order*

$$\left|\mathrm{GL}(3,\mathbb{F}_{13})\right| = (13^3-1)(13^3-13)(13^3-13^2) = 9726417792, \tag{152}$$

the centraliser of the generator is the unit group of the algebra it generates, of order 2016 *by Theorem 13.1, and therefore the conjugacy class of the generator has*

$$\frac{9726417792}{2016} = 4824612 \tag{153}$$

elements, the division being exact.

Proof. The order of the full linear group is the standard product of the numbers of ways to choose successive independent vectors. A matrix commutes with the generator exactly when it lies in the algebra generated by it, by Proposition 14.4, and it is invertible exactly when it is a unit of that algebra. The class size is the index of the centraliser. □

21.8 The order in the tower of depths

Theorem 21.9 (One factor of the prime for each further digit). *The order of the generator modulo the successive powers of the prime is*

$$\mathrm{ord}\big(T_3 \bmod 13^k\big) = 13^{\,k-1}\cdot 168, \tag{154}$$

with values 168, 2184, *and* 28392 *at the first three depths.*

Proof. The three values were computed directly. In general the order at depth k is a multiple of the order at depth one, which is prime to the characteristic, so it has the form $13^j \cdot 168$; by the depth law of Theorem 32.4 the difference $T_3^{168\cdot 13^j} - I$ has valuation exactly $j+1$, so the smallest exponent annihilating it modulo 13^k is the one displayed. □

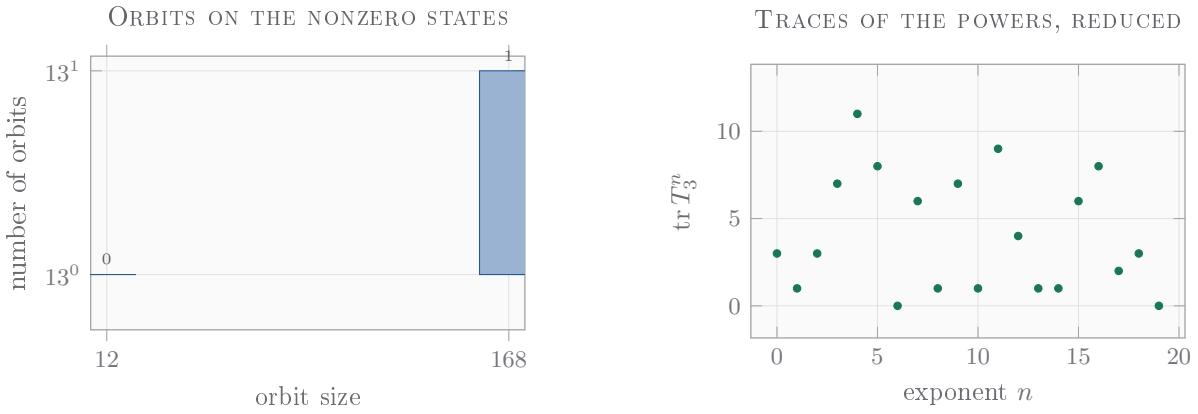


Figure 12: Left: the fourteen orbits of Theorem 21.1, thirteen of size one hundred sixty eight and one of size twelve, the short one being the nonzero multiples of the eigenvector. Right: the traces of the first twenty powers reduced modulo thirteen, obeying the recurrence of Theorem 21.5 with seeds three, one, three.

Role in the architecture. The state space carries fourteen orbits and a chain of four invariant subspaces, the powers carry an integer trace sequence and a determinant that first meets the prime at the twelfth step, and inside the ambient group the operator has a centraliser of two thousand and sixteen elements. The order grows by exactly one factor of the prime for each further digit of depth.

Structure 22. The cyclic group and its sixteen subgroups

The orbit is a cyclic group, and this structure lists every one of its subgroups explicitly, with the generator, the order, the index, and the role each plays elsewhere in the paper. Nothing is omitted: a cyclic group of order one hundred sixty eight has exactly sixteen subgroups, one for each divisor.

22.1 The group

Definition 22.1 (The orbit as a group)**.** The set of powers of the generator forms a group under matrix multiplication over $\mathbb{F}_{13}$,

$$\langle T_3 \rangle = \{\, T_3^n : 0 \le n \le 167 \,\} \;\cong\; C_{168}, \tag{155}$$

cyclic of order one hundred sixty eight by Theorem 20.1, with the identity $T_3^0 = I$ and the inverse of T_3^n given by T_3^{168-n}.

22.2 One subgroup for each divisor

Theorem 22.2 (The complete lattice)**.** *For each divisor d of 168 the group has exactly one subgroup of order d, namely*

$$C_d = \big\langle T_3^{168/d} \big\rangle, \qquad \big|C_d\big| = d, \qquad \big[\langle T_3\rangle : C_d\big] = \frac{168}{d}, \tag{156}$$

and the sixteen subgroups are exactly those listed below, each verified by computing the order of the stated generator.

subgroup	*generator*	*order*	*index*	*role in this paper*
C_1	$T_3^{168} = I$	1	168	*the trivial subgroup*
C_2	T_3^{84}	2	84	*the unique involution, Theorem 20.4*
C_3	$T_3^{56} = 3I$	3	56	*the scalars μ_3, Theorem 20.6*
C_4	T_3^{42}	4	42	*square root of the involution*
C_6	T_3^{28}	6	28	*the product C_2C_3*
C_7	T_3^{24}	7	24	*Sylow at seven, U_7*
C_8	T_3^{21}	8	21	*Sylow at two, U_2*
C_{12}	T_3^{14}	12	14	*the syntactic scale, Proposition 12.4*
C_{14}	T_3^{12}	14	12	*the positional clock, Theorem 48.1*
C_{21}	T_3^{8}	21	8	*the odd part times seven*
C_{24}	T_3^{7}	24	7	*the order of D on its image, Proposition 15.2*
C_{28}	T_3^{6}	28	6	*the group of the two ratios, Theorem 48.2*
C_{42}	T_3^{4}	42	4	*the squarefree part of the period*
C_{56}	T_3^{3}	56	3	*the cubic cosets, Proposition 20.5*
C_{84}	T_3^{2}	84	2	*the even powers*
C_{168}	T_3	168	1	*the whole orbit*

Proof. In a cyclic group of order n the subgroup generated by the power at exponent k has order $n/\gcd(n,k)$, and taking $k = n/d$ gives order d; uniqueness holds because a cyclic group has at most one subgroup of each order. The sixteen divisors are those of $168 = 2^3 \cdot 3 \cdot 7$, and each row of the table was checked by exponentiating the stated generator until the identity was reached. □

22.3 Containments and the shape of the lattice

Proposition 22.3 (Divisibility is containment)**.** *For divisors d and e of the period,*

$$C_d \subseteq C_e \iff d \mid e, \tag{157}$$

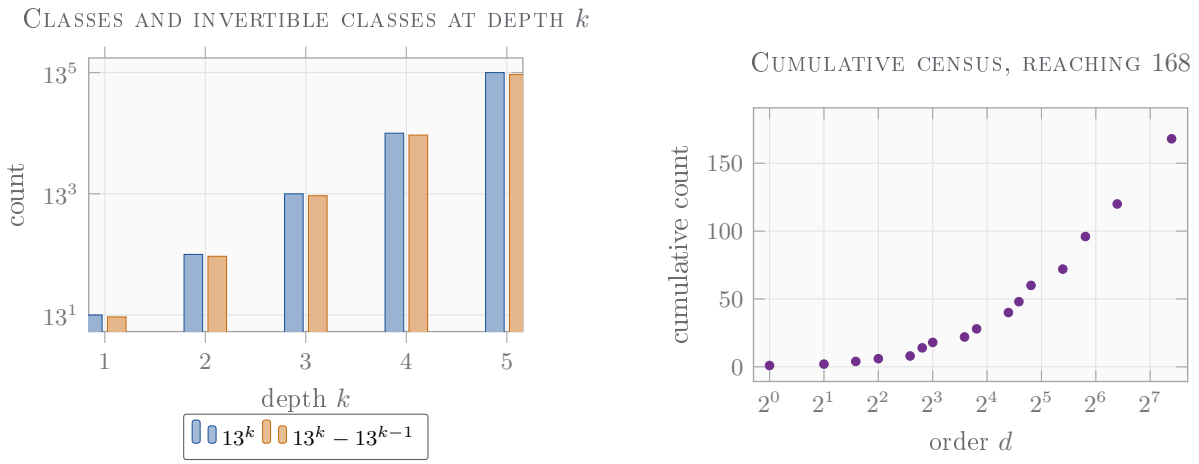


Figure 13: Left: at depth k there are 13^k classes of which $13^k - 13^{k-1}$ are invertible, by Proposition 39.2. Right: the cumulative number of elements of order dividing d, which reaches one hundred sixty eight at the last divisor, by Theorem 22.4.

so the lattice of subgroups is the divisor lattice of 168*, a product of three chains with four, two, and two members, corresponding to the exponents of* 2*,* 3*, and* 7*. Its maximal proper subgroups are* C_{84}*,* C_{56}*, and* C_{24}*, obtained by removing one prime each, and its minimal nontrivial subgroups are* C_2*,* C_3*, and* C_7*.*

Proof. In a cyclic group there is exactly one subgroup of each order dividing the group order, and a subgroup of order d is contained in one of order e exactly when d divides e, since the smaller is then generated by a power of the generator of the larger. The lattice of divisors of $2^3 \cdot 3 \cdot 7$ is the product of three chains of the stated sizes. □

22.4 Counting the elements of each order

Theorem 22.4 (Census by order)**.** *The number of elements of exact order* d *is* $\varphi(d)$*, and*

$$\sum_{d \mid 168} \varphi(d) = 168. \tag{158}$$

The sixteen values are $1, 1, 2, 2, 2, 6, 4, 4, 6, 12, 8, 12, 12, 24, 24, 48$ *in the order of the table above. In particular forty eight powers generate the whole group, and* $\varphi(2) = 1$ *gives a second proof that the involution is unique.*

Proof. In a cyclic group of order n the elements of exact order d, for d dividing n, are the generators of the unique subgroup of that order, of which there are $\varphi(d)$; summing over the divisors counts every element once, which is (158). Evaluating φ at the sixteen divisors gives the table. □

Role in the architecture. Every cycle available to the engine is one of sixteen, each with a named generator that is a single power of the one operator. Choosing a mechanism of a required period is a table lookup, and the containments say exactly which mechanisms refine which.

Structure 23. The exponents of vanishing trace

23.1 The list

Proposition 23.1 (Eighteen exponents). *The trace of a power vanishes at exactly eighteen exponents below the period, namely*

$$\begin{aligned} &6,\ 19,\ 22,\ 23,\ 29,\ 41,\ 62,\ 75,\ 78,\\ &79,\ 85,\ 97,\ 118,\ 131,\ 134,\ 135,\ 141,\ 153, \end{aligned} \tag{159}$$

and the list was obtained in two independent ways that agree: by evaluating the linear form $3c_0 + c_1 + 3c_2$ on the coefficient triples of Structure 6, and by forming each matrix power and taking its trace.

Proof. The list was obtained in two independent ways that agree: by evaluating the linear form of Proposition 3.4 on all one hundred sixty eight coefficient triples, and by forming each matrix power and taking its trace. □

23.2 Position of the list on the period

Remark 23.2 (Distribution). The eighteen exponents are neither equally spaced nor symmetric about the half period. They are disjoint from the three scalar anchors $0, 56, 112$ of Proposition 20.5, from the three balanced exponents $3, 59, 115$, and from the involution at 84, so the distinguished exponents of the orbit that carry a name are twenty four in number, leaving one hundred forty four exponents carrying none.

Role in the architecture. The trace is a timestamp of one residue, and it fails to distinguish a step only at eighteen of the one hundred sixty eight positions. The engine can therefore use the trace as a cheap partial index and knows exactly where it is uninformative.

Structure 24. The Frobenius involution on the index group

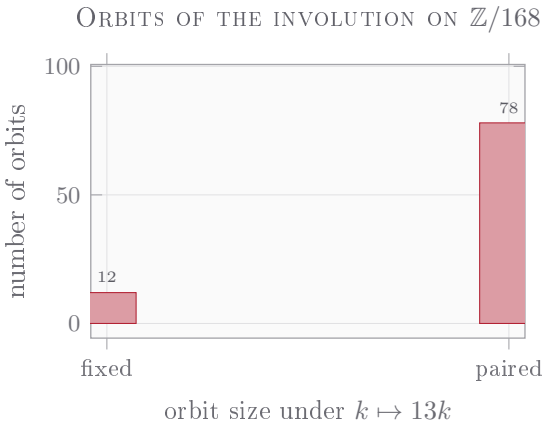


Figure 14: Twelve indices are fixed, namely the multiples of fourteen, and the remaining one hundred fifty six fall into seventy eight pairs, giving ninety orbits in all, by Theorem 24.1.

24.1 An involution of the indices

Theorem 24.1 (Multiplication by the prime is an involution). *Since*

$$13^2 = 169 \equiv 1 \pmod{168}, \tag{160}$$

the map $k \mapsto 13k$ on the index group $\mathbb{Z}/168$ satisfies $k \mapsto 13(13k) = 169k = k$, so it is an involution. Its fixed points are the twelve multiples of fourteen,

$$0,\ 14,\ 28,\ 42,\ 56,\ 70,\ 84,\ 98,\ 112,\ 126,\ 140,\ 154, \tag{161}$$

and the remaining one hundred fifty six indices fall into seventy eight pairs, so the involution has

$$12 + \frac{168 - 12}{2} = 12 + 78 = 90 \tag{162}$$

orbits in total.

Proof. The congruence (160) is arithmetic, and a map whose square is the identity is an involution. The fixed points satisfy $12k \equiv 0$ modulo 168, that is $k \equiv 0$ modulo 14, giving the twelve indices displayed; the rest pair off, giving (162). All three counts were confirmed by enumeration. □

24.2 Reading the fixed points

Remark 24.2 (The fixed set is the clock of fourteen). The twelve fixed indices are exactly the multiples of fourteen, that is the indices lying on the subgroup of order twelve generated by T_3^{14}. The number fourteen occurs here for the seventh time, as the modulus whose multiples are fixed by the Frobenius action on indices, and the number twelve is the order of the syntactic scale of Proposition 12.4.

Role in the architecture. Raising a state to the power of the prime acts on the index by an involution, so the operation is its own inverse and pairs the indices two by two, fixing only the twelve positions that lie on the clock of fourteen.

Structure 25. Three periods and their least common multiple

Theorem 25.1 (The three algebraic periods)**.** *Three operators of the construction repeat with three different periods:*

operator	*acts on*	*period*	*half turn*
the residue 7	*the rank one channel*	12	$7^6 = -1$
the torus generator $g = 3 + 2\theta$	*the rank two channel*	14	$g^7 = -1$
the difference operator D	*its own image*	24	$D^{12} = -I$ *on* $\operatorname{im} D$

and their least common multiple is the period of the generator,

$$\operatorname{lcm}(12, 14, 24) = 168 = \operatorname{ord}(T_3). \tag{163}$$

Each half turn is a sign, verified in its own setting, and the relation $D^3 = -2D$ of Proposition 15.2 gives the third row.

Proof. The first period is the order of seven in $\mathbb{F}_{13}^*$ and its sixth power is minus one; the second is Theorem 48.2; the third follows from $D^3 = -2D$ together with the order twelve of -2, and the two displayed actions on the image were checked directly. The least common multiple was computed and equals the order of the generator by Theorem 20.1. □

Role in the architecture. Three independent counters run at twelve, fourteen, and twenty four steps, each with an exact half turn given by a sign, and they come back together after one hundred sixty eight steps, which is the period of the whole construction.

Structure 26. Distinguished residues of the construction

Several residues recur in the construction and are collected here with their derivations. Each is an element of the prime field obtained by an exact computation; none is a measured quantity, and no reading beyond the arithmetic is attached to any of them.

residue	derivation	value	where
7	$\left(v_0^{\top} T_3 v_0\right)\left(v_0^{\top} v_0\right)^{-1} = 10 \cdot 7^{-1}$	7	the eigenvalue, (77)
7	2^{-1} in $\mathbb{F}_{13}$	7	the inverse of two, (60)
10	$13 - 3$, the characteristic less the dimension	10	
5	$10 \cdot 7$ reduced	5	
11	$\operatorname{tr} T_3 - \operatorname{tr} I = 1 - 3 = -2$	11	(13)
1	$\det T_3^k$ for every k	1	(7)

Remark 26.1 (On naming these residues). The two occurrences of seven in the first two rows arise from different computations, one a ratio built from the eigenvector and one the inverse of two, and they coincide because $10 \cdot 7^{-1} = 10 \cdot 2 = 20 \equiv 7$ and $2^{-1} = 7$. Attaching physical names to these residues is an assignment in the sense of Remark 55.4: the field admits no order by Proposition 1.3, so no residue can be a magnitude of any quantity, and only the arithmetic above is asserted.

Part V
The lattice of fields

Structure 27. The lattice of fields

The construction generates a definite finite list of fields, and this structure records all of them, their degrees, their discriminants, the behaviour of the working prime in each, and the top of the lattice. Nothing outside this list occurs anywhere in the paper.

27.1 The base and the cubic field

Remark 27.1 (One root, two names, and one field). The symbol α used for a root of χ over the rationals and the symbol η used for the generator of the ring $\mathbb{Z}_{13}[\eta]$ denote the same element: a root of the one defining cubic, taken over $\mathbb{Q}$ in the first case and over $\mathbb{Z}_{13}$ in the second. Accordingly

$$\alpha = \eta, \qquad \chi(\alpha) = \chi(\eta) = 0, \qquad K_3 = \mathbb{Q}(\alpha) = \mathbb{Q}(\eta), \tag{164}$$

and K_3 is the field generated by a root of the Tribonacci cubic, called here the cubic field of the construction. The two symbols are kept apart only to mark which ring the root is being taken in; no second element is introduced.

Definition 27.2 (The cubic field)**.** Let α be a root of χ. Since χ is irreducible over $\mathbb{Q}$ by Proposition 4.2, the field

$$K_3 = \mathbb{Q}(\alpha), \qquad [K_3 : \mathbb{Q}] = 3, \qquad \operatorname{disc}(\chi) = -44, \tag{165}$$

is a cubic field, not Galois over $\mathbb{Q}$, since the Galois group of the splitting field is the full symmetric group on three letters.

27.2 The discriminant of a quadratic field

Definition 27.3 (Field discriminant of a quadratic field)**.** For a squarefree integer $d \neq 1$ the field $\mathbb{Q}(\sqrt{d})$, obtained by adjoining a radix of d, has ring of integers generated by $\tfrac{1}{2}(1+\sqrt{d})$ when $d \equiv 1$ modulo four and by $\sqrt{d}$ otherwise, and its discriminant is accordingly

$$\operatorname{disc}\mathbb{Q}(\sqrt{d}) = \begin{cases} d, & d \equiv 1 \pmod 4, \\ 4d, & \text{otherwise.} \end{cases} \tag{166}$$

This is to be distinguished from the discriminant of the polynomial $x^2 - d$, which is $4d$ in every case; the two agree exactly when $d \not\equiv 1$ modulo four, as recorded in Proposition 29.1.

Proposition 27.4 (Seven quadratic fields, and their discriminants)**.** *The squarefree divisors $d > 1$ of $42 = 2 \cdot 3 \cdot 7$ give exactly seven quadratic fields, with discriminants*

$$\begin{aligned} &\operatorname{disc}\mathbb{Q}(\sqrt{2}) = 8, \qquad \operatorname{disc}\mathbb{Q}(\sqrt{3}) = 12, \qquad \operatorname{disc}\mathbb{Q}(\sqrt{6}) = 24, \\ &\operatorname{disc}\mathbb{Q}(\sqrt{7}) = 28, \qquad \operatorname{disc}\mathbb{Q}(\sqrt{14}) = 56, \quad \operatorname{disc}\mathbb{Q}(\sqrt{21}) = 21, \\ &\operatorname{disc}\mathbb{Q}(\sqrt{42}) = 168, \end{aligned} \tag{167}$$

each dividing the period, and each realised as the order of a power of the generator,

$$\operatorname{ord}\big(T_3^{168/\Delta}\big) = \Delta, \qquad \Delta \in \{8, 12, 24, 28, 56, 21, 168\}. \tag{168}$$

Only $d = 21$ satisfies $d \equiv 1$ modulo four, which is why its discriminant is 21 and not 84.

Proof. For squarefree d the discriminant is d when $d \equiv 1$ modulo four and $4d$ otherwise, and among the seven divisors only 21 is in the first case. Each displayed value divides $168 = 4 \cdot 42$, and the order statement follows from the divisor law of Theorem 20.1 with $\gcd(168, 168/\Delta) = 168/\Delta$; all seven were confirmed by direct exponentiation. $\square$

27.3 The quadratic resolvent

Proposition 27.5 (The imaginary quadratic field)**.** *The unique quadratic subfield of the splitting field of χ is*

$$\mathbb{Q}\big(\sqrt{\operatorname{disc}\chi}\big) = \mathbb{Q}(\sqrt{-44}) = \mathbb{Q}(\sqrt{-11}), \qquad \operatorname{disc} = -11, \tag{169}$$

an imaginary field, and it is not any of the seven fields of (167)*. The exact sequence attached to the splitting field is*

$$1 \longrightarrow A_3 \longrightarrow S_3 \longrightarrow \operatorname{Gal}\big(\mathbb{Q}(\sqrt{-11})/\mathbb{Q}\big) \longrightarrow 1. \tag{170}$$

Proof. For an irreducible cubic the fixed field of the alternating subgroup is generated by a radix of the discriminant, here -44, whose squarefree part is -11; since $-11 \equiv 1$ modulo four its field discriminant is -11. The sequence (170) is the quotient of the symmetric group by its alternating subgroup. That this field is none of the seven of (167) follows because those have positive radicands. $\square$

27.4 The biquadratic compositum

Proposition 27.6 (Degree four, and the discriminant)**.** *The compositum*

$$K_4 = \mathbb{Q}(\sqrt{2}, \sqrt{21}) = \mathbb{Q}(\sqrt{2}, \sqrt{42}), \qquad [K_4 : \mathbb{Q}] = 4, \qquad \operatorname{Gal}(K_4/\mathbb{Q}) \cong C_2 \times C_2, \tag{171}$$

has the three quadratic subfields $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{21})$, $\mathbb{Q}(\sqrt{42})$, *whose discriminants* 8, 21, 168 *have product*

$$8 \cdot 21 \cdot 168 = 28224 = 168^2 = \operatorname{disc} K_4, \tag{172}$$

the last equality by the conductor and discriminant formula, the four characters of the Klein group having conductors 1, 8, 21, 168.

Proof. The compositum of two distinct quadratic fields has degree four with group the Klein four group, and the three quadratic subfields correspond to the three subgroups of index two, here with radicands 2, 21, and their product 42. Their discriminants are given by Definition 27.3, and the conductor and discriminant formula multiplies the conductors of the four characters, namely 1, 8, 21, 168. □

27.5 The splitting field and the top of the lattice

Theorem 27.7 (The Galois closure of the whole construction)**.** *Let* L_6 *be the splitting field of* χ*, of degree six over* $\mathbb{Q}$ *with group* S_3*, and let* K_4 *be as in* (171)*. Then* $L_6 \cap K_4 = \mathbb{Q}$*, because* -11 *is not a rational square times any of* 2, 21, 42*, and therefore the compositum*

$$M = L_6 \cdot K_4, \qquad [M : \mathbb{Q}] = 6 \cdot 4 = 24, \qquad \operatorname{Gal}(M/\mathbb{Q}) \cong S_3 \times C_2 \times C_2, \tag{173}$$

is the top of the lattice: it is Galois over $\mathbb{Q}$ *and contains every field named in this paper.*

Proof. The splitting field of an irreducible cubic with non-square discriminant has degree six and group S_3. Its unique quadratic subfield is $\mathbb{Q}(\sqrt{-11})$ by Proposition 27.5, while every quadratic subfield of K_4 is one of $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{21}), \mathbb{Q}(\sqrt{42})$; the ratios $-11/2$, $-11/21$, $-11/42$ are not rational squares, so the two fields share no quadratic subfield and their intersection is $\mathbb{Q}$. For linearly disjoint Galois extensions the compositum has degree the product and group the direct product. □

The degrees, discriminants, groups, and the decomposition of the working prime in every field of the lattice are collected once, in Theorem 30.3.

27.6 Behaviour of the working prime in each field

Proposition 27.8 (Splitting and inertia at thirteen)**.** *The quadratic residues modulo thirteen are* $\{1, 3, 4, 9, 10, 12\}$*, and reduction of the radicands gives*

$$\begin{aligned} &42 \equiv 3, \quad 3 \equiv 3, \quad 14 \equiv 1 \quad \text{residues, so thirteen splits;} \\ &2 \equiv 2, \quad 21 \equiv 8, \quad 6 \equiv 6, \quad 7 \equiv 7, \quad -11 \equiv 2 \quad \text{nonresidues, so thirteen is inert.} \end{aligned} \tag{174}$$

In particular thirteen splits in $\mathbb{Q}(\sqrt{42})$*, whose discriminant is the period, and is inert in* $\mathbb{Q}(\sqrt{2})$*, whose radix generates the residue extension.*

Proof. A prime not dividing the discriminant splits in a quadratic field exactly when the radicand is a quadratic residue modulo that prime. Reducing each radicand modulo thirteen and comparing with the residue list of Definition 9.1 gives (174). □

27.7 The local field and the coincidence of two directions

Proposition 27.9 (The unramified quadratic extension)**.** *Both two and twenty one are nonresidues modulo thirteen, and* $21 \equiv 8 = 2 \cdot 4$ *with four a square, so*

$$\mathbb{Q}_{13}(\sqrt{21}) = \mathbb{Q}_{13}(\sqrt{2}), \tag{175}$$

the unique unramified quadratic extension of $\mathbb{Q}_{13}$, whose residue field is $\mathbb{F}_{169}$. Two directions that are distinct over $\mathbb{Q}$ therefore coincide over the working prime, which is the local content of the compositum K_4.

Proof. Both radicands reduce to nonresidues, so each generates a quadratic extension of the local field, and any two such extensions generated by units coincide with the unique unramified quadratic extension. Explicitly $21 \equiv 8 = 2 \cdot 4$ with four a square, so the two radicands differ by a square and generate the same field. Its residue field is the quadratic extension of the residue field, namely $\mathbb{F}_{169}$. □

27.8 The finite fields

Proposition 27.10 (Three finite fields, and why the cubic one is excluded)**.** *The finite fields occurring are*

$$\mathbb{F}_{13}, \qquad \mathbb{F}_{169} = \mathbb{F}_{13}[x]/(x^2 - 2), \qquad \mathbb{F}_{2197} = \mathbb{F}_{13^3}, \tag{176}$$

with multiplicative groups of orders

$$12, \qquad 168 = 13^2 - 1, \qquad 2196 = 13^3 - 1 = 2^2 \cdot 3^2 \cdot 61. \tag{177}$$

The cubic extension cannot host the generator with its period, since

$$168 \nmid 2196, \qquad \gcd(168, 2196) = 12, \tag{178}$$

so an element of order one hundred sixty eight does not lie in a group of order two thousand one hundred ninety six. This is consistent with Theorem 9.2: the algebra is a product of two fields and not a field of degree three.

Proof. The three fields are the residue field, its quadratic extension of Definition 9.1, and its cubic extension. Their multiplicative groups are cyclic of orders $q - 1$, $q^2 - 1$, $q^3 - 1$, giving (177). An element of order one hundred sixty eight exists in a cyclic group exactly when that number divides the group order, and $2196 = 12 \cdot 183$ is not a multiple of 168, since $\gcd(168, 2196) = 12$. □

27.9 The ring, which is not a field

Theorem 27.11 (The ring $\mathbb{Z}_{13}[\eta]$ and its reduction)**.** *The ring generated by the eigenvalue over the thirteen-adic integers is*

$$\mathbb{Z}_{13}[\eta] = \mathbb{Z}_{13}[x]/(\chi(x)), \tag{179}$$

and it is not a domain and not a valuation ring, since χ is reducible modulo thirteen: the two factors annihilate one another,

$$(\eta - 7)\big(\eta^2 + 6\eta + 2\big) = 0 \quad \text{in } \mathbb{Z}_{13}[\eta]/13. \tag{180}$$

By the Chinese remainder theorem the ring splits as a product of two local pieces,

$$\mathbb{Z}_{13}[\eta] \;\cong\; \mathbb{Z}_{13} \times W(\mathbb{F}_{169}), \qquad \mathbb{Z}_{13}[\eta]/13 \;\cong\; \mathbb{F}_{13} \times \mathbb{F}_{169}, \tag{181}$$

of residue orders thirteen and one hundred sixty nine, so the reduction has $13 \cdot 169 = 2197$ elements.

27.10 The tower of ring powers

Proposition 27.12 (Reductions at every depth)**.** *The reductions of the ring at the successive powers of the prime form a tower*

$$\mathbb{Z}/13 \longleftarrow \mathbb{Z}/13^2 \longleftarrow \mathbb{Z}/13^3 \longleftarrow \cdots \longleftarrow \mathbb{Z}_{13}, \tag{182}$$

with orders and unit counts

depth k	$\lvert\mathbb{Z}/13^k\rvert$	*units*	*note*
1	13	12	*the prime field*
2	169	156	*the size of a boundary class,* (225)
3	2197	2028	*the size of the state space,* (63)
4	28561	26364	*fourth Hensel level,* (219)
5	371293	342732	

and the number of units at depth k *is* $13^k - 13^{k-1}$*. The order of the generator rises by exactly one factor of the prime at the second level,* $\operatorname{ord}(T_3 \bmod 13^2) = 2184 = 13 \cdot 168$*, as in Theorem 32.4.*

Proof. The ring $\mathbb{Z}/13^k$ has 13^k elements, and a class is invertible exactly when its representative is prime to thirteen, which excludes the 13^{k-1} multiples of thirteen. The order at the second level is Theorem 32.4. □

27.11 Summary of the lattice

Remark 27.13 (What the lattice contains). The complete list is: one base field; nine quadratic fields, namely the seven over the divisors of the radicand together with the resolvent and, locally, their common unramified extension; one cubic field; one biquadratic compositum; one splitting field of degree six; one top of degree twenty four; three finite fields; one ring that is not a field, with its tower of reductions. No other field is used, no field of characteristic zero other than those listed occurs, and no completion other than the one at thirteen is taken.

Role in the architecture. The engine computes in the prime field and its quadratic extension and lifts through the tower of ring powers when depth is required. The remaining fields are not idle: each one carries a definite decomposition of the working prime, recorded in Structure 30, and that decomposition is what the field contributes. A field in which thirteen is inert supplies the residue field $\mathbb{F}_{169}$ and hence a second channel; a field in which it splits supplies two independent copies of the prime field and hence a pair of independent counters; the cubic field supplies one of each, which is exactly the two channel structure; and the Galois closure organises all of these into the orbits of one Frobenius.

27.12 Where each element of the construction lives

Proposition 27.14 (Everything closes inside the degree twenty four field)**.** *Over the rationals the three roots of* χ *lie in the splitting field* L_6*, the radices of two and of twenty one lie in* K_4*, and a radix of minus eleven lies in* L_6*; consequently all of them lie in the compositum*

$$M = L_6 \cdot K_4, \qquad [M : \mathbb{Q}] = 24, \qquad \operatorname{Gal}(M/\mathbb{Q}) \cong S_3 \times C_2 \times C_2, \tag{183}$$

and M *is the smallest field of the lattice containing all of them. Reducing modulo thirteen sends the roots to* $\lambda_1 = 7$ *in* $\mathbb{F}_{13}$ *and to* λ_2, λ_3 *in* $\mathbb{F}_{169}$*, and sends a radix of two to* θ*; a radix of minus one needs no reduction step, being already the residue five by Theorem 9.4.*

Proof. The splitting field contains all roots by definition and contains a radix of the discriminant, hence of minus eleven, by Proposition 27.5. The biquadratic field contains the two radices by Proposition 27.6. The compositum therefore contains all of them, and its degree and group are Theorem 27.7. Minimality holds because any field containing all the roots contains L_6, and any field containing both radices contains K_4. The reductions are Theorem 9.2 and Definition 9.1. □

Remark 27.15 (Finite and infinite kept apart). The field M is a number field of degree twenty four over the rationals and is infinite as a set. The finite objects of the construction are the residue field of thirteen elements, its quadratic extension of one hundred sixty nine, and the algebra $\mathbb{F}_{13}[T_3]$ of 2197 elements, which by Theorem 13.1 is not a field but a product of two. The passage between the two worlds is reduction modulo thirteen, and no statement of this paper mixes them.

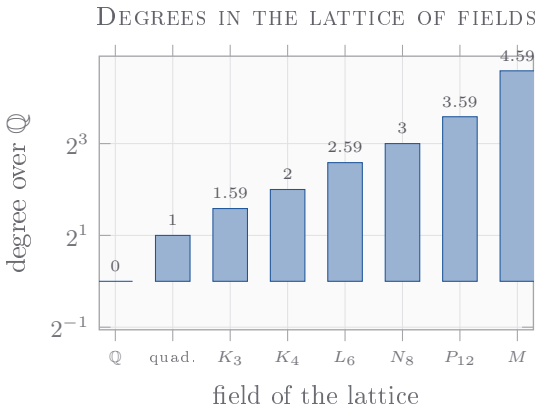


Figure 15: The degrees of the fields of the lattice, from the rationals to the Galois closure of degree twenty four, on a logarithmic scale in base two.

Structure 28. The eleven fields with their minimal polynomials

This structure lists every field of the lattice with a generating element, its minimal polynomial, its degree, its discriminant, the automorphism group of the extension, and what the field contributes. Each entry was verified by exact computation.

28.1 The table

Theorem 28.1 (The lattice, completely tabulated). *All eleven fields, with a generating polynomial, the degree, the discriminant, the group, and the decomposition of thirteen, appear in the single table of Theorem 30.3.*

Proof. Each row is established elsewhere: the quadratic rows by Definition 27.3 and Proposition 27.8, the cubic row by Proposition 4.2 and Theorem 9.2, the biquadratic row by Proposition 27.6, the sextic and octic rows by Theorem 28.3, and the two composita by Proposition 28.6. The decomposition data in the last two columns is Theorem 30.3. □

28.2 The two minimal polynomials of the composita

Proposition 28.2 (Rows seven and nine). *Put* $u = \sqrt{2} + \sqrt{21}$. *Then* $u^2 = 23 + 2\sqrt{42}$, *so* $(u^2 - 23)^2 = 4 \cdot 42 = 168$ *and*

$$u^4 - 46u^2 + 361 = 0, \qquad 529 - 168 = 361, \tag{184}$$

which is irreducible of degree four. Adjoining the third radix, the element $w = \sqrt{2}+\sqrt{21}+\sqrt{-11}$ *satisfies*

$$w^8 - 48w^6 + 2552w^4 - 17856w^2 + 976144 = 0, \tag{185}$$

irreducible of degree eight. Both polynomials were computed as minimal polynomials and agree with the displayed forms.

Proof. Squaring $u = \sqrt{2}+\sqrt{21}$ gives $23+2\sqrt{42}$, so $(u^2-23)^2 = 4\cdot 42$, which rearranges to (184); the polynomial is irreducible because u generates a field of degree four by Proposition 27.6. For w the minimal polynomial was computed and is the displayed one of degree eight, which is the degree of the field by Theorem 29.3. □

28.3 The discriminants of rows eight and nine

Theorem 28.3 (Two discriminants from the conductor and discriminant formula)**.** *The splitting field* L_6 *has group* S_3*, whose irreducible characters have conductors* 1, 11, *and* 44 *of degrees* 1, 1, *and* 2*, so*

$$\left|\operatorname{disc} L_6\right| = 1 \cdot 11 \cdot 44^2 = 21296 = 2^4 \cdot 11^3, \tag{186}$$

and since L_6 *contains* $\mathbb{Q}(\sqrt{-11})$ *it is totally imaginary with* $r_2 = 3$*, so the sign is* $(-1)^3$ *and*

$$\operatorname{disc} L_6 = -2^4 \cdot 11^3 = -21296. \tag{187}$$

The field N_8 *has group* $(\mathbb{Z}/2)^3$*, whose eight characters have conductors* 1 *together with the absolute discriminants of the seven quadratic subfields of Theorem 29.4, namely* 8, 21, 11, 168, 88, 231, 1848*, so*

$$\left|\operatorname{disc} N_8\right| = 1 \cdot 8 \cdot 21 \cdot 11 \cdot 168 \cdot 88 \cdot 231 \cdot 1848 = 11662935330816 = 2^{12}\,3^4\,7^4\,11^4, \tag{188}$$

and since N_8 *is totally imaginary with* $r_2 = 4$ *the sign is positive.*

Proof. For an abelian or a general Galois extension the discriminant is the product of the conductors of the irreducible characters raised to their degrees. For S_3 the trivial character has conductor one, the sign character cuts out the quadratic resolvent and has conductor $|-11|$, and the two dimensional character has conductor the absolute discriminant of the cubic field, which is 44; multiplying gives (186). For $(\mathbb{Z}/2)^3$ every nontrivial character is quadratic and cuts out one of the seven subfields, so its conductor is that subfield's absolute discriminant; multiplying the eight values gives (188), and the factorisation was computed. The signs follow from $(-1)^{r_2}$, with $r_2 = 3$ and $r_2 = 4$ respectively, both fields being totally imaginary because each contains a radix of a negative number. □

28.4 A correction concerning the cubic field

Proposition 28.4 (The cubic field is not Galois)**.** *The cubic field* $K_3 = \mathbb{Q}(\alpha)$ *is not a Galois extension of* $\mathbb{Q}$*, and its automorphism group is trivial rather than cyclic of order three. Indeed* $\operatorname{disc}\chi = -44$ *is not a rational square, so the Galois group of the splitting field is the full symmetric group on three letters, and a cubic field is Galois exactly when that group is the alternating one, that is exactly when the discriminant is a square.*

Proof. A cubic field is Galois over the rationals exactly when its Galois closure has degree three, that is when the Galois group of the defining polynomial is the alternating group, which happens exactly when the discriminant is a square. Here $\operatorname{disc}\chi = -44$ is not a square, so the group is the full symmetric group, the closure has degree six, and the cubic field has no nontrivial automorphism. □

Remark 28.5 (Where the group of order three does occur). A cyclic group of order three does occur in the construction, but not as the automorphism group of K_3: it is the alternating subgroup A_3 inside the Galois group of L_6, whose fixed field is $\mathbb{Q}(\sqrt{-11})$ by Proposition 27.5, and it is also the subgroup H of Theorem 29.3 whose fixed field has degree eight, and separately the group μ_3 of scalars of Theorem 20.6. These are three different groups of order three, and none of them acts on K_3.

28.5 The two composita

Proposition 28.6 (Degrees twelve and twenty four)**.** *The only quadratic subfield of L_6 is $\mathbb{Q}(\sqrt{-11})$, so $\mathbb{Q}(\sqrt{42})$ is not contained in L_6 and the compositum has degree*

$$[P_{12} : \mathbb{Q}] = 6 \cdot 2 = 12, \qquad \mathrm{Gal}(P_{12}/\mathbb{Q}) \cong S_3 \times C_2. \tag{189}$$

Likewise $L_6 \cap K_4 = \mathbb{Q}$, so

$$[M : \mathbb{Q}] = 6 \cdot 4 = 24, \qquad \mathrm{Gal}(M/\mathbb{Q}) \cong S_3 \times V_4, \tag{190}$$

which is Theorem 27.7. Both composita contain N_8 or not according as they contain all three radices: M does, and P_{12} does not, since it contains only $\sqrt{-11}$ and $\sqrt{42}$.

Proof. The splitting field has a unique quadratic subfield, namely the resolvent of Proposition 27.5, so it does not contain $\mathbb{Q}(\sqrt{42})$ and the compositum has degree $6 \cdot 2$ with group the direct product. The same argument with K_4, whose three quadratic subfields are all distinct from the resolvent, gives degree $6 \cdot 4$ and the group of Theorem 27.7. The last statement holds because N_8 is generated by the three radices, all of which lie in M while only two lie in P_{12}. □

Remark 28.7 (The two discriminants left open). The discriminants of P_{12} and of M are not computed in this paper. For these groups the conductor and discriminant formula requires the conductors of the two dimensional characters twisted by quadratic characters, which were not determined, and no value is asserted in their place.

28.6 What each field contributes, restated

Remark 28.8 (Roles). Reading the table against Theorem 30.3: row two supplies the radix of two and hence the residue field $\mathbb{F}_{169}$; row three is the field whose discriminant is twenty one and in which thirteen is inert; row four is the field whose discriminant is the period, and thirteen splits there, supplying two independent counters over the prime field; row five is the quadratic resolvent of the defining cubic; row six is the cubic field, which supplies one prime with residue degree one and one with residue degree two, that is the two channel structure itself; row seven is the biquadratic compositum, of discriminant the square of the period; row nine is the field of Theorem 29.3, whose seven quadratic subfields carry the period and the period times eleven among their discriminants; and rows ten and eleven are the two composita, the last being the Galois closure inside which every field of this paper sits.

Structure 29. The degree eight fixed field

The top of the lattice has degree twenty four, and one of its subgroups cuts out a field of degree eight generated by three quadratic radices. This structure identifies that field, lists its seven quadratic subfields with their discriminants, and records the behaviour of the working prime in each.

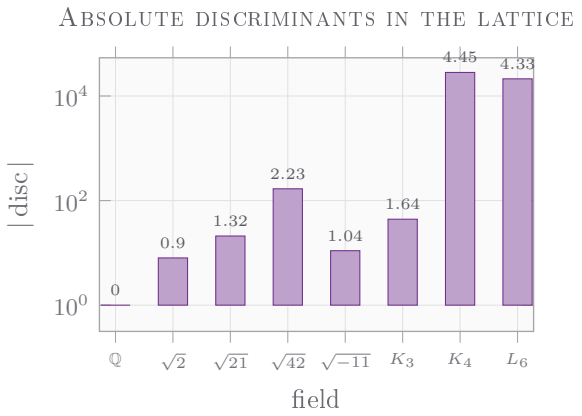


Figure 16: Absolute discriminants of eight fields of the lattice, on a logarithmic scale; the fourth is the period and the seventh is its square.

29.1 Polynomial and field discriminants are different invariants

Proposition 29.1 (The two discriminants)**.** *For a squarefree d the polynomial x^2-d has discriminant $4d$, whereas the field $\mathbb{Q}(\sqrt{d})$ has discriminant d when $d \equiv 1$ modulo four and $4d$ otherwise. The two therefore differ exactly when $d \equiv 1$:*

$$\begin{aligned} &\operatorname{disc}\big(x^2-2\big)=8, && \operatorname{disc}\mathbb{Q}(\sqrt{2})=8, && \textit{equal};\\ &\operatorname{disc}\big(x^2-21\big)=84, && \operatorname{disc}\mathbb{Q}(\sqrt{21})=21, && \textit{different, since } 21\equiv 1;\\ &\operatorname{disc}\big(x^2+11\big)=-44, && \operatorname{disc}\mathbb{Q}(\sqrt{-11})=-11, && \textit{different, since } -11\equiv 1. \end{aligned} \tag{191}$$

Proof. The discriminant of x^2-d is $4d$ by the formula for a quadratic, while the field discriminant is given by Definition 27.3. The two agree exactly when $d \not\equiv 1$ modulo four, which gives the three displayed cases. □

Remark 29.2 (A genuine agreement). The polynomial discriminant of x^2+11 equals the discriminant of the defining cubic,

$$\operatorname{disc}\big(x^2+11\big) = -44 = \operatorname{disc}(\chi), \tag{192}$$

and this is not accidental: the squarefree part of $\operatorname{disc}(\chi)$ is -11, so $\operatorname{disc}(\chi) = 4\cdot(-11)$, which is precisely the polynomial discriminant of x^2+11. The quadratic resolvent of χ is therefore $\mathbb{Q}(\sqrt{-11})$, of field discriminant -11, as in Proposition 27.5.

29.2 The subgroup and its fixed field

Theorem 29.3 (A field of degree eight with three quadratic generators)**.** *Let $M = L_6\cdot K_4$ be the top of the lattice, of degree twenty four with*

$$\operatorname{Gal}(M/\mathbb{Q}) \cong S_3\times C_2\times C_2, \qquad |S_3\times C_2\times C_2| = 6\cdot 2\cdot 2 = 24, \tag{193}$$

by Theorem 27.7. Let $H = A_3\times\{e\}$ be the subgroup of order three generated by a three cycle in the first factor, of index

$$[\operatorname{Gal}(M/\mathbb{Q}) : H] = \frac{24}{3} = 8. \tag{194}$$

Then the fixed field of H is

$$M^H = \mathbb{Q}\big(\sqrt{-11},\ \sqrt{2},\ \sqrt{21}\big), \qquad \big[M^H:\mathbb{Q}\big] = 8, \tag{195}$$

generated by three quadratic radices, with Galois group $(\mathbb{Z}/2)^3$ over $\mathbb{Q}$.

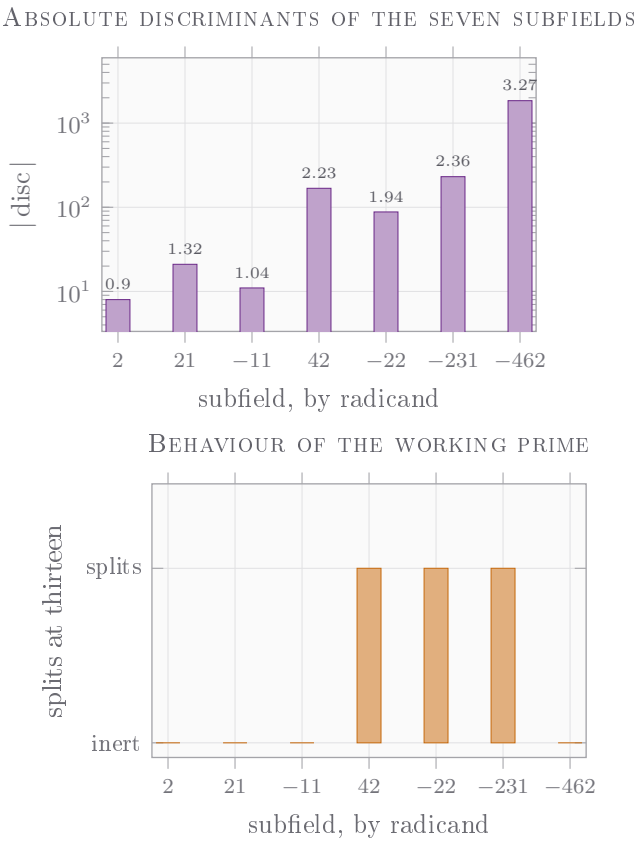


Figure 17: Left: the absolute discriminants 8, 21, 11, 168, 88, 231, 1848 of the seven quadratic subfields of Theorem 29.4, on a logarithmic scale. The fourth is the period and the seventh is the period times eleven, which is Theorem 29.5. Right: thirteen is inert in the three generators and in the triple product, and splits in the three double products.

Proof. The fixed field of the alternating subgroup inside the splitting field of a cubic is generated by a radix of the discriminant, so $L_6^{A_3} = \mathbb{Q}(\sqrt{-44}) = \mathbb{Q}(\sqrt{-11})$ by Remark 29.2. Since H acts trivially on the second and third factors, its fixed field contains all of $K_4 = \mathbb{Q}(\sqrt{2}, \sqrt{21})$, and therefore contains the compositum $\mathbb{Q}(\sqrt{-11}) \cdot K_4$. That compositum has degree $2 \cdot 4 = 8$ because $\mathbb{Q}(\sqrt{-11})$ and K_4 intersect in $\mathbb{Q}$, by the argument of Theorem 27.7, and eight is exactly the index (194), so the two fields coincide. A compositum of three independent quadratic extensions has group $(\mathbb{Z}/2)^3$. □

29.3 The seven quadratic subfields

Theorem 29.4 (Complete list, with discriminants)**.** *The field* (195) *has exactly seven quadratic subfields, one for each of the seven nontrivial characters of* $(\mathbb{Z}/2)^3$*, namely the fields generated by the products of the three radicands taken in every nonempty combination:*

combination	*radicand*	*field discriminant*	*behaviour of thirteen*
$\sqrt{2}$	2	8	*inert*
$\sqrt{21}$	21	21	*inert*
$\sqrt{-11}$	-11	-11	*inert*
$\sqrt{2}\sqrt{21}$	42	168	*splits*
$\sqrt{2}\sqrt{-11}$	-22	-88	*splits*
$\sqrt{21}\sqrt{-11}$	-231	-231	*splits*
$\sqrt{2}\sqrt{21}\sqrt{-11}$	-462	-1848	*inert*

The fourth row is the field whose discriminant is the period, by Proposition 31.5.

Proof. A compositum of three independent quadratic extensions has $2^3 - 1 = 7$ subgroups of index two, hence seven quadratic subfields, obtained by adjoining the product of any nonempty subset of the radicands and taking the squarefree part. The discriminants follow from Proposition 29.1: the radicands 21, -11, -231 are congruent to one modulo four and keep their value, while 2, 42, -22, -462 are congruent to two and are multiplied by four. The behaviour at thirteen is decided by the quadratic character of the reduced radicand, computed against the residue list of Definition 9.1. □

29.4 The number one thousand eight hundred forty eight

Theorem 29.5 (A discriminant, not merely a product)**.** *The product of the period with the absolute discriminant of the resolvent is*

$$168 \cdot 11 = 1848, \tag{196}$$

and this number is not merely a product: it is the absolute value of the field discriminant of the seventh subfield of Theorem 29.4,

$$\operatorname{disc} \mathbb{Q}\big(\sqrt{-462}\big) = -1848, \qquad -462 = 2 \cdot 21 \cdot (-11), \tag{197}$$

that is of the field generated by the product of all three radices. Its factorisation is $1848 = 2^3 \cdot 3 \cdot 7 \cdot 11$*, the period contributing* $2^3 \cdot 3 \cdot 7$ *and the resolvent the factor eleven.*

Proof. The radicand -462 is squarefree and congruent to two modulo four, so its field discriminant is $4 \cdot (-462) = -1848$ by Proposition 29.1, and $|-1848| = 1848 = 168 \cdot 11$ by (196). That -462 is the squarefree part of the product of the three radicands is the computation $2 \cdot 21 \cdot (-11) = -462$. □

Remark 29.6 (Why this is an improvement). A product of two counts is a count only when the two choices are independent, as recorded in Remark 61.2. Here no such independence was available, so the equality $1848 = 168 \cdot 11$ alone would have been a numerical remark. Theorem 29.5 upgrades it: the number is the discriminant of a specific field of the lattice, so it is the value of an invariant and not a bare product.

29.5 The working prime in the degree eight field

Proposition 29.7 (Three inert generators)**.** *Thirteen is inert in each of the three generating quadratic fields, since*

$$2 \equiv 2, \qquad 21 \equiv 8, \qquad -11 \equiv 2 \pmod{13}, \tag{198}$$

and none of 2, 8 *is a quadratic residue modulo thirteen. It splits in the three fields whose radicands are products of exactly two generators, and is inert in the field of the triple product. In particular the residue extension attached to each inert generator is the same field* $\mathbb{F}_{169}$*, since there is only one unramified quadratic extension of* $\mathbb{Q}_{13}$*, by Proposition 27.9.*

Proof. Each radicand was reduced modulo thirteen and compared with the residue list of Definition 9.1. The last statement is Proposition 27.9, since the unramified quadratic extension of the local field is unique. □

29.6 What the three generators are and are not

Remark 29.8 (An honest reading). The field (195) is generated by three quadratic radices, and the state space of the model has dimension three. These two occurrences of the number three are not identified here, and no map between the generators and the coordinates is exhibited: the dimension three comes from the degree of χ, while the three generators come from the three independent characters of $(\mathbb{Z}/2)^3$, and the two constructions are unrelated. Any reading of the generators as components of a state is an assignment in the sense of Remark 55.4, not a consequence of the algebra. What is proved here is the field, its degree, its seven subfields, their discriminants, and the behaviour of the working prime in each.

Role in the architecture. The construction sits inside one Galois extension of degree twenty four, and one subgroup of order three cuts out a field of degree eight whose seven quadratic subfields contain both the field whose discriminant is the period and the field whose discriminant is the period times the resolvent. All of these are bookkeeping: the engine computes in the prime field and its quadratic extension, and lifts through the tower of depths.

Structure 30. What each field contributes

Every field of the lattice carries a decomposition of the working prime, and that decomposition is the contribution of the field. This structure records it for all of them.

30.1 The three invariants of a prime in a field

Definition 30.1 (Ramification, residue degree, number of primes)**.** For a prime q and a field K of degree n over $\mathbb{Q}$, the prime factors in the ring of integers of K as a product of g distinct primes, each occurring e times and each with residue field of degree f over $\mathbb{F}_q$, subject to

$$\sum e_i f_i = n, \qquad \text{and for a Galois } K, \qquad e\,f\,g = n. \tag{199}$$

The prime is unramified when every e_i is one, inert when $g = 1$ and $f = n$, and split when $f = 1$.

30.2 Thirteen is unramified everywhere in the lattice

Proposition 30.2 (No ramification)**.** *Thirteen divides none of the discriminants of the lattice, namely*

$$8,\ 12,\ 24,\ 28,\ 56,\ 21,\ 168,\ -11,\ -88,\ -231,\ -1848,\ -44, \tag{200}$$

so it is unramified in every field of the lattice, and in each of them $e = 1$.

Proof. A prime ramifies in a field only if it divides the discriminant. Thirteen divides none of the values in (200), as each is a product of powers of two, three, seven, and eleven, so every ramification index is one. □

30.3 The complete table

Theorem 30.3 (Decomposition of thirteen in every field)**.** *With $e = 1$ throughout, the residue degree and the number of primes are*

field	*minimal polynomial*	*deg*	*discriminant*	*group*	*f*	*g*
$\mathbb{Q}$	$x-1$	1	1	*trivial*	1	1
$\mathbb{Q}(\sqrt{2})$	x^2-2	2	8	C_2	2	1
$\mathbb{Q}(\sqrt{3})$	x^2-3	2	12	C_2	1	2
$\mathbb{Q}(\sqrt{6})$	x^2-6	2	24	C_2	2	1
$\mathbb{Q}(\sqrt{7})$	x^2-7	2	28	C_2	2	1
$\mathbb{Q}(\sqrt{14})$	x^2-14	2	56	C_2	1	2
$\mathbb{Q}(\sqrt{21})$	x^2-21	2	21	C_2	2	1
$\mathbb{Q}(\sqrt{42})$	x^2-42	2	168	C_2	1	2
$\mathbb{Q}(\sqrt{-11})$	x^2+11	2	-11	C_2	2	1
$\mathbb{Q}(\sqrt{-22})$	x^2+22	2	-88	C_2	1	2
$\mathbb{Q}(\sqrt{-231})$	x^2+231	2	-231	C_2	1	2
$\mathbb{Q}(\sqrt{-462})$	x^2+462	2	-1848	C_2	2	1
$K_3=\mathbb{Q}(\alpha)$	x^3-x^2-x-1	3	-44	*trivial*	1 *and* 2	2
$K_4=\mathbb{Q}(\sqrt{2},\sqrt{21})$	x^4-46x^2+361	4	$28224=168^2$	V_4	2	2
L_6, *splitting field*	*see* (187)	6	$-21296=-2^4 11^3$	S_3	2	3
$N_8=K_4(\sqrt{-11})$	$x^8-48x^6+2552x^4-17856x^2+976144$	8	$2^{12}3^4 7^4 11^4$	$(\mathbb{Z}/2)^3$	2	4
$P_{12}=L_6\cdot\mathbb{Q}(\sqrt{42})$	*compositum*	12	*not determined*	$S_3\times C_2$	2	6
$M=L_6\cdot K_4$	*compositum*	24	*not determined*	$S_3\times V_4$	2	12

In every Galois row the product efg equals the degree, and the residue degree is two because the Frobenius has order two.

Proof. For a quadratic field the prime splits when the radicand is a quadratic residue modulo thirteen and is inert otherwise, and the residues are those of Definition 9.1; each row was obtained by reducing the radicand and testing membership. For the cubic field the factorisation type of χ modulo thirteen is $(1,2)$ by Theorem 9.2, which by Dedekind gives two primes with residue degrees one and two. For the Galois fields the decomposition group is generated by the Frobenius, whose order is the residue degree; in the degree eight field the Frobenius sends each of the three radices to its negative, which is consistent with the whole table, since it fixes exactly the three radicands that are products of two generators and negates the three generators and their triple product. Hence its order is two, and g is the degree divided by two in each Galois case. The consistency of the seven signs with the seven splitting behaviours was verified. □

30.4 What each kind of field supplies

Remark 30.4 (Reading the table). A field in which thirteen is inert has residue field $\mathbb{F}_{169}$ and therefore supplies the quadratic extension in which the rank two channel lives; the three generating radices of the degree eight field are all of this kind, and so is their triple product. A field in which thirteen splits has two primes, each with residue field $\mathbb{F}_{13}$, and therefore supplies a pair of independent counters over the prime field; the field whose discriminant is the period is of this kind. The cubic field supplies exactly one of each, which is the two channel structure of Theorem 10.1 seen arithmetically. The Galois closure has twelve primes above thirteen, permuted transitively by its group, with the decomposition group of each generated by one Frobenius of order two.

Structure 31. Local theory at thirteen

31.1 The Newton polygon

Proposition 31.1 (Newton polygon)**.** *All coefficients of χ are units at thirteen, so its Newton polygon is a single segment of slope zero and every root is a thirteen-adic unit; no root degenerates on reduction.*

Proof. The Newton polygon is the lower convex hull of the points $(i, v_{13}(a_i))$ formed from the coefficients. All four coefficients of χ are units, so all four points have valuation zero and the

hull is the flat segment from $(0,0)$ to $(3,0)$. A segment of slope zero corresponds to roots of valuation zero. $\square$

31.2 The quaternion algebra

Theorem 31.2 (Quaternion algebra and Brauer invariant)**.** *For two units the Hilbert symbol at thirteen is trivial, whereas with the uniformizer thirteen and the nonresidue two,*

$$(13,2)_{13} = -1, \tag{201}$$

so $(13,2)$ *is a division algebra over* $\mathbb{Q}_{13}$ *of Brauer invariant one half, and* $\mathbb{F}_{169}$ *is its unramified quadratic residue subfield.*

Proof. For an odd prime the symbol of two units is trivial, and for a uniformizer and a unit it is the Legendre symbol; here $2^{6} \equiv -1$, so the symbol is minus one, which characterises the division quaternion algebra, whose Brauer class has order two. $\square$

31.3 Witt classes

Proposition 31.3 (Witt classes)**.** *The form* $x^{2} - 2y^{2}$*, the norm form of* $\mathbb{F}_{169}$*, is anisotropic over* $\mathbb{F}_{13}$*; the form* $x^{2} - y^{2} + z^{2}$ *is isotropic at* $(1,1,0)$ *and has Witt index one.*

Proof. Anisotropy of the binary form is Theorem 48.1. The ternary form vanishes at $(1,1,0)$, so it is isotropic; a nondegenerate isotropic form of rank three over a field splits off one split plane, that is one copy of the binary form xy, and leaves an anisotropic part of rank one, so its Witt index is one. $\square$

31.4 The discriminant of the cubic field

Proposition 31.4 (The order generated by a root is maximal)**.** *The discriminant of the defining polynomial is* -44*, and for a cubic field the polynomial discriminant and the field discriminant differ by the square of the index of* $\mathbb{Z}[\alpha]$ *in the ring of integers. If that index were greater than one it would be two, forcing a field discriminant of* -11*; but the smallest absolute value of the discriminant of a cubic field with one real embedding is twenty three. Hence the index is one, the order* $\mathbb{Z}[\alpha]$ *is maximal, and*

$$\operatorname{disc} K_{3} = -44. \tag{202}$$

Proof. For a cubic field the polynomial discriminant and the field discriminant differ by the square of the index of $\mathbb{Z}[\alpha]$ in the ring of integers. If that index exceeded one it would be two, since $-44 = 4 \cdot (-11)$ has only the square factor four, and the field discriminant would then be -11. No cubic field has discriminant of absolute value smaller than twenty three, so this is impossible and the index is one. $\square$

31.5 Quadratic fields attached to the period

Proposition 31.5 (Quadratic fields attached to the period)**.** *For squarefree* d *the discriminant of* $\mathbb{Q}(\sqrt{d})$ *is* d *when* $d \equiv 1$ *modulo four and* $4d$ *otherwise. Then*

$$\operatorname{disc}\mathbb{Q}(\sqrt{2}) = 8, \quad \operatorname{disc}\mathbb{Q}(\sqrt{21}) = 21, \quad \operatorname{disc}\mathbb{Q}(\sqrt{42}) = 168, \tag{203}$$

$$8 \cdot 21 \cdot 168 = 28224 = 168^{2} = \operatorname{disc}\mathbb{Q}(\sqrt{2},\sqrt{21}), \tag{204}$$

the last by the conductor and discriminant formula. Thirteen splits in $\mathbb{Q}(\sqrt{42})$*, since* $42 \equiv 3 \equiv 4^{2}$*, and is inert in the other two, which coincide locally because* $21 \equiv 8 = 2 \cdot 4$ *with four a square.*

Each of the seven quadratic fields over a divisor of forty two has discriminant equal to the order of a power of the generator, the orders (168)*, The quadratic subfield of the splitting field of* χ *is* $\mathbb{Q}(\sqrt{-11})$*, from* $\operatorname{disc}\chi = -44$*, and is not the real field above.*

Proof. The discriminants follow from Definition 27.3, only 21 being congruent to one modulo four among the seven divisors. Each value divides 168, and the order statement is the divisor law of Theorem 20.1 applied to the exponent $168/\Delta$. The splitting behaviour is Proposition 27.8, and the last statement is Proposition 27.5. □

Part VI
Valuation, depth, and refinement

Structure 32. Valuation, depth, and the geometric weight

This module states the three admitted notions of Definition 1.2 as theorems.

32.1 The valuation

Theorem 32.1 (Strong triangle inequality, and the isosceles property)**.** *For integers* a, b, c*,*

$$v_{13}(a-c) \;\geq\; \min\big(v_{13}(a-b),\, v_{13}(b-c)\big), \tag{205}$$

and the minimum of the three valuations is attained at least twice. The inequality is not an equality in general: for $(a,b,c) = (0,1,13)$ *one has* $v_{13}(a-c) = 1 > 0 = \min(0,0)$*. Consequently the classes*

$$B_k(a) = a + 13^k\mathbb{Z}_{13} \tag{206}$$

are nested or disjoint and never partially overlapping. Both statements were verified by a deterministic sweep of sixty four thousand triples together with families built from powers of thirteen.

Proof. Write $a - c = (a-b) + (b-c)$. The valuation of a sum is at least the smallest of the valuations of the summands, which is (205). If the two summands had different valuations the sum would have exactly the smaller, so equality holds unless they agree; this gives the isosceles statement, since the minimum then occurs at both summands. The example shows equality can fail. Nesting of the classes follows from (205) as in Theorem 33.2. □

Remark 32.2 (The reading available here). The set $B_k(a)$ is a congruence class, that is an index set of states agreeing to depth k, and k is an integer count of fixed digits. Nothing in (205) or (206) is a measurement, and no comparison of two states is defined other than by their agreement depth.

32.2 A filtration, not a valuation

Proposition 32.3 (The valuation on the ring is a filtration)**.** *In* $\mathbb{Z}_{13}[\eta]/13$ *the two factors of the cubic annihilate one another,*

$$(\eta - 7)(\eta^2 + 6\eta + 2) = 0, \tag{207}$$

so the ring has zero divisors and is not a domain. Hence v_{13} *satisfies only*

$$v_{13}(ab) \;\geq\; v_{13}(a) + v_{13}(b), \tag{208}$$

with equality failing in general: it is a filtration and not a discrete valuation.

Proof. The displayed product vanishes because the two factors are the images of the coprime factors of χ, whose product is χ and therefore zero at η. Two nonzero elements with zero product are zero divisors, so the ring is not a domain, and for a product of two elements of valuation zero to have positive valuation the inequality (208) cannot be an equality. □

32.3 The depth law

Theorem 32.4 (The orbit depth law)**.** *The order of the generator rises by one factor of the prime at each precision,*

$$\operatorname{ord}(T_3 \bmod 13) = 168, \qquad \operatorname{ord}(T_3 \bmod 13^2) = 2184 = 13 \cdot 168, \tag{209}$$

and since $\gcd(168, 13) = 1$ *the reduction lifts uniquely to an element of order one hundred sixty eight over* $\mathbb{Z}_{13}$*. Moreover*

$$v_{13}\big(T_3^{168 \cdot 13^k} - I\big) = k + 1, \tag{210}$$

and for all exponents m, n*,*

$$v_{13}\big(T_3^m - T_3^n\big) = \begin{cases} 1 + v_{13}(m-n), & 168 \mid (m-n), \\ 0, & \text{otherwise.} \end{cases} \tag{211}$$

Proof. The two orders were computed by repeated multiplication modulo thirteen and modulo one hundred sixty nine. Since the order and the prime are coprime, Hensel gives a unique lift of the same order. The identity (210) was evaluated modulo 13^8 for $k = 0, 1, 2, 3$, and (211) was verified on a deterministic family of exponents, together with the complementary case for offsets not divisible by the period. □

32.4 The geometric weight

Proposition 32.5 (The geometric weight)**.** *The weight attached to an index is the group element*

$$w(k) = \eta^{-k} \in \mathbb{F}_{169}^*, \qquad w(k + 168) = w(k), \tag{212}$$

injective on a full period, so distinct indices within a period carry distinct labels. It is not a magnitude: by Proposition 1.3 the field carries no order, so two weights can be compared for equality but never for size.

Proof. Injectivity on a full period and the periodicity are Proposition 1.3. That no comparison of size is available is the absence of an order on the field, established there from $5^2 \equiv -1$. □

Proposition 32.6 (The Frobenius involution on the index group)**.** *Since* $13^2 = 169 \equiv 1$ *modulo one hundred sixty eight, the map* $k \mapsto 13k$ *is an involution of* $\mathbb{Z}/168$*; it has twelve fixed points, namely the multiples of fourteen, and the remaining indices fall into ninety orbits in total.*

Proof. This is Theorem 24.1: the square of the map is multiplication by $169 \equiv 1$, the fixed points are the solutions of $12k \equiv 0$, that is the multiples of fourteen, and the remaining indices pair off. □

Role in the architecture. Position and similarity are integer notions throughout. Two states are compared by the depth at which they agree, a state is refined by fixing one more digit, and the weight attached to an index is a label in a finite group. The depth law makes the comparison exact: two powers of the generator agree to depth one plus the valuation of the difference of their exponents, and to no depth at all when the period does not divide that difference.

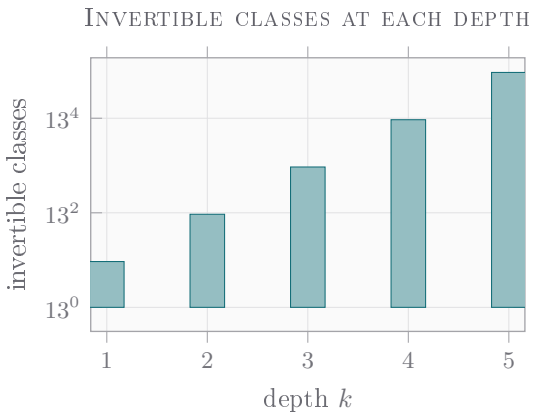


Figure 18: At depth k the invertible classes number $13^k - 13^{k-1}$, the classes divisible by the prime being excluded; the weight attached to a class of that depth is η^{-k}.

Structure 33. Classes are nested or disjoint

33.1 The classes

Definition 33.1 (Class at a depth)**.** For an integer a and a depth k the class of a at depth k is

$$B_k(a) = a + 13^k \mathbb{Z}_{13} = \{x : v_{13}(x - a) \geq k\}, \tag{213}$$

the set of integers agreeing with a in the first k digits of the radix thirteen. It is simultaneously closed and open in the sense that it is the complement of a union of such classes, and it is a coset of a subgroup, hence a congruence class and not a ball of any radius.

33.2 The dichotomy

Theorem 33.2 (Nested or disjoint, never partially overlapping)**.** *For any two classes, either one contains the other or they are disjoint:*

$$B_k(a) \cap B_l(b) \neq \emptyset \quad \text{and} \quad k \geq l \quad \Longrightarrow \quad B_k(a) \subseteq B_l(b). \tag{214}$$

A partial overlap is impossible. The statement was verified on 57600 *pairs formed from a deterministic sweep of representatives and depths.*

Proof. If the classes meet at x then $v_{13}(x - a) \geq k$ and $v_{13}(x - b) \geq l$, so by Theorem 32.1 we get $v_{13}(a - b) \geq \min(k, l) = l$, hence every y with $v_{13}(y - a) \geq k \geq l$ satisfies $v_{13}(y - b) \geq \min(v_{13}(y - a), v_{13}(a - b)) \geq l$, which is (214). □

Structure 34. The crystal and integer refinement

Definition 34.1 (Witt vectors, Frobenius, Verschiebung)**.** The ring of Witt vectors $W(\mathbb{F}_{13})$ has as elements the sequences of residues, with operations defined so that the ghost components

$$w_n = x_0^{13^n} + 13\, x_1^{13^{n-1}} + \cdots + 13^n x_n \tag{215}$$

add and multiply componentwise; the carries are integer polynomials, and $W(\mathbb{F}_{13}) \cong \mathbb{Z}_{13}$. The Frobenius F raises coordinates to the thirteenth power and the Verschiebung V shifts them, with

$$FV = VF = 13. \tag{216}$$

On $W(\mathbb{F}_{13})$ the Frobenius is the identity, since $a^{13} = a$ for every residue; the nontrivial Frobenius acts on the rank two channel only.

Proposition 34.2 (Teichmuller lift of the eigenvalue)**.** *The multiplicative group $\mathbb{F}_{13}^*$ is cyclic of order twelve, and*

$$\omega(7) \equiv 150 \pmod{169}, \qquad 150 \equiv 7 \pmod{13}, \qquad 150^{12} \equiv 1 \pmod{169}, \tag{217}$$

where 150 is the unique lift of seven of order twelve. Note that $T_3^{13} \neq T_3$ modulo thirteen: only the eigenvalue is its own representative.

Proof. The group of nonzero residues is cyclic of order twelve, so every element satisfies $a^{12} = 1$. Raising to the thirteenth power is the Teichmuller step, and modulo the square of the prime it already returns the multiplicative representative, giving $\omega(7) \equiv 7^{13} \equiv 150$. Direct computation gives $150^{12} \equiv 1$ with no smaller exponent, and among the thirteen residues congruent to seven only this one has order twelve. □

34.1 The Teichmuller representatives of every residue

Proposition 34.3 (The complete table of representatives)**.** *The Teichmuller representative of each residue modulo 169, obtained as a^{13}, is*

a	0	1	2	3	4	5	6	7	8	9	10	11	12
$\omega(a)$	0	1	80	146	147	70	19	150	99	22	23	89	168

and every entry is fixed by the map $x \mapsto x^{13}$ modulo 169, so the table is the multiplicative section of the reduction at this depth. The entry at seven is the lift of the eigenvalue used in Proposition 34.2, and the entry at twelve is $168 \equiv -1$.

Proof. Each entry is a^{13} reduced modulo the square of the prime, and each was checked to be fixed by a further application of the same map, which is the defining property of the multiplicative representative. □

Proposition 34.4 (Refinement is the Newton iteration in integers)**.** *The lifting step is*

$$x_{k+1} = x_k - f(x_k)\, f'(x_k)^{-1}, \tag{218}$$

with f' a formal derivative and the inverse a modular inverse, so no division outside the integers occurs. For χ at the eigenvalue, $\chi'(7) = 132 \equiv 2$ is a unit and

$$7 \to 33 \to 202 \to 26566 \quad \textit{exact modulo } 13,\ 169,\ 2197,\ 28561, \tag{219}$$

while the quadratic mode gives exact roots modulo $13^2, 13^4, 13^8$, doubling the number of fixed digits at each step. Inversion to any depth is the division free iteration

$$x_{n+1} = x_n(2 - a\, x_n), \qquad a\, x_{n+1} \equiv 1 \pmod{13^{2^{n+1}}}. \tag{220}$$

Proof. Writing a candidate root as $r + 13^k s$ and expanding, all terms beyond the first two are divisible by 13^{2k}, so modulo 13^{k+1} the equation becomes $f(r) + 13^k s f'(r) \equiv 0$, one linear congruence for s, uniquely solvable because $f'(r)$ is a unit. Taking the inverse modulo the prime adds one digit at each step; taking it modulo the current power doubles the number of correct digits. Each displayed value was checked to be an exact root at its modulus and to reduce to its predecessor. □

Role in the architecture. Refinement adds digits and never leaves the integers, and the number of fixed digits either increases by one or doubles at each step. The Verschiebung is the single operation that discards a digit, and its composition with Frobenius is multiplication by the prime, so loss of depth is explicit and unique.

Part VII
Where the data enters: tree, channels, orbit, pyramid

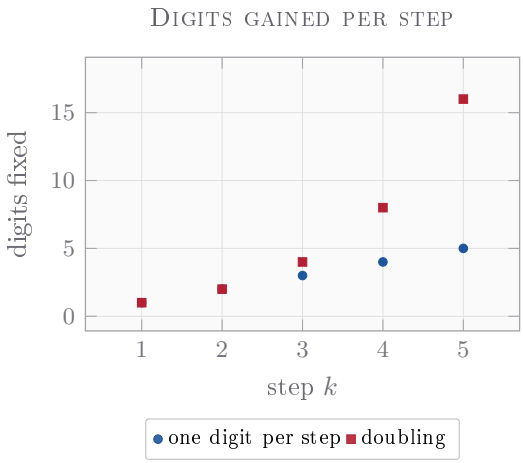


Figure 19: Refinement adds one digit per step in the multiplicative mode and doubles the number of fixed digits in the quadratic mode, by Proposition 34.4.

Structure 35. The order in which the data enters

The construction has several structures that could each be taken as the entry point, and this structure settles the order by counting. The conclusion is that the data enters at the tree, through the digits of the radix thirteen, and that the pyramid acts later, on the orbit, since its levels are subgroups of the orbit and an orbit position exists only once a state exists.

35.1 The five stages

Theorem 35.1 (The pipeline, with its counts)**.** *Every datum passes through the following five stages, in this order, and no other order is available:*

$$\textit{token} \longrightarrow \textit{address} \longrightarrow \textit{state} \longrightarrow \textit{channels} \longrightarrow \textit{orbit position} \longrightarrow \textit{levels}, \tag{221}$$

with the counts

	stage	*mechanism*	*count*
1	*token*	*a byte*	256
2	*address*	*digits of the radix thirteen, Proposition 36.3*	2197 *classes at depth three*
3	*state*	*the same triple read in the ring,* (107)	2197
4	*channels*	*the two projectors, Theorem 10.1*	13 *and* 169
5	*orbit position*	*a residue modulo the period*	168
6	*levels*	*the three subgroups, Theorem 43.2*	$24 \cdot 7 = 56 \cdot 3 = 84 \cdot 2$

Proof. Stage two is the unique digit expansion of Proposition 36.3, verified for all two hundred fifty six bytes; stage three is the identification of a coordinate triple with a ring element, Definition 14.1; stage four is Theorem 10.1; stage five is a residue modulo one hundred sixty eight. The order cannot be permuted, because each stage consumes the output of the previous one: the digits are needed to name a state, the state is needed before a projector can be applied, and an orbit position is a residue attached to a state. □

35.2 Why the pyramid cannot come first

Corollary 35.2 (The pyramid acts on the orbit, not on the token)**.** *The three levels of Theorem 43.2 are the subgroups $\langle T_3^{24} \rangle$, $\langle T_3^{56} \rangle$, $\langle T_3^{84} \rangle$ of the orbit, of orders seven, three, and two. A coset of such a subgroup is a set of orbit positions, so the pyramid is defined only after stage five. In particular the pyramid does not chunk tokens and does not act on the tree; what it coarse grains is the position on the cycle of one hundred sixty eight steps.*

Proof. The three levels are cosets of subgroups of the orbit by Theorem 43.2, and a coset is a set of orbit positions. An orbit position is a residue attached to a state, which by Theorem 35.1 exists only after the third stage, so no level of the pyramid is defined before then. □

35.3 What the tree does and what the pyramid does

Remark 35.3 (Two different kinds of grouping). The tree groups *addresses* by shared leading digits, and its classes are nested or disjoint by Theorem 33.2. The pyramid groups *orbit positions* by cosets of a subgroup, and its classes are all of the same size by Proposition 43.3. The first grouping is by agreement of digits and has classes of two different sizes at the boundary, namely one hundred fifty six and one hundred sixty nine by Theorem 36.6; the second is by a subgroup and has classes of one size. They are unrelated groupings of two different sets.

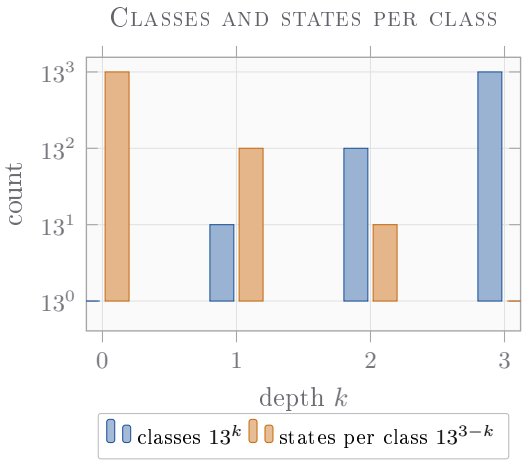


Figure 20: At every depth the number of classes times the number of states in each is $13^3 = 2197$. Fixing one further digit multiplies the classes by thirteen and divides the contents by thirteen.

35.4 The pipeline drawn

The five stages of Theorem 35.1 are shown in Figure 21, with the count carried at each stage.

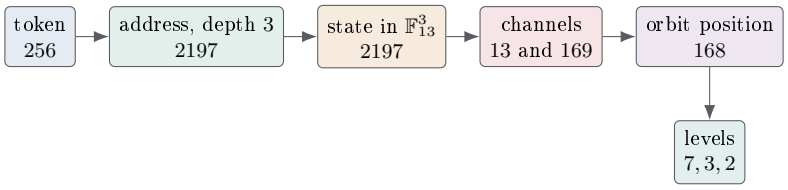


Figure 21: The five stages of Theorem 35.1. The data enters at the address, that is at the tree, through the digits of the radix thirteen; the pyramid of levels acts last, on the orbit position, by Corollary 35.2.

35.5 Nested and disjoint classes drawn

Figure 22 shows the classes at three successive depths, each splitting into thirteen classes of the next depth, with no partial overlap anywhere, by Theorem 33.2.

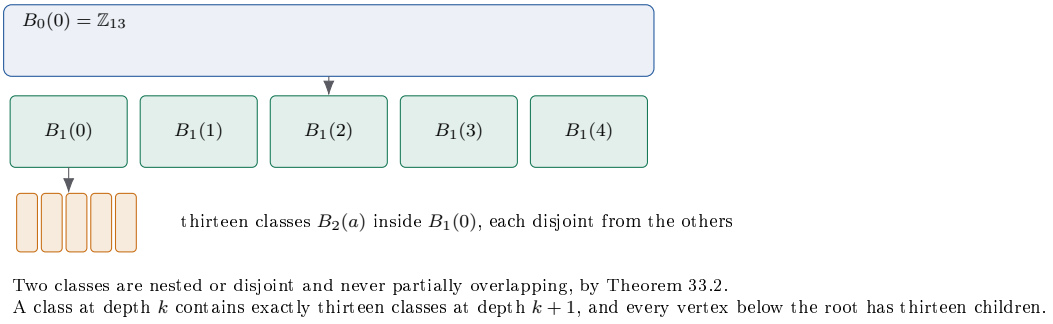


Figure 22: Classes at successive depths. Each class of depth k splits into thirteen classes of depth $k+1$, and two classes of any depths either contain one another or are disjoint. Depth is the integer valuation and no other comparison is defined.

35.6 The pyramid drawn

Figure 23 shows the three levels of Theorem 43.2 as partitions of the orbit positions into cosets, with index times order equal to the period at every level.

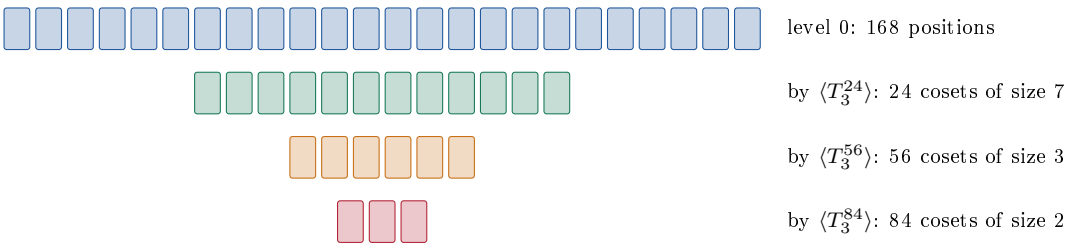


Figure 23: The pyramid of Theorem 43.2. Each level is a partition of the one hundred sixty eight orbit positions into cosets of one of the three subgroups, with index times order equal to the period at every level, and each level reversible by Proposition 43.3. The widths shown are schematic; the counts are the exact ones stated.

35.7 The tree with its fixed end drawn

Figure 24 shows the tree to depth two, with the unique end fixed by the generator marked in colour, by Theorem 38.3.

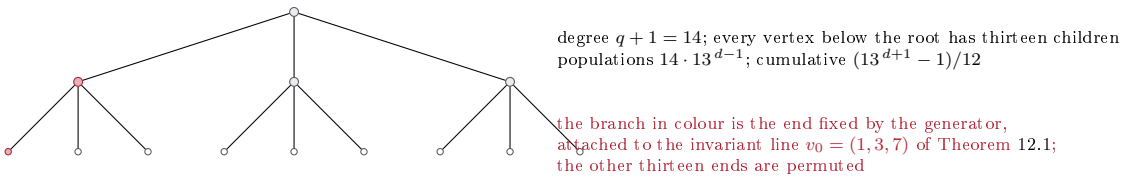


Figure 24: The tree of Structure 39, drawn to depth two with three of the fourteen directions shown. The subtree below any vertex is a copy of the whole, by Proposition 39.5, and the coloured branch is the unique fixed end of Theorem 38.3.

Role in the architecture. Tokenisation writes a byte as three digits and thereby names a vertex of the tree; the same triple is at once an address, a state, and a ring element. Only afterwards is a channel projected, an orbit position taken, and a level of the pyramid applied. The engine therefore has one entry point and one exit, and every intermediate object is three residues.

Structure 36. Addresses, the tree, and the boundary

Definition 36.1 (Leaf map)**.** A symbol is written in the radix thirteen, and its digits address a vertex of the tree whose vertices are the classes modulo 13^k and whose edges fix one further digit. The depth of a vertex is the integer k.

Proposition 36.2 (Population and ends)**.** *The tree is regular of degree* $13+1=14$ *and carries*

$$14 \cdot 13^{\,d-1} \tag{222}$$

vertices at depth d*, so the first three depths carry* 14*,* 182*, and* 2366 *vertices; its ends form the projective line over the prime field, of fourteen points. Agreement of two addresses is the integer count of shared leading digits, and the strong triangle inequality of Theorem 32.1 states that of two agreement counts the joint count is the smaller.*

Proof. A vertex at depth k is a class modulo 13^k, and it contains exactly thirteen classes modulo 13^{k+1}, so every vertex below the root has thirteen children and one parent, giving degree fourteen. The population at depth d is therefore $14 \cdot 13^{d-1}$. An end is a compatible choice of one class at every depth, and these correspond to the points of the projective line over the residue field. □

36.1 Tokenisation in the radix thirteen

Proposition 36.3 (A byte is three digits, with no dictionary)**.** *Every byte* $b \in \{0, \dots, 255\}$ *is written uniquely as*

$$b = a_0 + 13a_1 + 169a_2, \qquad a_i \in \mathbb{F}_{13}, \tag{223}$$

so the coordinate triple of a token is produced by two divisions with remainder and no learned vocabulary. For example 255 *has digits* $(8, 6, 1)$*. The address space of* 2197 *states holds all two hundred fifty six bytes with* 1941 *states unused, and by* (111) *the same count is the order of the algebra, so a token, an address, and a ring element occupy the same three residues.*

Proof. Uniqueness of the digits is division with remainder in the radix thirteen, and the identity was checked for all two hundred fifty six values. □

36.2 Vertices to a given depth

Proposition 36.4 (Cumulative population)**.** *The number of vertices of the tree to depth* k *is*

$$\frac{13^{\,k+1}-1}{12}, \tag{224}$$

giving 14 *at depth one,* 183 *at depth two, and* 2380 *at depth three. The value at depth two coincides with the number of points of the projective plane of Proposition 58.24, both being* $(13^3-1)/(13-1)$*.*

Proof. Summing the populations of Proposition 36.2 over depths one to k gives a geometric sum with ratio thirteen, namely $14(13^k-1)/12+1$, which simplifies to (224). The stated values are the evaluations at $k = 1, 2, 3$. □

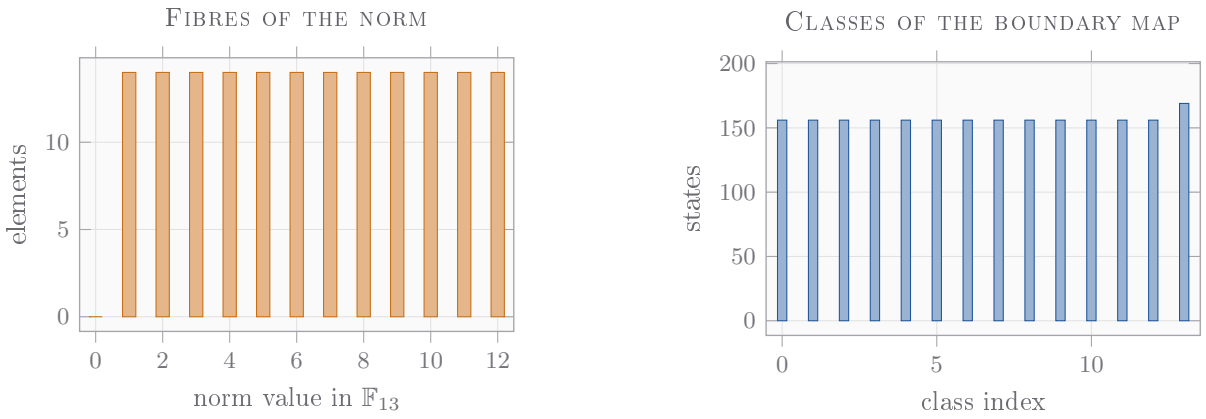


Figure 25: Left: the value zero carries no nonzero element, which is the anisotropy of Theorem 48.1, and each of the twelve nonzero values carries exactly fourteen, so $168 = 14 \cdot 12$. Right: thirteen classes of one hundred fifty six states and one of one hundred sixty nine, totalling 2197 by Theorem 36.6.

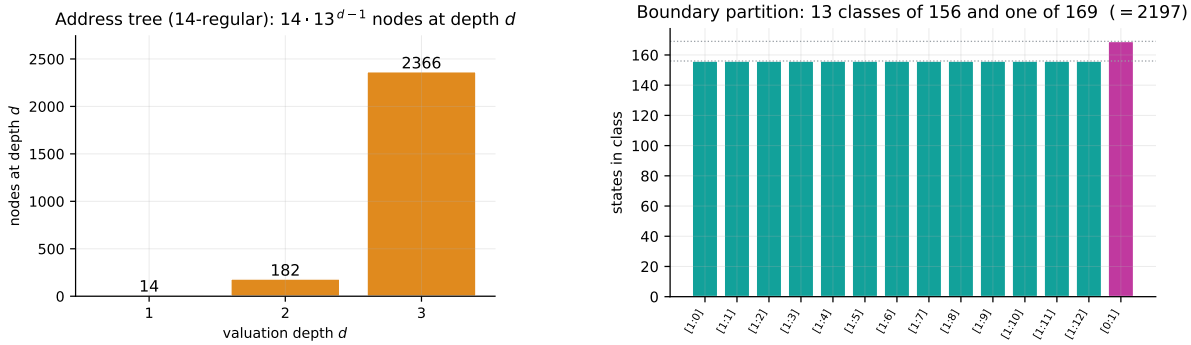


Figure 26: Left: the tree is regular of degree fourteen and carries $14 \cdot 13^{d-1}$ vertices at depth d, the depth being the integer valuation. Right: the boundary partition, thirteen classes of one hundred fifty six states and one of one hundred sixty nine.

Proposition 36.5 (Address budget). *Since* $13^2 = 169 < 256 \le 13^3 = 2197$, *two digits do not suffice for a byte and three do; the three digit space of* 2197 *states holds a byte with* 1941 *states to spare, and by* (111) *it is the same count as the order of the algebra, so an address and a ring element occupy the same space.*

Proof. The two inequalities are arithmetic, and the identification of the counts is (111), both sides being 13^3. □

Theorem 36.6 (The boundary partition). *Sending a state with nonzero first coordinate to* $[1 : a_1 a_0^{-1}]$ *and otherwise to* $[0 : 1]$ *partitions the state space into fourteen classes of two sizes,*

$$13 \text{ classes of } 156, \qquad 1 \text{ class of } 169, \qquad 13 \cdot 156 + 169 = 2197. \tag{225}$$

The classes are therefore not of equal size.

Proof. Over a finite base point the first coordinate has twelve nonzero choices and the third is free, giving $12 \cdot 13 = 156$; over the remaining point the first coordinate vanishes and the other two are free, giving 169. □

36.3 The boundary listed point by point

Theorem 36.7 (The orbit of fourteen is a projective line, and it misses the fixed point)**.** *The orbit of size fourteen in the decomposition* $183 = 1 + 14 + 3 \cdot 56$ *of Theorem 40.1 consists of the following points, listed in the order in which the generator visits them, each written with its normalised representative:*

$$\begin{aligned}&(0,1,11),\ (1,1,7),\ (1,3,3),\ (1,10,2),\ (1,8,6),\ (1,12,11),\ (1,7,8),\\ &(1,6,10),\ (1,5,12),\ (1,0,9),\ (1,4,1),\ (1,2,5),\ (1,9,4),\ (1,11,0).\end{aligned} \tag{226}$$

These fourteen points are exactly the points of one projective line, namely the line

$$9x + 11y + 12z = 0, \tag{227}$$

so the boundary orbit is a line, of the $q + 1 = 14$ *points a line carries. The fixed point* $(1, 3, 7)$ *does not lie on it, since* $9 + 33 + 84 = 126 \equiv 9 \neq 0$*.*

Proof. The orbit was computed by applying the generator repeatedly to a representative and normalising at each step, returning to the start after fourteen steps. The line through the first two points was obtained as their cross product, and all fourteen points were checked to satisfy (227). A projective line over a field of q elements carries $q + 1$ points, so the orbit exhausts it. Substituting the fixed point into the equation gives a nonzero value. □

Corollary 36.8 (A line and a point off it)**.** *The generator preserves a line and a point not on that line, and these are the only invariant objects of the projective action: the point is the class of the eigenvector and the line is spanned by the rank two channel. The three remaining orbits, each of fifty six points, cover the* $183 - 15 = 168$ *points that are neither the fixed point nor on the line.*

Proof. The point and the line are invariant by Theorem 36.7. That there are no others is Theorem 21.2, where the invariant subspaces form a chain of four, of which the two proper ones are the eigenline and the invariant plane. The count is $183 - 1 - 14 = 168 = 3 \cdot 56$. □

36.4 The layer map in one step

Proposition 36.9 (From the deepest layer to the balanced state)**.** *For every residue* k*,*

$$T_3\,(0, 0, k)^{\top} = (k, k, k)^{\top}, \tag{228}$$

so a state supported on the last coordinate alone is carried in one step to the balanced state with all three coordinates equal. In the notation of Definition 58.1 this raises the layer from the deepest to the shallowest in a single application.

Proof. Applying the companion form to $(0, 0, k)$ gives $(k,\ 0 + k,\ 0 + k) = (k, k, k)$, and this was checked for several residues. □

36.5 The construction assembled in one line

Proposition 36.10 (From one representative to the full orbit)**.** *The objects of this paper assemble in the following order, each count established where indicated:*

$$\begin{aligned}13\ \textit{representatives} &\longrightarrow 2197\ \textit{states} && \textit{Proposition 34.3, (111)},\\ 2197\ \textit{states} &\longrightarrow 14\ \textit{boundary classes} && \textit{Theorem 36.6},\\ 14\ \textit{classes} &\longrightarrow 168\ \textit{orbit positions} && \textit{Theorem 20.1},\\ 168 &\longrightarrow 8 \cdot 3 \cdot 7 && \textit{Theorem 44.1},\end{aligned} \tag{229}$$

and the three factors of the last row are realised by the three Sylow subgroups $\langle T_3^{21} \rangle$, $\langle T_3^{56} \rangle$, $\langle T_3^{24} \rangle$, which are also the three orders appearing in the presentation of Definition 48.15 together with the exponent two.

Proof. Each row is the reference given; the last row is the coprime factorisation of the period, and the three subgroups have the stated orders by Proposition 43.1. □

Remark 36.11 (Where the subgroups of order seven live). A group of order seven occurs in the multiplicative group of the field of sixty four elements, since $63 = 9 \cdot 7$. That field has characteristic two and is not among the fields of this construction, which are $\mathbb{F}_{13}$, $\mathbb{F}_{169}$, and the algebra of 2197 elements, of characteristic thirteen throughout. The subgroups of order seven that this paper uses are $\langle T_3^{24} \rangle$ inside the orbit, the subgroup of index two inside the torus of order fourteen, and the cyclic group generated by the third element of the presentation (270). None of them involves the field of sixty four elements.

Structure 37. The tree of the rank one channel

37.1 The tree as a combinatorial object

Definition 37.1 (Vertices, edges, depth)**.** The tree attached to the rank one group over the local field has as vertices the classes modulo the powers of thirteen, and an edge joins a class modulo 13^k to each of the thirteen classes modulo 13^{k+1} contained in it. It is regular of degree $13 + 1 = 14$. The depth of a vertex is the integer k. No distance is defined on the tree, and none is used: the only comparison between two ends is the integer count of their shared leading digits.

37.2 Self-similarity by integer counts

Proposition 37.2 (Branching and population)**.** *Every vertex below the root has thirteen descendants, so the population at depth d is $14 \cdot 13^{d-1}$ and the cumulative population to depth k is $(13^{k+1} - 1)/12$. The subtree below any vertex is isomorphic to the subtree below any other vertex of the same depth, so the tree is self-similar in the exact sense that these two integer counts, the branching number and the population, are the same at every vertex.*

Proof. Every vertex below the root has thirteen children by Proposition 36.2, so the rooted subtree below any two vertices of the same depth is described by the same branching number, and the populations follow by the same geometric sum. □

Remark 37.3 (Self-similarity recorded by two integers). Self-similarity is recorded by the branching number thirteen and the population counts above. No exponent expressing a dimension is formed, since forming one would require a quantity that the construction does not admit.

37.3 Galleries counted, not measured

Proposition 37.4 (The count of edges)**.** *Between two vertices of the tree there is a unique path without repetition, and the number of edges on it is an integer count. For two ends the count of shared leading digits determines the vertex at which their paths separate. Both quantities are counts of objects and neither is a measurement.*

Proof. A tree has a unique path without repetition between any two vertices, so the number of its edges is well defined, and it is a count of edges. For two ends the classes coincide down to the depth at which their digit strings first differ, which is the vertex where the paths separate. □

Structure 38. The tree seen from the invariant line

38.1 Depth as the only comparison

Definition 38.1 (Classes along the line)**.** A state of the invariant line is refined by fixing digits in the radix thirteen, so at depth k it is a class modulo 13^k. Two states of the line are compared only by the integer count of digits at which they agree, and no other comparison is defined.

38.2 The population at each depth

Proposition 38.2 (Counts along the tower)**.** *The classes at depth k number 13^k, of which $13^k - 13^{k-1}$ are invertible, and the vertices of the tree of Structure 37 at depth d number $14 \cdot 13^{d-1}$, with cumulative count $(13^{d+1} - 1)/12$. At depth two the cumulative count is 183, which is also the number of points of the projective plane by Proposition 58.24.*

Proof. The counts are Proposition 39.2 and Proposition 36.2, and the coincidence at depth two is the identity $(13^3 - 1)/(13 - 1) = 183$ of Proposition 58.24. □

38.3 The generator fixes one end

Theorem 38.3 (A single fixed end)**.** *The generator fixes exactly one end of the tree, namely the one attached to the invariant line of Theorem 12.1, and it permutes the remaining thirteen ends. The count follows from Theorem 40.1, where the projective action has exactly one fixed point, and from Proposition 12.5, which identifies that point with the class of v_0.*

Proof. An end fixed by the generator is a projectively fixed direction, and by Theorem 40.1 there is exactly one such point, the class of v_0, identified in Proposition 12.5. The remaining thirteen ends are therefore permuted among themselves. □

Remark 38.4 (A permutation of fourteen objects). That an end is fixed and the others are permuted is a statement about a finite permutation of fourteen objects. Nothing is displaced by an amount, and the only integer attached to a pair of ends is the count of digits at which they agree.

Role in the architecture. The engine has one direction of storage that the dynamics never carries away, and every other direction turns. Refinement along that direction is the addition of a digit, and the whole comparison structure on it is a count of agreeing digits.

Structure 39. The tree in full: balls, boundary, and self-similarity

The rank one channel carries a tree of rank one, and this structure gives it completely: the vertices as classes, the balls as cosets, the boundary as the projective line, the branching, and the sense in which the tree repeats itself. Every quantity is a count.

39.1 Vertices as classes at a depth

Definition 39.1 (The vertex set)**.** For each depth $k \geq 0$ the vertices at that depth are the classes modulo 13^k, of which there are 13^k, and an edge joins a class modulo 13^k to each of the thirteen classes modulo 13^{k+1} it contains. The resulting graph is a tree, since a class at depth $k+1$ lies in exactly one class at depth k.

Proposition 39.2 (Counts at each depth)**.** *The numbers of classes and of invertible classes at depth k are*

$$\left|\mathbb{Z}/13^k\right| = 13^k, \qquad \left|(\mathbb{Z}/13^k)^*\right| = 13^k - 13^{k-1}, \tag{230}$$

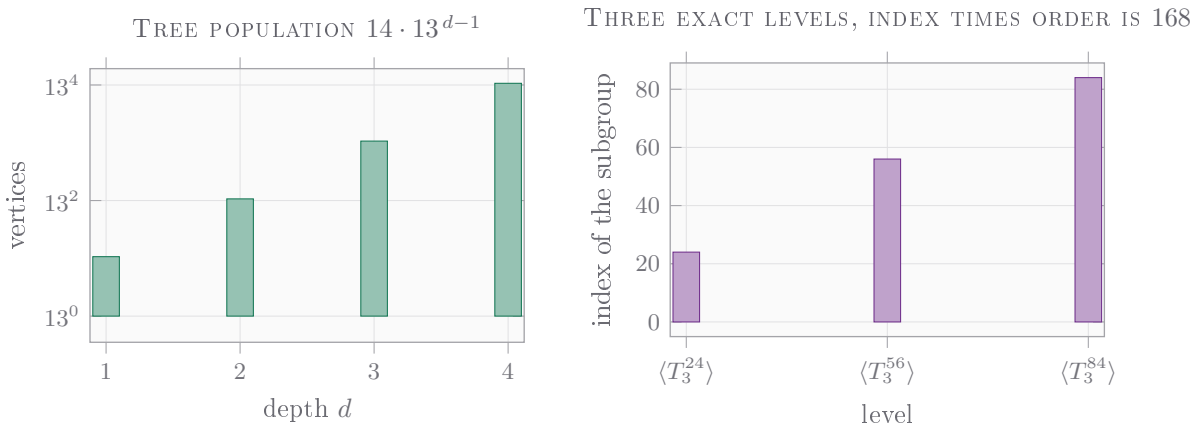


Figure 27: Left: the number of vertices of the tree at each depth, on a logarithmic scale in the radix thirteen, so that equal steps correspond to fixing one further digit. Right: the indices 24, 56, 84 of the three subgroups of the pyramid of Theorem 43.2, whose orders are 7, 3, 2.

so the first depths carry $13, 169, 2197, 28561$ *classes with* $12, 156, 2028, 26364$ *invertible ones. The value* 169 *at depth two is the size of the semantic channel, and the value* 2197 *at depth three is the size of the state space, by* (63).

Proof. The ring $\mathbb{Z}/13^k$ has 13^k elements, and a class is invertible exactly when its representative is prime to thirteen, which excludes the 13^{k-1} multiples of thirteen. □

39.2 Balls are cosets

Proposition 39.3 (The ball of a state)**.** *The ball of a at depth k is the additive coset*

$$B_k(a) = a + 13^k \mathbb{Z}_{13}, \tag{231}$$

which is a class of the subgroup $13^k\mathbb{Z}_{13}$, hence is closed under addition of that subgroup and is simultaneously closed and open. By Theorem 33.2 two balls are nested or disjoint, so the balls at all depths form a tree ordered by containment, and this tree is the one of Definition 39.1.

Proof. The set of x with $v_{13}(x - a) \geq k$ is exactly $a + 13^k\mathbb{Z}_{13}$, a coset of the subgroup $13^k\mathbb{Z}_{13}$. Two cosets of subgroups of a chain are nested or disjoint by Theorem 33.2, and containment of the subgroups orders them, which is the tree of Definition 39.1. □

39.3 Branching and the population

Theorem 39.4 (Regular branching)**.** *Every vertex below the root has exactly thirteen children, and the tree is regular of degree*

$$q + 1 = 14. \tag{232}$$

The populations are

$$14 \cdot 13^{d-1} \text{ vertices at depth } d, \qquad \frac{13^{d+1} - 1}{12} \text{ vertices to depth } d, \tag{233}$$

giving $14, 182, 2366, 30758$ *by depth and* $14, 183, 2380, 30941$ *cumulatively. The cumulative value at depth two, namely* 183*, is the number of points of the projective plane by Proposition 58.24.*

Proof. This is Proposition 36.2 together with the geometric sum of Proposition 36.4; the four displayed values are the evaluations at depths one to four. □

39.4 Self-similarity, stated by counts

Proposition 39.5 (Every subtree is a copy)**.** *The subtree hanging below any vertex at depth d is isomorphic, as a rooted tree, to the subtree below any other vertex at the same depth, and to the whole tree below its root: each is a rooted tree in which every vertex has thirteen children. The self-similarity is therefore complete and is recorded by two integers, the branching number thirteen and the degree fourteen.*

Proof. A rooted tree in which every vertex has thirteen children is determined up to isomorphism by that number, and by Proposition 36.2 every vertex below the root has thirteen children, so all such subtrees are isomorphic. □

Remark 39.6 (The branching number and the populations). A quantity of the form "the exponent relating the branching number to a scaling" is not defined here and is not needed. The construction records the branching number, the degree, and the populations (233), all of which are integers, and it forms no ratio of logarithms and no exponent expressing a dimension, since such a value would not lie in any ring the construction admits.

39.5 The boundary

Theorem 39.7 (The ends and the projective line)**.** *The ends of the tree, that is the infinite descending chains of balls, are in bijection with the points of the projective line over the residue field,*

$$\left|\mathrm{P}^1(\mathbb{F}_{13})\right| = q + 1 = 14, \tag{234}$$

and the generator fixes exactly one of them, the end attached to the invariant line of Theorem 12.1, permuting the remaining thirteen, by Theorem 38.3. The fourteen ends are also the fourteen classes of the boundary map of Theorem 36.6, and the same number counts the torus of Theorem 48.1, by Corollary 48.4.

Proof. The ends are the compatible sequences of classes, which correspond to the points of the projective line over the residue field, of which there are $q + 1$. The statement about the fixed end is Theorem 38.3, and the identifications with the boundary classes and the torus are Corollary 48.4. □

39.6 Comparison of two ends

Definition 39.8 (Agreement count)**.** Two ends are compared by the integer number of initial edges they share, equivalently the greatest depth at which their balls coincide. By Theorem 58.48 the three agreement counts of any three ends have the property that the smallest occurs at least twice.

Role in the architecture. Storage is indexed by balls, which are congruence classes, and refinement is the fixing of one further digit. Two entries are compared by the depth at which they first differ, so retrieval halts at that digit, and the tree repeats itself at every vertex, so the same lookup procedure applies at every depth.

Structure 40. The projective action

40.1 The scalars act trivially

Theorem 40.1 (Projective order and orbit decomposition)**.** *Since μ_3 consists of scalars it acts trivially on the projective plane, and the image of T_3 there has order*

$$\left[\langle T_3\rangle : \mu_3\right] = 56, \tag{235}$$

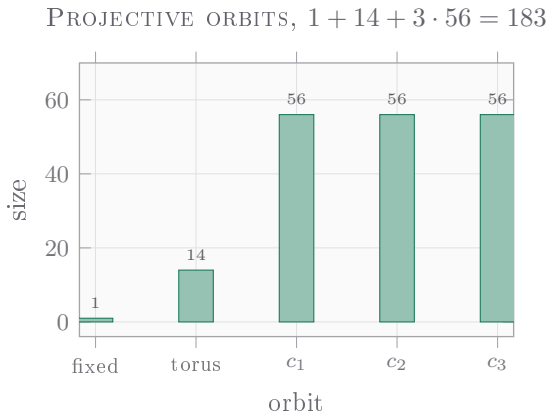


Figure 28: The five orbits of the generator on the projective plane, Theorem 40.1: one fixed point, one orbit of the size of the torus, and one orbit of size fifty six for each coset of the scalars.

that is, fifty six is the least k for which T_3^k is scalar. Acting on the one hundred eighty three points of $\mathrm{PG}(2,\mathbb{F}_{13})$ *the generator has exactly five orbits, of sizes*

$$56,\ 56,\ 56,\ 14,\ 1, \qquad \textit{so} \qquad 183 = 3 \cdot 56 + 14 + 1 = 168 + 14 + 1, \tag{236}$$

matched to the channels as follows: the fixed point is the class of $v_0 = (1,3,7)$, lying in $\operatorname{im} P_1$*; the orbit of size fourteen lies entirely in* $\operatorname{im} P_{23}$ *and is the projective line of that plane; the three orbits of size fifty six exhaust the remaining one hundred sixty eight points, one for each coset of μ_3.*

Proof. The kernel of the map to the projective group is the group of scalars inside the orbit, which is μ_3, so the image has order $168/3 = 56$, and testing successive powers confirms fifty six is least. Enumerating the action on all one hundred eighty three points returns the five orbits. The fixed point is the class of the eigenvector, since $T_3 v_0 = 7 v_0$; applying P_1 to every point of the fourteen element orbit gives zero, and a rank two space has a projective line of $q + 1 = 14$ points. □

Role in the architecture. Projectively the engine sees one fixed direction, one boundary cycle of fourteen, and three generic cycles of fifty six, so a projective computation runs on a clock of fifty six while the factor three is carried separately as one residue.

Structure 41. The index map and its collisions

41.1 The map

Definition 41.1 (A chamber index)**.** Define, for a state (a_0, a_1, a_2),

$$\mathrm{ch}(a_0, a_1, a_2) = \big(a_2 \bmod 13\big) \cdot 168 + \big(a_0 + 13 a_1 + 169 a_2\big) \bmod 168. \tag{237}$$

41.2 The image and the collisions

Theorem 41.2 (Exactly thirteen collisions, and the size of the image)**.** *The map* (237) *takes exactly*

$$2184 = 13 \cdot 168 = 13^3 - 13 \tag{238}$$

distinct values on the 2197 *states, so exactly thirteen states are identified with others, and the identifications are*

$$\mathrm{ch}(0, 0, a_2) = \mathrm{ch}(12, 12, a_2) \qquad \textit{for each } a_2, \tag{239}$$

one collision for each of the thirteen values of the third coordinate. Both statements were verified by tabulating the map on all states.

Proof. The map was tabulated on all states, giving exactly 2184 distinct values and therefore thirteen identifications. The displayed identity holds because the two arguments differ by $12 + 13 \cdot 12 = 168$ in the second summand, which vanishes modulo the period, while the first summand depends only on the third coordinate. □

Remark 41.3 (What 2184 is and is not). The number 2184 occurs here as the size of the image of (237), and it is also the order of the generator modulo 13^2 by Theorem 32.4, and it equals $13^3 - 13$. It is not a count of chambers: the spherical residue has 2562 chambers by Proposition 58.7, and the affine building is infinite, as recorded in Remark 58.8.

Structure 42. Stationary states and the ladder

42.1 The states of the address line

Definition 42.1 (The ladder)**.** For $a \in \mathbb{F}_{13}$ let $v_a = (0, a, a)$, the thirteen states of the address line $\ker D$ of Theorem 15.1.

42.2 Four properties, all verified

Theorem 42.2 (The ladder and its release)**.** *The thirteen states satisfy*

(i) an index in arithmetic progression,

$$\mathrm{ch}(v_a) = 182\,a \quad (0 \le a \le 11), \qquad \mathrm{ch}(v_{12}) = 2016, \tag{240}$$

with step $182 = 13 \cdot 14 = p(p+1)$*;*

(ii) a syntactic part proportional to the eigenvector,

$$P_1 v_a = 4a\, v_0; \tag{241}$$

(iii) a vanishing total,

$$\sum_{a=0}^{12} v_a = 0, \qquad \textit{hence} \qquad \sum_{a=0}^{12} P_1 v_a = 0; \tag{242}$$

(iv) release after one turn of the positional clock,

$$D\,T_3^{14} v_a \neq 0 \qquad \textit{for every } a \neq 0. \tag{243}$$

Proof. Each item was computed over $\mathbb{F}_{13}$ for all thirteen states. For (i) the index formula gives $182a$ directly, the last value being the stated exception; for (ii) the coefficient four is $u \cdot (0,1,1) = 7 + 10 = 17 \equiv 4$ by Theorem 12.2; (iii) is the sum $\sum_a a = 78 \equiv 0$ modulo thirteen in each coordinate; and (iv) was verified for each nonzero a. □

Remark 42.3. Item (iv) says that a state parked on the address line does not remain there once the positional clock of Theorem 48.1 has completed one turn of fourteen steps, since the difference operator no longer annihilates it. The address line is therefore stationary for the projections but not for the dynamics.

Structure 43. The Sylow structure and the renormalisation pyramid

43.1 The three Sylow subgroups

Proposition 43.1 (Orders and generators). *The orbit has exactly three Sylow subgroups, of orders eight, three, and seven, generated respectively by T_3^{21}, T_3^{56}, and T_3^{24}, and they intersect pairwise in the identity, since their orders are pairwise coprime. Their internal product is the whole orbit.*

Proof. A cyclic group has one subgroup of each order dividing its own, so the Sylow subgroups are those of orders eight, three, and seven, generated by the powers at the complementary exponents by the divisor law of Theorem 20.1. Their pairwise intersections are trivial because their orders are pairwise coprime, and the product of the orders is the group order, so their internal product is everything. □

43.2 The three exact levels

Theorem 43.2 (The pyramid). *Quotienting the orbit by each Sylow subgroup in turn gives three levels of coarse graining, of exact integer factors,*

$$\underbrace{24 \cdot 7}_{\text{by } \langle T_3^{24} \rangle} = \underbrace{56 \cdot 3}_{\text{by } \langle T_3^{56} \rangle} = \underbrace{84 \cdot 2}_{\text{by } \langle T_3^{84} \rangle} = 168, \tag{244}$$

with subgroup orders seven, three, and two and indices twenty four, fifty six, and eighty four. The middle level is the projective level of Theorem 40.1, its subgroup being the group of scalars.

Proof. The three subgroup orders are $\operatorname{ord}(T_3^{24}) = 7$, $\operatorname{ord}(T_3^{56}) = 3$, $\operatorname{ord}(T_3^{84}) = 2$ by Theorem 20.1, and the index is the quotient of the group order by the subgroup order. Cosets of a subgroup partition the group and, in a cyclic group, all have the size of that subgroup. □

43.3 Reversibility of each level

Proposition 43.3 (No information is lost). *Each level is reversible: a coset of a subgroup of order m determines its elements once one representative is fixed, and the representative is recovered by multiplying by the generator of that subgroup. Hence coarse graining and refinement are mutually inverse and neither uses an approximation.*

Proof. A coset of a subgroup of order m is the set of products of one representative with the m elements of the subgroup, so fixing a representative recovers the coset elementwise by multiplying by the powers of its generator. Coarse graining and refinement are therefore mutually inverse. □

Role in the architecture. An extended context is summarised by one exact factor per level, and the summary of a summary is again a coset. A block covering one full period carries no residue at all, by Proposition 20.3, so a nonzero summary records only the deviation from a complete turn.

Structure 44. The remainder decomposition of the index

44.1 The index group splits

Theorem 44.1 (Three independent counters). *The index group of the orbit splits as*

$$\mathbb{Z}/168 \;\cong\; \mathbb{Z}/8 \times \mathbb{Z}/3 \times \mathbb{Z}/7, \qquad n \longmapsto \big(n \bmod 8,\ n \bmod 3,\ n \bmod 7\big), \tag{245}$$

and the map is a bijection, verified on all one hundred sixty eight indices. A position on the orbit is therefore three small residues rather than one large one.

Proof. The three moduli are pairwise coprime with product one hundred sixty eight, so the map is a ring homomorphism with trivial kernel, hence a bijection between sets of the same size; this was also confirmed by tabulating all indices. □

44.2 The idempotents as the glue

Proposition 44.2 (Reassembly)**.** *The three residues are reassembled by the idempotents of* (137)*,*

$$n \equiv 105\,(n \bmod 8) + 112\,(n \bmod 3) + 120\,(n \bmod 7) \pmod{168}, \tag{246}$$

each coefficient being congruent to one modulo its own part and to zero modulo the others, with $105 + 112 + 120 \equiv 1$*.*

Proof. Each coefficient is congruent to one modulo its own part and to zero modulo the other two, so the right side agrees with n modulo each of eight, three, and seven, hence modulo their product by Theorem 44.1. □

44.3 Cost of carrying a position

Remark 44.3 (Three counters of bounded size). Advancing a position by one step advances three counters of periods eight, three, and seven independently, and no counter exceeds seven. Comparing two positions for equality is three comparisons of residues, and no arithmetic beyond addition modulo a small number occurs.

Part VIII
The transform

Structure 45. The number theoretic transform

45.1 The transform and its size

Proposition 45.1 (An exact transform of size twelve)**.** *Two is a primitive root modulo thirteen, its successive powers being*

$$1, 2, 4, 8, 3, 6, 12, 11, 9, 5, 10, 7, \tag{247}$$

which exhaust $\mathbb{F}_{13}^{*}$*. Hence the prime field carries an exact transform of size twelve whose roots of unity are the powers of two. No root outside the prime field and no transcendental value is required, and the inverse transform uses the Fermat inverse of Structure 59.*

Proof. The listed powers of two exhaust the twelve nonzero residues, so two has order twelve and is a primitive root. A root of unity of that order and an invertible scalar N are all that a transform of size N requires, and both are present. □

45.2 Twiddle factors as distinguished powers

Definition 45.2 (The three twiddle factors)**.** The twiddle factors of the transform on the orbit are the three Sylow generators of Theorem 20.2,

$$W_8 = T_3^{21} \text{ (order 8)}, \qquad W_3 = T_3^{56} = 3I \text{ (order 3)}, \qquad W_7 = T_3^{24} \text{ (order 7)}, \tag{248}$$

so the middle factor is a scalar and costs one multiplication by the residue three.

45.3 The Cooley and Tukey factorisation

Theorem 45.3 (Splitting the transform along the period)**.** *The period factors into pairwise coprime parts,*

$$168 = 8 \cdot 3 \cdot 7, \qquad \gcd(8,3) = \gcd(8,7) = \gcd(3,7) = 1, \tag{249}$$

so a transform over the full orbit decomposes into three nested transforms of sizes eight, three, and seven, in any order, with the twiddle factors (248)*. Every power of the generator factors through the three,*

$$T_3^n = W_8^a \, W_3^b \, W_7^c, \tag{250}$$

with exponents determined by the residues of n modulo eight, three, and seven.

Proof. The factorisation of the period and the coprimality are arithmetic. Since the three subgroups generated by the twiddles have pairwise coprime orders and their product has order $8{\cdot}3{\cdot}7 = 168$, they generate the whole cyclic orbit internally, so every element has a representation (250); the representation was exhibited by direct search for a sample of exponents including 0, 1, 5, 42, 84, and 167. □

Role in the architecture. Convolution over the orbit is performed by three small exact transforms and never by a single large one, and one of the three stages is multiplication by a scalar. No rounding occurs at any stage, since every twiddle is a power of the one operator.

Structure 46. The transform in full

The transform of the construction is arithmetic. This structure gives it completely: the roots, the available sizes, the inverse, the orthogonality, and the convolution rule, over both fields, together with the stages of the factorisation.

46.1 Definition of the transform

Definition 46.1 (Transform of size N)**.** Let F be $\mathbb{F}_{13}$ or $\mathbb{F}_{169}$ and let $\omega \in F^*$ have order exactly N, with N invertible in F. The transform of a vector $a = (a_0, \ldots, a_{N-1})$ over F and its inverse are

$$\widehat{a}_i = \sum_{j=0}^{N-1} a_j \, \omega^{ij}, \qquad a_j = N^{-1} \sum_{i=0}^{N-1} \widehat{a}_i \, \omega^{-ij}, \tag{251}$$

all sums being finite and all entries elements of F.

46.2 Orthogonality and inversion

Theorem 46.2 (The transform is invertible)**.** *The rows of the transform are orthogonal in the exact sense*

$$\sum_{j=0}^{N-1} \omega^{(i-k)j} = \begin{cases} N, & i \equiv k \pmod{N}, \\ 0, & \text{otherwise}, \end{cases} \tag{252}$$

so the second formula of (251) *inverts the first. Over $\mathbb{F}_{13}$ with $N = 12$ and $\omega = 2$ this was checked on all one hundred forty four pairs of indices, and the inverse was checked to recover the input.*

Proof. For $i \equiv k$ every term is one and the sum is N. Otherwise $\omega^{i-k} \neq 1$ and the sum is the geometric series $(\omega^{(i-k)N} - 1)(\omega^{i-k} - 1)^{-1}$, whose numerator vanishes since $\omega^N = 1$ while the denominator is invertible. □

46.3 The convolution rule

Theorem 46.3 (Products become pointwise). *For the cyclic convolution* $(a * b)_i = \sum_j a_j b_{i-j}$, *indices modulo* N,

$$\widehat{(a * b)}_i = \widehat{a}_i \, \widehat{b}_i, \tag{253}$$

so a convolution is computed by two transforms, N *products, and one inverse transform. The rule was verified over* $\mathbb{F}_{13}$ *at size twelve and over* $\mathbb{F}_{169}$ *at sizes fourteen and one hundred sixty eight.*

Proof. Substituting the definition and exchanging the two finite sums gives $\sum_{j,k} a_j b_k \omega^{i(j+k)}$, which factors as the product of the two transforms at the index i. The rule was verified at the three sizes stated. □

46.4 The available sizes

Theorem 46.4 (Every divisor of the period is an admissible size). *Over* $\mathbb{F}_{169}$ *a transform of size* N *exists exactly when* N *divides* 168, *since the multiplicative group is cyclic of that order and contains an element of each such order, and the scalar* N *is invertible for every one of them, because thirteen divides no divisor of* 168. *Over* $\mathbb{F}_{13}$ *the admissible sizes are the divisors of twelve,*

$$\{1, 2, 3, 4, 6, 12\} \subset \mathbb{F}_{13}, \qquad \{1, 2, 3, 4, 6, 7, 8, 12, 14, 21, 24, 28, 42, 56, 84, 168\} \subset \mathbb{F}_{169}. \tag{254}$$

In particular the size one hundred sixty eight is admissible, with $168 \equiv 12$ *invertible and inverse twelve, and a root of that order is* $1 + 2\theta$.

Proof. A root of order N exists in a cyclic group of order M exactly when N divides M, and here M is twelve or one hundred sixty eight. The scalar N is invertible unless thirteen divides N, and no divisor of $168 = 2^3 \cdot 3 \cdot 7$ is divisible by thirteen. The element $1 + 2\theta$ was found to have order one hundred sixty eight, and the transform of that size was verified to invert and to satisfy the convolution rule. □

Remark 46.5 (Every admissible size is available). One might expect the largest size to fail because the scalar N vanishes in the field. That happens only when thirteen divides N, and here $168 \equiv 12$, which is invertible. No admissible size of this construction is obstructed.

46.5 The stages

Proposition 46.6 (Three coprime stages). *Since* $168 = 8 \cdot 3 \cdot 7$ *with pairwise coprime factors, a transform of the full size decomposes into three nested stages of sizes eight, three, and seven, whose scalars are invertible with*

$$8^{-1} = 5, \qquad 3^{-1} = 9, \qquad 7^{-1} = 2 \quad \text{in } \mathbb{F}_{13}, \tag{255}$$

and whose roots are the three twiddle factors (248), *namely* T_3^{21}, $T_3^{56} = 3I$, *and* T_3^{24}, *of orders eight, three, and seven.*

Proof. The three factors are pairwise coprime with product the period, so the transform factors into three nested stages by the remainder theorem, and each scalar is invertible since thirteen divides none of them; the inverses were computed. The roots of the stages are the powers of the generator of the corresponding orders, which are the three twiddle factors. □

Role in the architecture. A convolution over any admissible size is three transforms and a pointwise product, with every root a power of the one operator and every scalar invertible. Nothing is approximated and no size is unavailable, so the transform is exact at every stage.

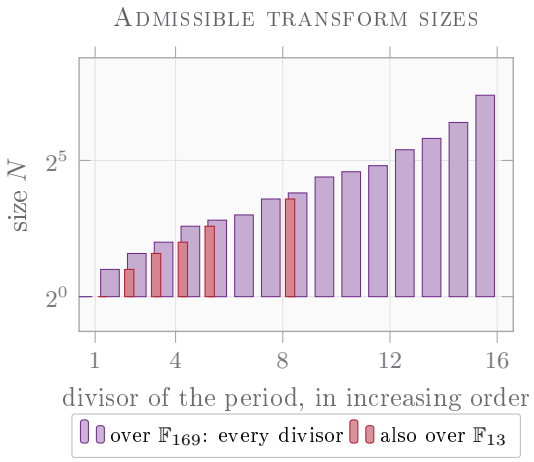


Figure 29: The sixteen divisors of the period are all admissible over the quadratic extension, since thirteen divides none of them; over the prime field only the six divisors of twelve are available, by Theorem 46.4.

Structure 47. The transform is arithmetic, not analytic

Remark 47.1 (The transform used, and why it suffices). The transform of Structure 45 is the number theoretic transform: its roots of unity are the powers of two in $\mathbb{F}_{13}^*$, listed in (247), and all of its arithmetic is exact. The transform over a continuum, whose roots of unity lie outside every finite field and whose evaluation requires values that this construction does not admit, is not used anywhere and is not needed: the factorisation (249) supplies the same decomposition into nested stages, with the twiddle factors (248) being powers of the one operator. No notion of a rate of repetition per unit of anything occurs; the only invariant attached to a stage is the order of its twiddle factor, which is an integer.

Part IX

Arithmetic of the residue extension and the counting identities

Structure 48. Norm, torus, and the two distinguished elements

48.1 The norm and its kernel

Theorem 48.1 (Norm, anisotropy, and the exact sequence)**.** *The norm is*

$$\mathrm{N}(a+b\theta) = (a+b\theta)(a-b\theta) = a^2 - 2b^2 = x^{14} \in \mathbb{F}_{13}^*, \tag{256}$$

an identity verified on all one hundred sixty eight nonzero elements. The form is anisotropic: $\mathrm{N}(x) = 0$ *only for* $x = 0$*. The norm surjects onto* $\mathbb{F}_{13}^*$ *with kernel the group* T *of order fourteen,*

$$1 \longrightarrow T \longrightarrow \mathbb{F}_{169}^* \xrightarrow{\mathrm{N}} \mathbb{F}_{13}^* \longrightarrow 1, \qquad 168 = 14 \cdot 12, \tag{257}$$

and each of the twelve nonzero fibres has exactly fourteen elements.

Proof. Fermat fixes the prime field and (53) gives $\theta^{13} = -\theta$, so σ is the nontrivial automorphism and $x\sigma(x) = x^{14} = a^2 - 2b^2$. If $a^2 = 2b^2$ with $b \neq 0$ then two is a residue, contradicting Definition 9.1. The norm attains all twelve nonzero residues, so the kernel has order fourteen. □

48.2 The two distinguished elements

Theorem 48.2 (The two elements of order twenty eight). *With $\varphi = 7 + 8\theta$ and $\delta = 1 + \theta$,*

$$\varphi^2 = \varphi + 1, \qquad \delta^2 = 2\delta + 1, \qquad \mathrm{N}(\varphi) = \mathrm{N}(\delta) = -1, \qquad \operatorname{ord}\varphi = \operatorname{ord}\delta = 28, \tag{258}$$

they generate one and the same cyclic group with $\delta = \varphi^{23}$, and

$$\delta^2 = 3 + 2\theta = g, \qquad \mathrm{N}(g) = 1, \qquad \operatorname{ord}(g) = 14, \qquad g^7 = -1, \tag{259}$$

so the group of order fourteen is the square of δ. The norm splits the group of order twenty eight,

$$\mathrm{N}(\varphi^k) = (-1)^k, \qquad \langle\varphi\rangle = T \sqcup (-1) \cdot T, \qquad 14 + 14 = 28. \tag{260}$$

The classical expression holds verbatim with a radix in place of a real root,

$$(3\theta)^2 = 18 \equiv 5, \qquad 2^{-1} = 7, \qquad \varphi = (1 + 3\theta) \cdot 7 = 7 + 8\theta. \tag{261}$$

Proof. Squaring gives $\varphi^2 = 49 + 128 + 112\theta \equiv 8 + 8\theta = \varphi + 1$ and $\delta^2 = 3 + 2\theta = 2\delta + 1$, with norms $49 - 128 \equiv -1$ and $1 - 2 = -1$; both orders were computed as twenty eight. A cyclic group has one subgroup of each order, so the two coincide, and $\delta = \varphi^{23}$ with $\gcd(23, 28) = 1$. The element g has norm one and order fourteen, hence generates the torus. Multiplicativity of the norm gives the shell splitting, and the classical expression follows from $(3\theta)^2 = 5$ and $2^{-1} = 7$. □

48.3 The ratio of the conjugate roots

Theorem 48.3 (The eigenvalue ratio explains the orbit of size fourteen). *The ratio of the conjugate roots satisfies*

$$\frac{\lambda_2}{\lambda_3} = 8 + 5\theta, \qquad \mathrm{N}\left(\frac{\lambda_2}{\lambda_3}\right) = 1, \qquad \operatorname{ord}\left(\frac{\lambda_2}{\lambda_3}\right) = 14, \qquad \frac{\lambda_2}{\lambda_3} = g^3, \tag{262}$$

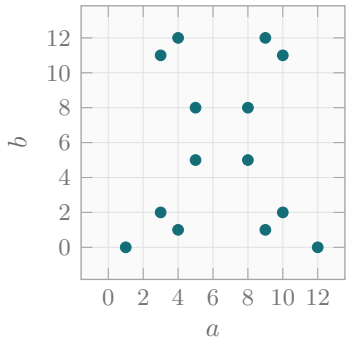
so the ratio lies in T, and the size of the projective orbit inside the rank two channel is the order of that ratio.

Proof. Computing $\lambda_2\lambda_3^{-1}$ in $\mathbb{F}_{169}$ gives $8 + 5\theta$, of norm $64 - 50 = 14 \equiv 1$, so it lies in T; its powers return the identity first at fourteen, and comparison with the powers of g identifies it as g^3. On the projective line of the rank two channel the generator acts by multiplication by the ratio, so an orbit there has size equal to the order of the ratio. □

Corollary 48.4 (Why three groups of fourteen agree). *The kernel T of the norm, the projective line of the rank two channel, and the set of ends of the tree of Structure 36 all have fourteen elements, and the agreement is structural: the first two are identified by Theorem 48.3, and the last two are the projective line of a rank two space over the prime field, which has $q + 1 = 14$ points in either presentation.*

Proof. The first two are identified by Theorem 48.3, the ratio lying in the torus and acting on the projective line of the rank two channel with orbit of its own order. The second and third are the projective line of a rank two space over the residue field, which has $q + 1$ points in either presentation. □

The fourteen positions of the torus



The group of order twenty eight, split by the norm

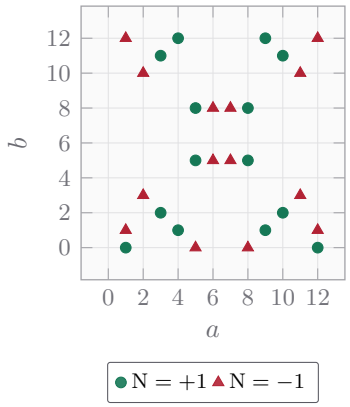


Figure 30: Left: the fourteen elements of $T = \ker \mathrm{N}$, generated by $g = 3 + 2\theta$, in the coordinates $a + b\theta$. Right: the twenty eight powers of φ, fourteen of norm one forming the torus and fourteen of norm minus one forming its single coset, by Theorem 48.2.

48.4 The metallic elements and the group of order twenty eight

Definition 48.5 (Metallic elements)**.** For a positive integer n the metallic element of index n is a solution of

$$x^2 = nx + 1 \tag{263}$$

in $\mathbb{F}_{169}$, when one exists. The indices one, two, three, and four are called golden, silver, bronze, and copper.

Theorem 48.6 (Which indices give elements of order twenty eight)**.** *Solving* (263) *over* $\mathbb{F}_{169}$ *for* n *from one to six gives*

index	*solutions*	*order*	*norm*
$n = 1$, *golden*	$7 + 5\theta$, $7 + 8\theta$	28	-1
$n = 2$, *silver*	$1 + \theta$, $1 - \theta$	28	-1
$n = 3$, *bronze*	8	4	-1
$n = 4$, *copper*	$2 + 3\theta$, $2 - 3\theta$	28	-1
$n = 5$	7, 11	12	10, 4
$n = 6$	9, 10	3, 6	3, 9

so exactly three indices, namely one, two, and four, give elements of order twenty eight, while the bronze index gives the residue eight of order four, lying already in the prime field. All the elements of order twenty eight have norm minus one.

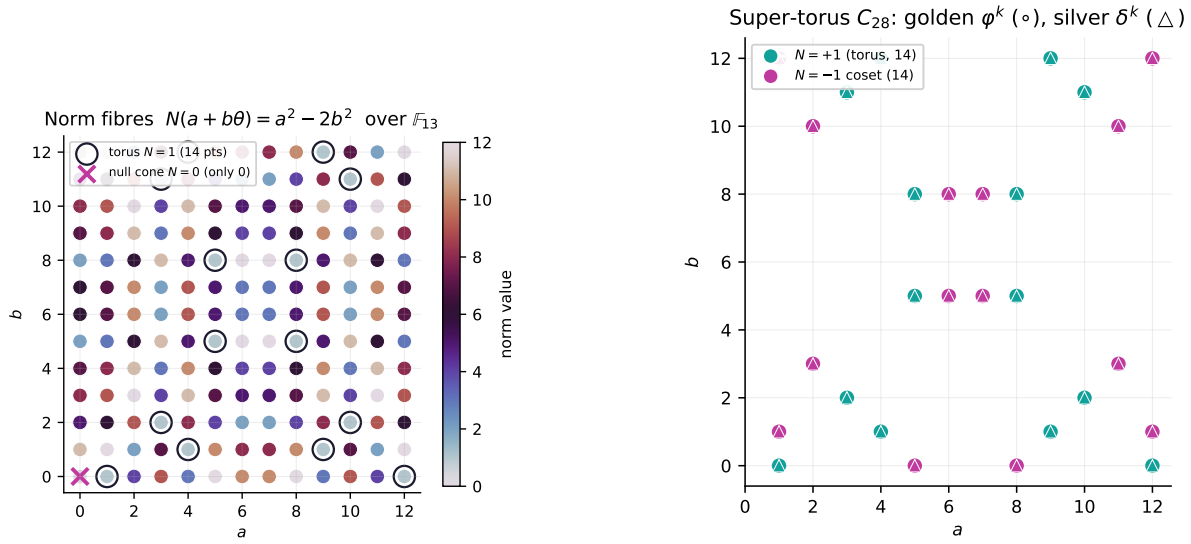


Figure 31: Left: the norm form $\mathrm{N} = a^2 - 2b^2$ on the grid of pairs; it is anisotropic, the locus $\mathrm{N} = 0$ being the origin alone, marked by a cross, and the ringed points the fourteen elements of T. Right: the group of order twenty eight, powers of φ as circles and of δ as triangles; even powers carry norm one and odd powers norm minus one.

Proof. Equation (263) was solved by exhaustive substitution over all one hundred sixty nine elements, and the orders and norms were computed. For $n = 3$ the discriminant $n^2 + 4 = 13$ vanishes modulo thirteen, so the quadratic has a repeated root, which is the single solution listed; for $n = 5$ and $n = 6$ the discriminants are $29 \equiv 3$ and $40 \equiv 1$, both squares, so the roots lie in the prime field and their orders divide twelve. □

Theorem 48.7 (One group of order twenty eight, containing the torus)**.** *The elements of order exactly twenty eight number twelve, and they all generate the same subgroup*

$$C_{28} \subset \mathbb{F}_{169}^{*}, \qquad |C_{28}| = 28, \qquad \big[C_{28} : \ker \mathrm{N}\big] = \frac{28}{14} = 2, \tag{264}$$

which contains the torus of order fourteen as its subgroup of index two. The golden, silver, and copper elements therefore all lie in this one group, and none of them lies in the torus, all having norm minus one.

Proof. A cyclic group has one subgroup of each order dividing its own, so all elements of order twenty eight generate the same subgroup, of that order, and there are $\varphi(28) = 12$ of them. The torus has order fourteen by Theorem 48.1 and fourteen divides twenty eight, so it is the subgroup of index two, consisting of the squares; membership was confirmed elementwise. An element of norm minus one is not in the kernel of the norm. □

Corollary 48.8 (The metallic elements are square roots of torus elements)**.** *For each metallic element x of order twenty eight, the square x^2 has order fourteen and lies in the torus. In particular the silver element satisfies*

$$\delta^2 = 2\delta + 1 = 3 + 2\theta = g, \tag{265}$$

the generator of the torus of Theorem 48.2.

Proof. Squaring halves the order in a cyclic group when the order is even, giving fourteen, and the squares of a group of order twenty eight form its subgroup of index two, which is the torus.

The displayed identity is the defining relation of the silver element together with the computation of Theorem 48.2. □

Remark 48.9 (One group, not several). The three metallic elements of order twenty eight do not generate three separate structures: they generate one group of order twenty eight, whose square subgroup is the single torus of order fourteen. The relation between the two is the norm, which sends the larger onto $\{\pm 1\}$ with the torus as kernel, so the twenty eight elements split into fourteen of norm one and fourteen of norm minus one, as recorded in Theorem 48.2.

48.5 Two factorisations of the period as a product of two

Proposition 48.10 (Twenty eight times six)**.** *Besides the five factorisations of Theorem 53.2, the period factors as*

$$168 = 28 \cdot 6, \tag{266}$$

where twenty eight is the order of the golden and silver elements, Theorem 48.2, and is also the period of the Fibonacci sequence modulo thirteen, computed to be twenty eight. The complementary factor six is the order of the coordinate group and of the Weyl group of type A_2.

Proof. The product is arithmetic. The order twenty eight is Theorem 48.2, and the Fibonacci period was obtained by iterating the recurrence until the pair $(0, 1)$ recurred, at step twenty eight. The order six is $|S_3|$. □

Remark 48.11 (The two twenty eights). That the order of the ratios and the Fibonacci period agree is an equality of counts: the first is the order of an element of $\mathbb{F}_{169}^{*}$, the second the period of a sequence over $\mathbb{F}_{13}$, and no map between them is exhibited here. The Tribonacci period, by contrast, is one hundred sixty eight and is identified with the order of the operator by Proposition 49.1, since the same matrix generates both.

Remark 48.12 (The rank two part carries a block, and the block has two invariants). The rank two part of the spectral decomposition is not a scalar multiple of its projector: the identity is

$$T_3 = 7P_1 + T_3P_{23}, \tag{267}$$

and no residue c satisfies $T_3 = 7P_1 + cP_{23}$, since the restriction of the operator to the rank two image has the matrix (66), of trace seven and determinant two, which is not scalar. A single eigenvalue there exists only after adjoining a radix of two, and then there are two of them, exchanged by the Frobenius.

48.6 The three numbers two, three, and seven

Proposition 48.13 (Read as degrees and as an eigenvalue)**.** *The three numbers occurring in the period admit the following reading, in which each is a quantity already defined:*

$$\begin{aligned} 2 &= \big[\mathbb{F}_{169} : \mathbb{F}_{13}\big], && \text{the degree of the quadratic step,} \\ 3 &= \deg \chi, && \text{the degree of the defining cubic,} \\ 7 &= \lambda_1, && \text{the eigenvalue in the prime field,} \end{aligned} \tag{268}$$

and their product is the squarefree part of the period, with

$$2 \cdot 3 \cdot 7 = 42, \qquad 4 \cdot 42 = 168. \tag{269}$$

Proof. The first two are the degrees stated in Definition 9.1 and Proposition 4.2; the third is Theorem 9.2. The two products are arithmetic. □

Remark 48.14 (The reading is a reading). Proposition 48.13 records that three quantities of the construction happen to be the integers two, three, and seven. It does not assert that the triple carries any further structure, and in particular it makes no use of the triangle group with those exponents, whose defining data are angles and therefore outside what this construction admits, as recorded in Remark 50.4. What is proved about that group elsewhere in this paper is only the order of its smallest nontrivial quotient, one hundred sixty eight, and the rational defect $1/42$.

48.7 The group with exponents two, three, and seven, defined by a presentation

Definition 48.15 (The von Dyck group, given algebraically)**.** Let $\Delta(2,3,7)$ be the group with two generators and three relations,

$$\Delta(2,3,7) = \langle\, x,\ y \ \mid\ x^2 = 1,\ y^3 = 1,\ (xy)^7 = 1 \,\rangle. \tag{270}$$

This is a definition by generators and relations only. It names no angle, no side, and no ambient space, and it is therefore admissible here, unlike the description of the same group as the symmetries of a triangle, which requires angles and a geometry that this construction does not admit.

Theorem 48.16 (Its smallest nontrivial quotient has the order of the generator)**.** *The group* (270) *admits a surjection onto* $\mathrm{PSL}(2,\mathbb{F}_7)$*, realised by the two classes*

$$x = \begin{bmatrix} 0 & 1 \\ 6 & 0 \end{bmatrix}, \qquad y = \begin{bmatrix} 0 & 1 \\ 6 & 1 \end{bmatrix}, \qquad \mathrm{ord}(x) = 2, \quad \mathrm{ord}(y) = 3, \quad \mathrm{ord}(xy) = 7, \tag{271}$$

which together generate the whole group, and

$$\big|\mathrm{PSL}(2,\mathbb{F}_7)\big| = \frac{|\,\mathrm{SL}(2,\mathbb{F}_7)|}{2} = \frac{336}{2} = 168 = \mathrm{ord}(T_3). \tag{272}$$

Proof. The two matrices were exhibited by searching $\mathrm{SL}(2,\mathbb{F}_7)$, and their orders in the quotient by the two scalars were computed to be two, three, and seven respectively, so the assignment respects the three relations of (270) and extends to a homomorphism. Closing the set of products of the two classes returns one hundred sixty eight elements, so the homomorphism is onto. The order of the quotient is the order of $\mathrm{SL}(2,\mathbb{F}_7)$, which is $(7^2-1)\cdot 7 = 336$, divided by the two scalars. □

Remark 48.17 (The status of the agreement in orders). Theorem 48.16 identifies a group of order one hundred sixty eight, and Theorem 53.1 records that the generator of this paper also has order one hundred sixty eight. The two groups are not isomorphic: $\mathrm{PSL}(2,\mathbb{F}_7)$ is simple and nonabelian, while the orbit of the generator is cyclic, as recorded in Remark 57.4. The agreement is of orders, and belongs with the equalities of counts. What the presentation adds is that the three exponents entering the period,

$$2\cdot 3\cdot 7 = 42, \qquad 4\cdot 42 = 168, \tag{273}$$

occur here as the orders of three specific group elements, which is an algebraic statement and needs no geometry to be made.

48.8 Three statements that do not hold

Remark 48.18 (Three statements in their correct form). The following three assertions suggest themselves and are false; each is recorded with the statement that replaces it.

(i) *The half period is minus the identity.* It is not: computing gives

$$T_3^{84} = \begin{bmatrix} 1 & 1 & 7 \\ 6 & 2 & 8 \\ 1 & 7 & 9 \end{bmatrix} \neq -I, \qquad \text{while} \qquad T_3^{84} = 2P_1 - I = P_1 - P_{23} \tag{274}$$

does hold, by Theorem 20.4. The half period acts as the identity on the rank one channel and as minus the identity on the rank two channel, not as minus the identity on all of the state space.

(ii) *The rank two image is orthogonal to the eigenvector.* It is not, for the standard bilinear form: $v_0 \cdot v_1 = 9$ and $v_0 \cdot v_2 = 10$, both nonzero. The correct statement is that the rank two image is the kernel of the left eigenvector,

$$\operatorname{im} P_{23} = \{\, v : u \cdot v = 0 \,\}, \qquad u = (1, 7, 10), \tag{275}$$

which follows from $P_1 v = (u \cdot v) v_0$ of Theorem 11.2. No bilinear form is needed, and none is used.

(iii) *The conjugate eigenvalues have no well defined order.* They do: both λ_2 and λ_3 have order one hundred sixty eight in $\mathbb{F}_{169}^*$ and are generators of that group, by Theorem 58.40, while $\lambda_1 = 7$ has order twelve.

Proposition 48.19 (The two ways of writing the rank one projector agree)**.** *Since* $-7 \equiv 6$ *modulo thirteen,*

$$7\big(T_3^2 - 7T_3 + 2I\big) = 7\big(T_3^2 + 6T_3 + 2I\big) = I + 3T_3 + 7T_3^2 = P_1, \tag{276}$$

so the interpolation form and the coefficient form are the same matrix.

Proof. The two middle expressions agree because their coefficients agree modulo thirteen, and expanding either and reducing gives the coefficient triple $(1, 3, 7)$, which is (60). □

Proposition 48.20 (The period as the order of a second group)**.** *The period is also the order of the full linear group in dimension three over the field of two elements,*

$$\big|\mathrm{GL}(3, \mathbb{F}_2)\big| = (2^3 - 1)(2^3 - 2)(2^3 - 4) = 7 \cdot 6 \cdot 4 = 168, \tag{277}$$

which is a fifth occurrence of the number, alongside the four of Theorem 53.1. That group is isomorphic to $\mathrm{PSL}(2, \mathbb{F}_7)$ *of Theorem 48.16, and as recorded there it is simple and nonabelian, so the agreement with the cyclic orbit is one of orders only.*

Proof. The order is the standard product of the numbers of ways to choose successive independent vectors over the field of two elements, evaluated above. □

Structure 49. The Tribonacci sequence and the interaction kernel

49.1 The sequence and its period

Proposition 49.1 (Period of the sequence)**.** *Let* Trib *be the sequence with seeds*

$$\mathrm{Trib}(0) = 0, \qquad \mathrm{Trib}(1) = 0, \qquad \mathrm{Trib}(2) = 1, \tag{278}$$

and the recurrence (25)*. Its first values modulo thirteen are*

$$0,\ 0,\ 1,\ 1,\ 2,\ 4,\ 7,\ 0,\ 11,\ 5,\ 3,\ \ldots \tag{279}$$

and its period modulo thirteen is exactly

$$168 = \operatorname{ord}(T_3), \tag{280}$$

the same as the order of the generator, because the sequence is generated by the companion matrix of χ.

Proof. The recurrence is generated by T_3 acting on the column of three consecutive terms, so the sequence returns to its seeds exactly when the matrix returns to the identity, which is at one hundred sixty eight by Theorem 20.1; the period was confirmed by direct iteration. □

49.2 The kernel

Definition 49.2 (Interaction kernel)**.** For indices $i \neq j$ the interaction kernel is

$$G(i,j) = \operatorname{Trib}\big(|i-j|-1\big) \cdot \eta^{-v_{13}(i-j)} \ \in \ \mathbb{F}_{169}, \tag{281}$$

where η is an element of $\mathbb{F}_{169}^{*}$ of order one hundred sixty eight, for instance $\eta = 1+2\theta$, and v_{13} is the valuation of Structure 32. The first factor is an orbital term depending on the separation of the indices and the second is the geometric weight of Proposition 32.5, depending on the depth at which the two indices agree.

49.3 Symmetry, periodicity, and vanishing

Proposition 49.3 (Properties of the kernel)**.** *The kernel is symmetric,*

$$G(i,j) = G(j,i), \tag{282}$$

since both $|i-j|$ and $v_{13}(i-j)$ are unchanged by exchanging the indices; it is periodic of period one hundred sixty eight in the separation, by Proposition 49.1 and Proposition 32.5; and it vanishes exactly when thirteen divides the orbital term,

$$G(i,j) = 0 \iff \operatorname{Trib}\big(|i-j|-1\big) \equiv 0 \pmod{13}, \tag{283}$$

which occurs in particular for adjacent and next but one indices, where the orbital term is $\operatorname{Trib}(0) = \operatorname{Trib}(1) = 0$. The weight factor never vanishes, being a unit.

Proof. Both $|i-j|$ and $v_{13}(i-j)$ are unchanged when the indices are exchanged, which gives the symmetry. Periodicity follows from Proposition 49.1 for the first factor and Proposition 32.5 for the second. A product vanishes exactly when a factor does, and the weight is a unit, so only the orbital term can vanish. □

Remark 49.4 (The kernel takes values in a field). The value of G is an element of a finite field, not a quantity. Two values may be compared for equality and multiplied, but not ordered, so the kernel selects by exact coincidence and by depth and never by degree of closeness.

49.4 Depth enters only through the weight

Proposition 49.5 (Separation of the two factors)**.** *Writing $i-j = 13^{k}u$ with u prime to thirteen, the weight factor is η^{-k} and depends only on the depth k, while the orbital factor depends only on the separation $|i-j|$. The two factors are therefore independent, and the kernel is constant on the set of index pairs sharing both a separation and a depth. On the range of separations up to two hundred the kernel takes twenty one distinct values.*

Proof. Writing $i - j = 13^k u$ with u prime to thirteen, the valuation is k by definition, so the weight is η^{-k} and depends only on k, while the first factor depends only on the separation. The count of distinct values on the stated range was obtained by evaluation. □

Role in the architecture. The interaction of two positions is one ring product: an orbital residue read from the table of Structure 6 multiplied by a weight read from the depth. No comparison of every position with every other is performed, since positions interact only when they lie in a common class, and the whole kernel is exact.

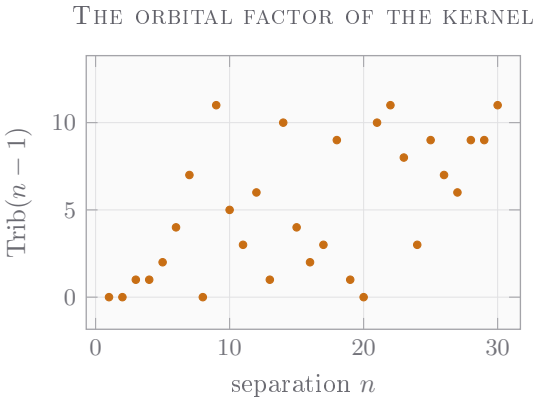


Figure 32: The orbital factor of the kernel as a function of the separation of the two indices; it vanishes wherever the prime divides the Tribonacci value, in particular at the first two separations.

Structure 50. The eigenvector on the conic

The coefficient triple of the rank one projector, the eigenvector of the generator, and a point of the Pythagorean conic are one and the same triple. This structure records that and the further identities around it.

50.1 One triple, three descriptions

Theorem 50.1 (The eigenvector is a point of the conic)**.** *The rank one projector has coefficient triple* $(1, 3, 7)$*, since* $P_1 = I + 3T_3 + 7T_3^2$ *by* (60)*; the eigenvector of the generator for the eigenvalue seven is* $v_0 = (1, 3, 7)$ *by Theorem 12.1; and this same triple satisfies the Pythagorean relation over the prime field,*

$$1^2 + 3^2 = 10 \equiv 49 = 7^2 \pmod{13}. \tag{284}$$

Hence the eigenvector is one of the one hundred sixty eight nonzero points of the conic of Theorem 53.1, and it lies in the stratum with $b \neq 0$*.*

Proof. The two descriptions of the triple are (60) and (77), and the relation was evaluated over $\mathbb{F}_{13}$: $1 + 9 = 10$ and $49 = 3 \cdot 13 + 10$. Since $b = 3 \neq 0$ the point lies outside the stratum $b = 0$ of Theorem 53.1 where the twenty four solutions sit. □

Remark 50.2 (Why the coincidence is structural). The triple is not chosen: the coefficients of P_1 are forced by the factorisation of χ, and the eigenvector is forced by the eigenvalue. That the same triple satisfies the conic relation is a further identity of that one triple and is verified above; it identifies a definite point of the conic, so it belongs among the identifications of Remark 61.1 rather than among the equalities of counts.

50.2 The prime as a sum of two squares

Proposition 50.3 (Two squares)**.** *The working prime is a sum of two squares,*

$$2^2 + 3^2 = 4 + 9 = 13, \tag{285}$$

which is possible because $13 \equiv 1$ *modulo four. Reduced modulo the prime this reads* $2^2 + 3^2 \equiv 0$*, so the triple* $(2, 3, 0)$ *lies on the conic with third coordinate zero, and so does* $(3, 2, 0)$*.*

Proof. The identity is arithmetic, and a prime is a sum of two squares exactly when it is congruent to one modulo four, which thirteen is. Reducing modulo the prime gives $2^2 + 3^2 \equiv 0$, so the two displayed triples satisfy the conic relation with third coordinate zero. □

Remark 50.4 (An identity of integers, and its reduction). A reading of (285) in which the three numbers are sides subtending angles is not available here: the construction admits no angle, as recorded in Definition 1.1, and the field admits no order by Proposition 1.3. What (285) asserts is an identity of integers, and its reduction gives a point of the conic. Separately, $\det\big(T_3|_{\operatorname{im} P_{23}}\big) = 2 = \theta^2$ by (66), which is an identity of residues and not a statement about any angle.

50.3 Triples over an alphabet of four letters

Theorem 50.5 (Nine of sixty four)**.** *Among the sixty four triples* (a, b, c) *with entries in* $\{0, 1, 2, 3\}$*, exactly nine satisfy the conic relation modulo thirteen, namely*

$$\begin{aligned} &(0,0,0),\ (0,1,1),\ (0,2,2),\ (0,3,3),\ (1,0,1),\\ &(2,0,2),\ (3,0,3),\ (2,3,0),\ (3,2,0), \end{aligned} \tag{286}$$

of which seven have a zero in the first or second place and satisfy the relation trivially as $b^2 = c^2$ *or* $a^2 = c^2$*, while the last two are the reductions of* (285) *and are the only ones in which all three entries are nonzero except for the third.*

Proof. All sixty four triples were tested and exactly the nine listed satisfy $a^2 + b^2 \equiv c^2$. For the seven with a zero entry the relation reduces to an equality of two squares; the remaining two are $(2, 3, 0)$ and $(3, 2, 0)$, where $4 + 9 = 13 \equiv 0$. □

50.4 Two triples with a distinguished position

Proposition 50.6 (A vanishing first coordinate and a vanishing third)**.** *The triple* $(0, 3, 2)$ *has vanishing first coordinate, so by Theorem 36.6 its boundary class is the exceptional one,*

$$\operatorname{bd}(0, 3, 2) = [0 : 1], \tag{287}$$

the class of one hundred sixty nine states rather than one hundred fifty six. It does not lie on the conic, since $0 + 9 - 4 = 5 \neq 0$*. The triple* $(3, 2, 0)$ *has vanishing third coordinate and does lie on the conic, since* $9 + 4 = 13 \equiv 0$*; it is one of the nine of Theorem 50.5.*

Proof. The first triple has vanishing first coordinate, so the boundary map sends it to the exceptional class by Theorem 36.6, and $0 + 9 - 4 = 5$ is nonzero, so it is not on the conic. The second has $9 + 4 = 13 \equiv 0$, so it is on the conic, and it appears in the list of Theorem 50.5. □

Remark 50.7 (Names attached to these triples). Attaching external names to $(0, 3, 2)$ and $(3, 2, 0)$, whether from biology or from any other subject, is an assignment in the sense of Remark 55.4. What is proved is the arithmetic just stated: one has a vanishing first coordinate and therefore the exceptional boundary class, and the other has a vanishing third coordinate and lies on the conic.

50.5 The eigenvectors of the rank two channel

Theorem 50.8 (The basis is not a basis of eigenvectors)**.** *The vectors* $v_1 = (6, 1, 0)$ *and* $v_2 = (3, 0, 1)$ *span the rank two image, but neither is an eigenvector of the generator: for every residue* c *one has* $T_3 v_i \neq c\, v_i$*, checked for all thirteen values and both vectors. The eigenvectors appear only over the extension: with* $\lambda_2 = 10 + 7\theta$*,*

$$w_2 = (9 + 7\theta)\, v_1 + v_2 = \big(5 + 3\theta,\ 9 + 7\theta,\ 1\big), \qquad T_3 w_2 = \lambda_2\, w_2, \tag{288}$$

and its conjugate $w_3 = \sigma(w_2)$ *satisfies* $T_3 w_3 = \lambda_3 w_3$*. Both were verified in* $\mathbb{F}_{169}$*.*

Proof. The two vectors lie in the image of the rank two projector by Proposition 10.3, and the eigenvector test was carried out over all residues. Forming the displayed combination in $\mathbb{F}_{169}$ with $\theta^2 = 2$ and applying the generator returns λ_2 times the vector; applying the Frobenius to both sides gives the statement for the conjugate. □

Remark 50.9 (The rank two channel carries a pair, available only over the extension). Theorem 50.8 is the concrete form of Remark 9.7: over the prime field the rank two channel carries the pair of conjugate eigenvalues together, so only their sum seven and product two are available, which are the trace and determinant of (66). The individual eigenvectors exist only after adjoining a radix of two.

Role in the architecture. The eigenvector of the address channel is a point of the conic, the working prime is a sum of two squares whose reduction gives two further points, and the content channel acquires eigenvectors only over the quadratic extension. All three facts are identities of the same small set of triples, and each was verified by exact computation.

Structure 51. Three Pythagorean triples and what they carry

Three triples occur in the construction, one over the integers, one over the prime field, and one over the quadratic extension. This structure states all three, verifies each, and records what each one carries.

51.1 The three triples

Theorem 51.1 (One relation at three levels)**.** *The Pythagorean relation* $a^2 + b^2 = c^2$ *holds in the three following instances, each at a different level of the construction:*

$$2^2 + 3^2 = 13 \qquad \text{over } \mathbb{Z}, \tag{289}$$

$$1^2 + 3^2 \equiv 7^2 \pmod{13} \qquad \text{over } \mathbb{F}_{13}, \tag{290}$$

$$1^2 + 1^2 = 2 = \theta^2 \qquad \text{over } \mathbb{F}_{169}. \tag{291}$$

The first says that the working prime is a sum of two squares, which is possible because $13 \equiv 1$ *modulo four. The second is the eigenvector of Theorem 50.1, whose entries are the coefficients of the rank one projector. The third says that a radix of two is the third side of a triple with two unit sides, and* 2 *is the determinant of the generator restricted to the rank two channel by* (66)*.*

Proof. The first is $4 + 9 = 13$. The second was verified in Theorem 50.1, where $1 + 9 = 10$ and $49 \equiv 10$. The third is the defining relation $\theta^2 = 2$ of Definition 9.1 read as a Pythagorean identity with $a = b = 1$, and the value two is the determinant computed in Proposition 10.3. □

51.2 What each triple carries

triple	level	carries	reference
$(2,3,13)$	$\mathbb{Z}$	the working prime as a sum of two squares	Proposition 50.3
$(1,3,7)$	$\mathbb{F}_{13}$	the eigenvector v_0 and the coefficients of P_1	Theorem 50.1
$(1,1,\theta)$	$\mathbb{F}_{169}$	the radix of two, and det on the rank two channel	(66)

51.3 The two conjugate eigenvalues are a Galois pair

Proposition 51.2 (Frobenius exchanges them)**.** *The two eigenvalues of the rank two channel are exchanged by the Frobenius and the first is fixed,*

$$\sigma(\lambda_2) = \lambda_2^{13} = \lambda_3, \qquad \sigma(\lambda_3) = \lambda_2, \qquad \sigma(\lambda_1) = \lambda_1 = 7, \tag{292}$$

so λ_2 and λ_3 form one orbit of the Galois group of $\mathbb{F}_{169}$ over $\mathbb{F}_{13}$ and λ_1 forms another. This is why only their sum and product, namely seven and two, are defined over the prime field, and it is the reason the rank two channel is indecomposable there.

Proof. The Frobenius fixes the residue field pointwise and sends the radix of two to its negative, so it fixes λ_1 and exchanges λ_2 with λ_3; the three powers were computed. Only symmetric functions of an exchanged pair are fixed, and for a pair those are the sum and the product. □

51.4 The three exponents

Proposition 51.3 (The defect of the triangle group)**.** *The three exponents two, three, and seven satisfy*

$$\tfrac{1}{2} + \tfrac{1}{3} + \tfrac{1}{7} = \tfrac{41}{42}, \qquad 1 - \tfrac{41}{42} = \tfrac{1}{42}, \tag{293}$$

a comparison and a difference of rational numbers naming no angle. Their product is $2 \cdot 3 \cdot 7 = 42$, the squarefree part of the period, and

$$168 = 4 \cdot 42 = 2^3 \cdot 3 \cdot 7. \tag{294}$$

Proof. The sum of the three fractions is $\frac{21+14+6}{42} = \frac{41}{42}$, and subtracting from one gives $\frac{1}{42}$. The product of the three exponents is forty two, and multiplying by four gives the period. □

51.5 Three finite sets carried by the same arithmetic

Proposition 51.4 (Alphabet, triples of four letters, exponents)**.** *The following three counts are those of finite sets produced by the construction, and each is recorded with what is proved about it:*

count	*factorisation*	*proved statement*	*reference*
26	$2 \cdot 13$	$\mathbb{Z}/26 \cong \mathbb{Z}/2 \times \mathbb{Z}/13$, *idempotents* 13 *and* 14	*Proposition 56.1*
64	4^3	*exactly nine of the triples lie on the conic*	*Theorem 50.5*
42	$2 \cdot 3 \cdot 7$	*the squarefree part of the period;* $168 = 4 \cdot 42$	(294)

Proof. Each row is the statement referenced: the first is Proposition 56.1, the second Theorem 50.5, and the third the factorisation of the period. □

Remark 51.5 (What the three counts assert). The three counts above are counts of sets the construction produces, and the arithmetic attached to each is proved. Identifying those sets with an alphabet, with a genetic code, or with any other external system is an assignment in the sense of Remark 55.4, and no such identification is used anywhere in this paper.

Role in the architecture. One relation appears at three levels: over the integers it exhibits the working prime as a sum of two squares, over the prime field it is the eigenvector of the address channel, and over the quadratic extension it is the radix of two that builds the content channel. The two conjugate eigenvalues form one Galois orbit, which is why the content channel has no scalar over the prime field.

Structure 52. Register of the distinguished numbers

Every integer that recurs in this paper is listed here with its origins and the place each is established. A number appears once, with all of its origins together, so that the reader can see which agreements are identifications and which are equalities of counts.

52.1 The register

number	origins	established in
0	the zero idempotent; the first coordinate of the zero state	Theorem 12.27
1	the identity; the multiplicative representative of one; the rank of P_1	Proposition 34.3
2	the degree of the residue extension; the determinant of the operator on the rank two channel; the exponent of the involution in the presentation	Definition 9.1, (66)
3	the degree of χ; the dimension of the state space; the order of the scalars μ_3; the determinant of the Cartan matrix, hence the number of vertex types	Proposition 4.2, Theorem 20.6, Proposition 58.11
7	the eigenvalue λ_1; the order of $U_7 = T_3^{24}$; the number of quadratic subfields of the degree eight field; the common size of the nontrivial orbits of U_7 on the projective plane	Theorem 9.2, Theorem 29.4, Theorem 12.12
8	the discriminant of $\mathbb{Q}(\sqrt{2})$; the order of $U_2 = T_3^{21}$; the two part of the period	Definition 27.3, Theorem 20.2
9	the inverse of three and the square of three in $\mathbb{F}_{13}$; the third scalar anchor $T_3^{112} = 9I$	Proposition 20.5
12	the order of $\mathbb{F}_{13}^*$; the order of the eigenvalue seven; the number of indices fixed by the Frobenius involution	Proposition 12.4, Theorem 24.1
13	the working prime; the branching number of the tree; the number of collisions of the index map	Structure 8, Theorem 41.2
14	$q+1$: the order of the torus; the number of boundary classes; the degree of the tree; the thickness of the residue; the size of the projective orbit; the size of the stabiliser of the eigenline	Theorem 58.34, Theorem 12.23
21	the discriminant of $\mathbb{Q}(\sqrt{21})$; the index of U_2	Theorem 30.3
24	the order of θ in $\mathbb{F}_{169}^*$; the order of the difference operator on its image; the index of U_7; the degree of the top field	Proposition 9.6, Theorem 27.7
28	the order of the golden and silver elements	Theorem 48.2
42	$2{\cdot}3{\cdot}7$: the squarefree part of the period; the denominator of the defect	Proposition 51.3
44	the absolute discriminant of χ and of the cubic field	Proposition 31.4
56	the index of μ_3; the size of each generic projective orbit	Theorem 40.1
84	the half period; the involution $T_3^{84} = P_1 - P_{23}$	Theorem 20.4
105, 112, 120	the three idempotents of the remainder decomposition	Theorem 44.1
137	the integer norm $13^2 - 2 \cdot 4^2$, reducing to the eigenvalue	Proposition 53.6
168	the order of the operator; the discriminant of $\mathbb{Q}(\sqrt{42})$; the order of $\mathrm{PSL}(2,\mathbb{F}_7)$; the order of $\mathrm{GL}(3,\mathbb{F}_2)$; the number of nonzero points of the conic	Theorem 53.1, Proposition 48.20
169	13^2: the size of the residue extension; the size of the rank two image	Definition 9.1

number	origins	established in
183	q^2+q+1: the points of the projective plane, and its lines	Proposition 58.24
2197	13^3: the states; the order of the algebra; the apartments through a chamber; the classes at depth three	Theorem 58.38
2562	$14 \cdot 183$: the chambers of the spherical residue	Proposition 58.7

52.2 Two entries that do not belong

Remark 52.1 (Numbers sometimes proposed for this register). Two numbers are sometimes offered for the list and are not in it.

(i) Sixty three and sixty four. The multiplicative group of the field of sixty four elements is cyclic of order sixty three and does contain a subgroup of order seven, but that field has characteristic two, whereas every field of this construction has characteristic thirteen; moreover $64 \equiv 12$ and $63 \equiv 11$ modulo thirteen, and neither divides the period. They are recorded in Remark 12.14 as absent, not as external context.

(ii) The thickness thirteen. The thickness of the residue is $q+1 = 14$ and not $q = 13$, by Theorem 58.19; thirteen is the branching number of the tree, that is the number of children of a vertex, while the degree, which counts the parent as well, is fourteen.

Remark 52.2 (The order of the radix of two). In the row for two above, the number two is the degree of the extension and the determinant on the rank two channel. It is not the order of θ, which is twenty four by Proposition 9.6; that value appears in the row for twenty four, alongside the order of the difference operator on its image, with which it coincides.

Role in the architecture. Every number the engine uses is one of the entries above, and each is either the order of a group element, the size of a set of states, the count of a set of simplices, or a discriminant. No entry is a magnitude, and none arises from a measurement.

Structure 53. The counting identities

53.1 One number, four sources

Theorem 53.1 (Four independent occurrences of one number)**.** *The following quantities, of independent definition, are equal:*

$$\operatorname{ord}(T_3) \;=\; \operatorname{disc}\mathbb{Q}(\sqrt{42}) \;=\; \big|\mathrm{PSL}(2,7)\big| \;=\; \#\{(a,b,c)\neq 0 : a^2+b^2\equiv c^2\} \;=\; 168. \tag{295}$$

The conic count splits along the three roots as

$$24 + 72 + 72 = 168, \tag{296}$$

with $b = 0$ giving twenty four solutions, and b a nonzero square or nonsquare giving seventy two each.

Proof. The order is Theorem 20.1; 42 is squarefree and congruent to two modulo four, so the discriminant is $4 \cdot 42$; the order of $\mathrm{PSL}(2,7)$ is $\frac{1}{2} \cdot 7 \cdot 48$. The form $a^2+b^2-c^2$ is nondegenerate over $\mathbb{F}_{13}$, its projective conic has fourteen points, each lifting to twelve nonzero triples. If $b = 0$ then $a = \pm c$ with $c \neq 0$, and the remaining one hundred forty four split evenly by the quadratic character of b. □

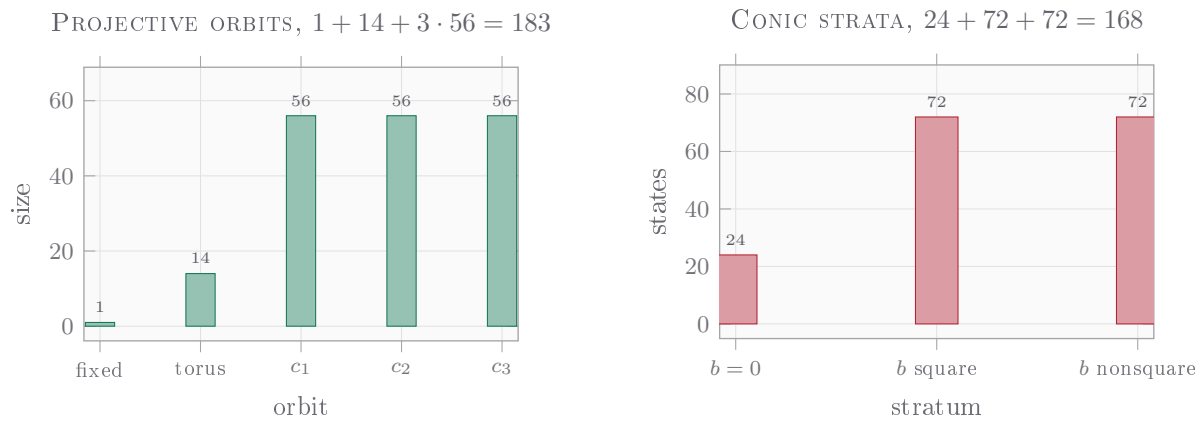


Figure 33: Left: the five orbits of the generator on the projective plane, Theorem 40.1. Right: the three strata of the conic count, Theorem 53.1, indexed by the three roots of the defining cubic.

53.2 Five factorisations

Theorem 53.2 (Five factorisations, each a subgroup)**.**

$$168 = 14 \cdot 12 = 3 \cdot 56 = 24 \cdot 7 = 84 \cdot 2 = 2^3 \cdot 3 \cdot 7, \tag{297}$$

and each factor is the order or the index of a subgroup already present:

factorisation	*subgroup*	*order*	*index*
$14 \cdot 12$	$T = \ker \mathrm{N}$	14	12
$3 \cdot 56$	$\mu_3 = \langle T_3^{56} \rangle$	3	56
$24 \cdot 7$	$\langle T_3^{24} \rangle$	7	24
$84 \cdot 2$	$\langle T_3^{84} \rangle$	2	84
$2^3 \cdot 3 \cdot 7$	*the three prime power parts*	$8, 3, 7$	$21, 56, 24$

Proof. Each row pairs a subgroup with its index, and the product of the order with the index is the group order by Lagrange's theorem. The orders are those of Theorem 48.1, Theorem 20.6, and Theorem 43.2, and the last row is the prime factorisation with the three Sylow subgroups. □

Corollary 53.3 (The projective count assembled)**.**

$$183 = \underbrace{1}_{\text{fixed line}} + \underbrace{14}_{\text{orbit of } T} + \underbrace{3 \cdot 56}_{\text{cosets of } \mu_3} = 1 + 14 + 168. \tag{298}$$

Proof. The three summands are the three orbit sizes of Theorem 40.1, and their total is the point count of the projective plane. □

53.3 The state count in two shapes

Theorem 53.4 (The state count in two shapes)**.**

$$2197 = 13 \cdot 169 \qquad \textit{and} \qquad 2197 = 13 \cdot 156 + 169, \tag{299}$$

the first the channel splitting of Theorem 10.1, preserved by the dynamics, the second the boundary partition of Theorem 36.6, not preserved. The two are not refinements of one another, and the equality is of totals only.

Proof. The first identity is Theorem 10.1, the two images meeting only at zero and spanning the space; the second is Theorem 36.6. Both totals are 13^3. □

53.4 Which counts are divisible by the working prime

Proposition 53.5 (Divisibility by thirteen)**.** *Among the principal counts of the construction,*

count	*value*	*divisible by thirteen*
period of the generator	168	*no*
points of the projective plane	183	*no*
chambers of the spherical residue	2562	*no*
states of the space	2197	*yes*
apartments of the residue	938119	*yes*
order of $\mathrm{SL}(3,\mathbb{F}_{13})$	810534816	*yes*

The three that are divisible are those built from q^3, and the three that are not are built from $q\pm1$ and from q^2+q+1, none of which is a multiple of the prime.

Proof. Each value was reduced modulo thirteen. Structurally, $2197 = q^3$, the apartment count has the factor q^2 of (333), and the group order has the factor q^3 of (321); while $168 = q^2-1$, $183 = q^2+q+1$, and $2562 = (q+1)(q^2+q+1)$ are each congruent to a nonzero residue. □

Role in the architecture. The operator can be reached from either end: from the cubic, as the matrix that satisfies it, or from the substitution, as the matrix that counts letters. The word produced by the substitution has exactly $2n+1$ factors of each size and one special factor on each side, so its combinatorial content is as tightly constrained as the algebra, and the two are related by one integral change of basis.

53.5 An integer norm reducing to the eigenvalue

Proposition 53.6 (An integer norm reducing to the eigenvalue)**.** *In $\mathbb{Z}[\theta]$ with $\theta^2 = 2$,*

$$\mathrm{N}(13+4\theta) = 13^2 - 2\cdot 4^2 = 169 - 32 = 137, \qquad 137 \equiv 7 \equiv \lambda_1 \pmod{13}, \tag{300}$$

and the same residue arises intrinsically, since $\mathrm{N}(4\theta) \equiv 7$. The class of seven is shared by all thirteen integers congruent to seven, so no single such integer is distinguished by the algebra.

Proof. Both evaluations are direct: $169 - 32 = 137$ and $-2\cdot 16 = -32 \equiv 7$. The reduction of 137 is seven, and reduction modulo thirteen is constant on each residue class, so no integer of that class is distinguished. □

Structure 54. The number fourteen

54.1 Six occurrences

Theorem 54.1 (One number, six roles)**.** *The number $14 = q+1$ occurs in the construction in six ways, each verified:*

(i) the degree of the tree of Structure 37;

(ii) the number of points of the projective line over the prime field;

(iii) the order of the torus $\ker \mathrm{N}$*, Theorem 48.1;*

(iv) the size of the projective orbit inside the rank two channel, Theorem 40.1;

(v) the order of the power T_3^{12}, since $\mathrm{ord}(T_3^{12}) = 168/\gcd(168,12) = 14$*;*

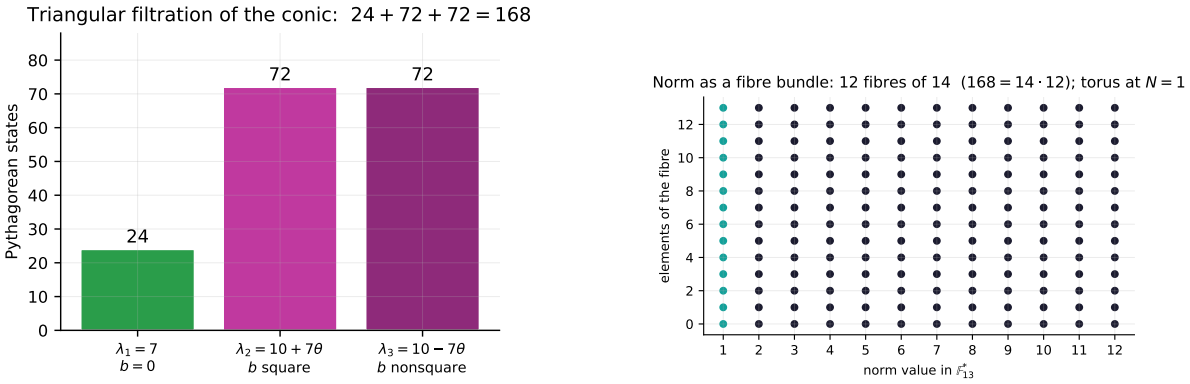


Figure 34: Left: the three strata of the conic count, twenty four with $b = 0$ and seventy two each for b a square or a nonsquare. Right: the norm as a bundle over $\mathbb{F}_{13}^{*}$, twelve fibres of fourteen, that is $168 = 14 \cdot 12$.

(vi) the number of chambers of the spherical residue containing a fixed panel, which is $q + 1$.

Items (ii), (iii), and (iv) are identified with one another by Corollary 48.4, and (i) and (vi) are the two panel counts of the same residue field. Item (v) is the divisor law of Theorem 20.1.

Proof. Each item is the statement referenced, and (v) was computed directly. □

Role in the architecture. A position, a boundary point, a tree direction, and a panel are counted by the same number, and in three of the six cases the sets are identified by an explicit map rather than by a count. The engine therefore carries one counter modulo fourteen wherever any of these appear.

Structure 55. The four distinguished ring elements

55.1 The elements and their identities

Proposition 55.1 (Three basis elements and their sum)**.** *In the ring of Definition 14.1 the three basis elements and their sum,*

$$1 = (1,0,0), \qquad \eta = (0,1,0), \qquad \eta^2 = (0,0,1), \qquad 1 + \eta + \eta^2 = (1,1,1), \tag{301}$$

satisfy the two identities

$$1 + \eta + \eta^2 = (1,1,1), \qquad \eta \cdot \eta^2 = \eta^3 = 1 + \eta + \eta^2, \tag{302}$$

the second being the defining relation (21) *itself, so the product of the two nonunit generators equals their sum with the unit. Moreover the first four images of the first basis vector under the generator are exactly these four elements,*

$$e_1,\ T_3 e_1,\ T_3^2 e_1,\ T_3^3 e_1 \ = \ (1,0,0),\ (0,1,0),\ (0,0,1),\ (1,1,1), \tag{303}$$

after which the orbit continues with $(1,2,2)$ *and leaves the set.*

Proof. The sum is componentwise. The product is $\eta \cdot \eta^2 = \eta^3$, which the defining relation rewrites as $\eta^2 + \eta + 1$. The images of e_1 are the successive columns produced by the companion form, and the fourth is obtained from (21); all four were computed over $\mathbb{F}_{13}$. □

55.2 The complement structure

Proposition 55.2 (A Klein four group on the four elements)**.** *Labelling the four elements of* (301) *by the pairs*

$$(0,0), \qquad (0,1), \qquad (1,0), \qquad (1,1) \tag{304}$$

in $(\mathbb{Z}/2)^2$*, the operation of adding* $(1,1)$ *is an involution without fixed points that exchanges the four labels in two pairs. The group* $(\mathbb{Z}/2)^2$ *has order four and is not cyclic, and it appears in the chain of subgroups*

$$S_6 \supset S_4 \supset V_4, \qquad |S_6| = 720 = 2^4 \cdot 3^2 \cdot 5, \quad |S_4| = 24, \quad |V_4| = 4. \tag{305}$$

Proof. Adding the pair with both entries one exchanges each label with another and fixes none, since the group has exponent two and no element equals its own translate by a nonzero element. The chain of subgroups and their orders is standard for the symmetric group on six letters. □

55.3 Triples in the radix four, reduced

Proposition 55.3 (Distribution of the sixty four triples)**.** *Writing a triple of the four labels as* $b_1 + 4b_2 + 16b_3$ *and reducing modulo thirteen distributes the sixty four triples as*

$$64 = 5 \cdot 12 + 4, \tag{306}$$

that is twelve residues receive five triples each and one residue receives four. The distribution was computed exhaustively.

Proof. All sixty four triples were reduced and tallied, giving twelve residues with five preimages and one with four, which is (306). □

55.4 What is derived and what is assigned

Remark 55.4 (The identities are algebra, the names are not). The identities (302), (303), and (306) are consequences of the defining relation and were verified exactly. Attaching external names to the four elements, whether of letters, symbols, or biochemical bases, is an *assignment* made by the user of the architecture and is not a consequence of the algebra: the ring supplies four distinguished elements and one involution on them, and nothing in the construction selects what those four elements should denote. Statements of the form "a set of twenty objects reduces to seven modulo thirteen" are arithmetic remarks about the number twenty and carry no structural content, since every integer congruent to twenty modulo thirteen shares that property. In this paper such readings are recorded as encodings, never as results.

Role in the architecture. The ring offers four distinguished elements, a product rule that identifies the product of two of them with the sum of all, and an involution of order two on their labels. Any four symbol alphabet can therefore be carried natively in three residues, and the choice of alphabet is external to the construction.

Structure 56. Encodings by the remainder theorem

56.1 A set of twenty six labels

Proposition 56.1 (Splitting a set of twenty six)**.** *Since* $\gcd(2, 13) = 1$*, the remainder theorem gives*

$$\mathbb{Z}/26 \;\cong\; \mathbb{Z}/2 \times \mathbb{Z}/13, \qquad n \longmapsto (n \bmod 2,\ n \bmod 13), \tag{307}$$

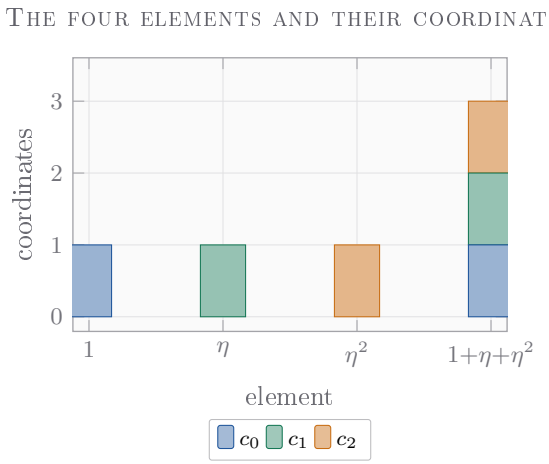


Figure 35: The three basis elements and their sum, whose coordinates are the first four images of the first basis vector under the operator, by Proposition 55.1.

a bijection verified on all twenty six indices, with idempotents

$$e_2 = 13, \qquad e_{13} = 14, \qquad e_2 + e_{13} \equiv 1 \pmod{26}, \tag{308}$$

since $13 \equiv 1$ *modulo two and* 0 *modulo thirteen, while* $14 \equiv 0$ *modulo two and* 1 *modulo thirteen. A set of twenty six labels is therefore carried as a parity together with a residue modulo the working prime.*

Proof. The two moduli are coprime, so the map is a bijection by the remainder theorem, confirmed by tabulating all indices. Each idempotent is congruent to one modulo its own factor and to zero modulo the other, and their sum is one modulo twenty six. □

Remark 56.2. As in Remark 55.4, the factorisation $26 = 2 \cdot 13$ is arithmetic. Which set of twenty six labels is carried is an assignment, and no property of the alphabet is derived here.

Structure 57. External symmetry context

57.1 The symmetric group on six letters

Proposition 57.1 (Orders in the chain)**.** *The symmetric group on six letters has order* $720 = 2^4 \cdot 3^2 \cdot 5$ *and is the unique symmetric group admitting an outer automorphism. It contains* S_4 *of order* 24 *and the Klein four group of order four, giving the chain* (305)*. The Galois group of* χ *over the rationals is the symmetric group on three letters, which embeds in* S_6 *as the stabiliser of a partition into two blocks of three.*

Proof. The order is the factorial of six, and its factorisation is displayed. The chain is the stabiliser of a point followed by the Klein subgroup, of orders twenty four and four. The Galois group of χ embeds as stated because it acts faithfully on three roots. □

57.2 The braid group on three strands

Proposition 57.2 (Presentation, centre, and quotient)**.** *The braid group on three strands is*

$$B_3 = \langle \sigma_1, \sigma_2 \mid \sigma_1\sigma_2\sigma_1 = \sigma_2\sigma_1\sigma_2 \rangle, \tag{309}$$

with centre generated by Δ^2 *where* $\Delta = \sigma_1\sigma_2\sigma_1$*, and quotient by the centre isomorphic to the modular group. There is a surjection* $B_3 \to S_3$ *onto the Galois group of* χ*.*

Proof. These are the standard facts about the braid group on three strands: the presentation is by definition, the centre is generated by the square of the half twist, the quotient by the centre is the modular group, and the surjection onto the symmetric group sends each generator to the corresponding transposition. □

57.3 Why the braid relation cannot hold on the orbit

Theorem 57.3 (Failure of the relation)**.** *The generator and its transpose do not satisfy the braid relation,*

$$T_3\,T_3^{\top}\,T_3 \;\neq\; T_3^{\top}\,T_3\,T_3^{\top}, \tag{310}$$

verified over $\mathbb{F}_{13}$*, and* $T_3^{\top}$ *is not a power of* T_3*, verified against all one hundred sixty eight exponents. Moreover no pair of distinct powers of the generator can satisfy the relation, since in a commutative group* $aba = a^2b$ *and* $bab = ab^2$*, so the relation reduces to* $a = b$*. The orbit is cyclic, hence commutative, while* B_3 *is not, so no injective homomorphism from* B_3 *into the orbit exists.*

Proof. Both products were computed over $\mathbb{F}_{13}$ and differ. Comparing the transpose with all powers of the generator shows it is not among them. In a commutative group the two sides of the relation reduce to a^2b and ab^2, which agree only when $a = b$, so the relation carries no information there; since the orbit is cyclic and the braid group is not abelian, no injective homomorphism exists. □

57.4 A numerical coincidence, recorded as such

Remark 57.4 (Order one hundred sixty eight in two unrelated groups). The modular group has $\mathrm{PSL}(2,7)$ of order one hundred sixty eight among its finite quotients, and this is the order of the orbit. The two are not isomorphic: $\mathrm{PSL}(2,7)$ is simple and nonabelian while the orbit is cyclic. The agreement is of numbers only, and belongs to the equalities of counts of Remark 61.2 and not to the identifications of Remark 61.1.

Part X
The model layer, status, and verification

Structure 58. Holographic processing: the two carriers and the pipeline

The two channels of Theorem 10.1 carry two different geometric objects, one of rank one and one of rank two, and their ranks add to the rank of the state space. This structure assembles them, states the ultrametric facts they rest on in one place, and gives the pipeline that moves a state through them.

58.1 The valuation on states

Definition 58.1 (Valuation of an integer and of a state)**.** For a nonzero integer n the valuation is

$$v_{13}(n) = \max\{\,k : 13^k \mid n\,\}, \qquad v_{13}(0) = \infty, \tag{311}$$

and for a state the valuation is the smallest valuation of its entries,

$$v_{13}\big((x,y,z)\big) = \min\big(v_{13}(x), v_{13}(y), v_{13}(z)\big), \tag{312}$$

with the same convention at the zero state. Both were checked against the ideal theoretic definition on sample states.

58.2 The three ultrametric facts used

Proposition 58.2 (Inequality, classes, and the dichotomy at equal depth)**.** *For all x, y, z,*

$$v_{13}(x-z) \;\geq\; \min\big(v_{13}(x-y),\, v_{13}(y-z)\big), \tag{313}$$

the classes

$$B_k(a) = \{\, x : v_{13}(x-a) \geq k \,\}, \qquad B_{k+1}(a) \subseteq B_k(a), \tag{314}$$

are closed and open, and two of them are nested or disjoint by Theorem 33.2. At equal depth the dichotomy is sharper:

$$B_k(a) \cap B_k(b) \neq \emptyset \implies B_k(a) = B_k(b), \tag{315}$$

so classes of one depth are either identical or disjoint, never merely nested. This was verified on a deterministic sweep of four thousand eight hundred pairs.

Proof. The inequality is Theorem 32.1. For (315), if the two classes share a point then $v_{13}(a-b) \geq k$ by (313), and then $x \in B_k(a)$ implies $v_{13}(x-b) \geq \min(v_{13}(x-a), v_{13}(a-b)) \geq k$, so the two sets contain one another. □

58.3 The first carrier: the tree of classes

Theorem 58.3 (The rank one channel carries a tree)**.** *Taking the classes* (314) *as vertices and the inclusions $B_{k+1}(a) \subseteq B_k(a)$ as edges gives a tree in which every vertex has one parent and thirteen children, hence of degree*

$$13 + 1 = 14. \tag{316}$$

Its ends are the infinite descending chains $B_0 \supset B_1 \supset B_2 \supset \cdots$, of which there are fourteen, and the generator fixes exactly one of them, the end attached to the eigenvector $v_0 = (1, 3, 7)$ of Theorem 12.1, by Theorem 38.3. This object is one dimensional in the sense that it is carried by the rank one image $\operatorname{im} P_1$*, on which the generator acts by the single residue seven.*

Proof. A class at depth k contains exactly thirteen classes at depth $k+1$ and lies in exactly one at depth $k-1$, which gives the degree and the tree property. The remaining statements are the references given. □

Remark 58.4 (Self-similar, and stated by counts). Every subtree below a vertex is a copy of the whole by Proposition 39.5, so the object repeats at every depth. As recorded in Remark 39.6, this repetition is stated by the branching number thirteen and the populations $14 \cdot 13^{\,d-1}$, and no exponent expressing a dimension is formed, since such a value would not lie in any ring the construction admits.

58.4 The second carrier: the building of the rank two channel

Theorem 58.5 (The rank two channel carries a building)**.** *The rank two image, which exists because two is a nonresidue and the residue extension $\mathbb{F}_{169} = \mathbb{F}_{13}[\theta]$ is a field by Definition 9.1, carries the affine building of type $\widetilde{A}_2$ of Definition 58.6. A chamber is a chain of lattice classes*

$$L_0 \supset L_1 \supset L_2 \supset 13L_0, \qquad [L_i : L_{i+1}] = 13, \tag{317}$$

so the three successive indices multiply to

$$13 \cdot 13 \cdot 13 = 2197 = [L_0 : 13L_0], \tag{318}$$

which is also the number of states by (111)*. The spherical residue at a vertex has one hundred eighty three points, one hundred eighty three lines, and two thousand five hundred sixty two*

chambers by Proposition 58.7, and is thick of thickness fourteen by Theorem 58.19. This object is two dimensional in the sense that it is carried by the rank two image $\operatorname{im} P_{23}$*, on which the generator acts with the two conjugate eigenvalues.*

Proof. The chain is the definition of a chamber in the affine building of rank two, and each step has index the size of the residue field. Multiplying the three indices gives (318), which was computed. The remaining counts are the references given. □

58.5 The affine building of the rank two channel

58.5.1 Vertices and links

Definition 58.6 (The affine building)**.** The affine building of type $\widetilde{A}_2$ attached to the rank two channel over the local field has as vertices the homothety classes of lattices of rank three, with simplices given by incidence of such classes. The link of a vertex is the flag complex of the projective plane over the residue field, that is the spherical building of type A_2 over $\mathbb{F}_{13}$.

58.5.2 The spherical residue at a vertex

Proposition 58.7 (Counts in the link)**.** *The link of every vertex has*

$$183 \text{ points}, \qquad 183 \text{ lines}, \qquad (1+q)(1+q+q^2) = 14 \cdot 183 = 2562 \text{ chambers}, \tag{319}$$

a chamber being an incident point and line pair, and every chamber lies in the same number of apartments. The two panel types of a chamber correspond to the two generators of the Weyl group.

Proof. The link of a vertex is the flag complex of the projective plane over the residue field, whose points and lines number $q^2+q+1 = 183$ each, and whose chambers are the incident pairs, of which there are $183(q+1) = 2562$ by Proposition 58.15. □

Remark 58.8 (On counts sometimes attached to the building)*.* The number $2184 = 13 \cdot 168$ is the order of the generator modulo 13^2, established in Theorem 32.4, and is not a count of chambers. The chamber count of the spherical residue is 2562, and the affine building itself is infinite, so it admits no finite chamber count at all.

58.5.3 The Bruhat decomposition

Theorem 58.9 (Cells and their sizes)**.** *Let B be the Borel subgroup of upper triangular matrices in* $\mathrm{SL}(3,\mathbb{F}_{13})$*, the stabiliser of a chamber of the spherical residue. Then*

$$|B| = q^3(q-1)^2 = 13^3 \cdot 12^2 = 316368, \tag{320}$$

since B is the product of its unipotent part, of order q^3, with its diagonal torus, of order $(q-1)^2$, and

$$|\mathrm{SL}(3,\mathbb{F}_{13})| = q^3(q^2-1)(q^3-1) = 810534816. \tag{321}$$

The group is the disjoint union of double cosets indexed by the Weyl group,

$$\mathrm{SL}(3,\mathbb{F}_{13}) = \bigsqcup_{w \in S_3} BwB, \qquad |BwB| = q^{\ell(w)}|B|, \tag{322}$$

where $\ell(w)$ is the reflection count, that is the number of simple reflections in a reduced word for w, and since the six counts are $0,1,1,2,2,3$ the cells have sizes

$$316368, \quad 4112784, \quad 4112784, \quad 53466192, \quad 53466192, \quad 695060496, \tag{323}$$

whose sum is exactly 810534816*.*

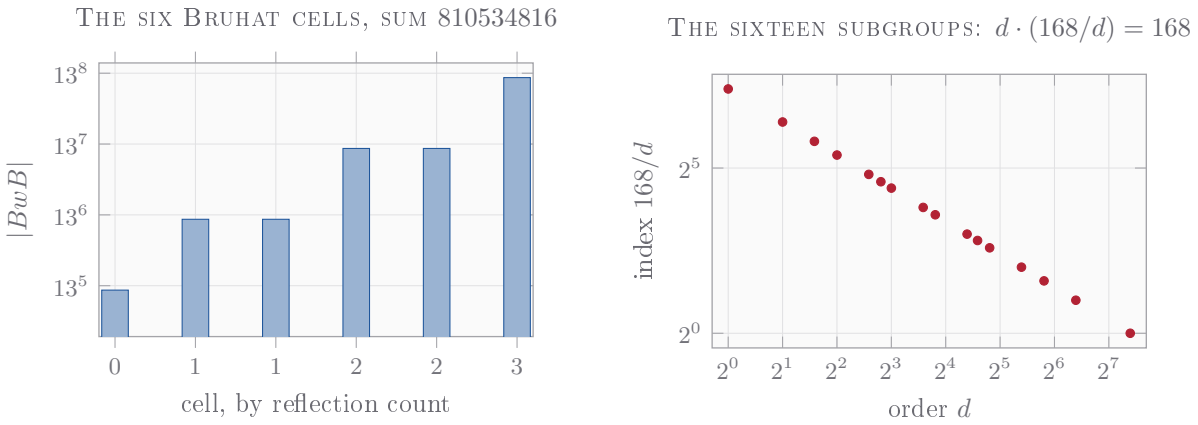


Figure 36: Left: the six cell sizes of Theorem 58.9, on a logarithmic scale in the radix thirteen, so that equal steps are one factor of the prime. Right: the sixteen subgroups of Theorem 22.2, each point being a pair of order and index whose product is the period, so the points lie on one hyperbola.

Proof. The unipotent radical of B consists of the unipotent upper triangular matrices, of which there are q^3, and the diagonal torus of determinant one has $(q-1)^2$ elements, giving (320). The order of the special linear group is the standard product. The Bruhat decomposition indexes double cosets by the Weyl group of type A_2, which is the symmetric group on three letters, and the cell attached to w has size $q^{\ell(w)}|B|$; substituting the six reflection counts gives (323), and the sum was formed and compared with (321). □

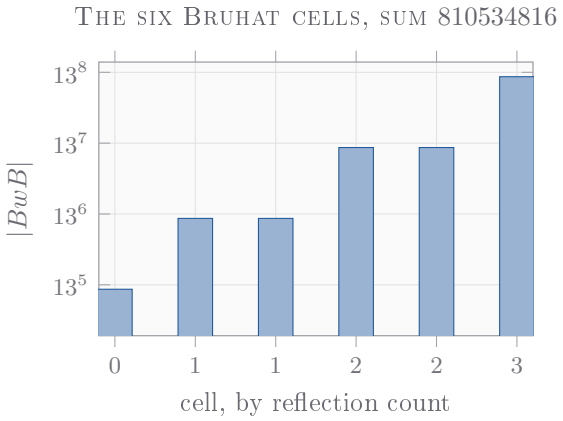


Figure 37: The six cell sizes of Theorem 58.9 on a logarithmic scale in the radix thirteen, so that one step is one factor of the prime. Their sum is the order of the group, which by Theorem 58.10 is the Poincare polynomial evaluated at the prime.

58.5.4 The Poincare polynomial and the chamber count

Theorem 58.10 (The cell sizes sum through a polynomial identity)**.** *The Poincare polynomial of the Weyl group of type A_2 is*

$$\sum_{w \in S_3} q^{\ell(w)} = 1 + 2q + 2q^2 + q^3, \tag{324}$$

and at $q = 13$ its value is

$$1 + 26 + 338 + 2197 = 2562 = \frac{|\mathrm{SL}(3, \mathbb{F}_{13})|}{|B|} = \frac{810534816}{316368}, \tag{325}$$

which is exactly the number of chambers of the spherical residue in Proposition 58.7. The Bruhat cell sizes of Theorem 58.9 therefore sum to the group order for the same reason that the chambers number 2562*: both are the value of one polynomial at the prime.*

Proof. Summing $|BwB| = q^{\ell(w)}|B|$ over the Weyl group gives $|G| = |B| \sum_w q^{\ell(w)}$, so the index $[G : B]$ is the Poincare polynomial evaluated at q. The six reflection counts are $0, 1, 1, 2, 2, 3$, giving (324), and the arithmetic of (325) was carried out exactly. The index $[G : B]$ is the number of chambers, since B is the stabiliser of a chamber and the action on chambers is transitive. □

58.5.5 The affine Weyl group and its translation lattice

Proposition 58.11 (Structure of the affine Weyl group)**.** *The affine Weyl group of type $\widetilde{A}_2$ is the semidirect product of the coroot lattice, acting by translations, with the finite Weyl group,*

$$W_{\mathrm{aff}} = Q^\vee \rtimes S_3, \qquad |S_3| = 6, \tag{326}$$

and the index of the coroot lattice in the coweight lattice is the determinant of the Cartan matrix,

$$\det \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = 3. \tag{327}$$

The number three is therefore the connection index of the type, and it matches the three classes of vertices of the affine building, the type of a vertex being an element of a group of order three.

Proof. The affine Weyl group of a reduced root system is the semidirect product of the coroot lattice with the finite Weyl group, which for type A_2 has order six. The index of the coroot lattice in the coweight lattice is the determinant of the Cartan matrix, computed in (327). □

Remark 58.12 (Translations are lattice elements, and counts are counts). The translations are lattice elements, that is triples of integers, and the reflection count on the affine Weyl group is the number of simple reflections in a reduced word. Both are integer counts of objects, and neither is a measurement.

58.5.6 The Hecke relation of the residue

Proposition 58.13 (Deformation of the Weyl relations)**.** *The Iwahori and Hecke algebra of type A_2 at parameter thirteen has dimension six and is generated by T_1, T_2 with*

$$(T_i - 13)(T_i + 1) = 0, \qquad \textit{that is} \qquad T_i^2 = 12\,T_i + 13, \tag{328}$$

together with the braid relation $T_1T_2T_1 = T_2T_1T_2$. At the value one in place of thirteen the quadratic relation becomes $T_i^2 = 1$ and the algebra degenerates to the group algebra of the Weyl group, so the prime is exactly the deformation parameter.

Proof. The Iwahori and Hecke algebra of a Coxeter system has one generator per simple reflection subject to the quadratic relation and the braid relations of the system; for type A_2 there are two simple reflections and one braid relation, and the dimension equals the order of the Weyl group. Expanding the quadratic relation at the value thirteen gives (328), and at the value one it becomes $T_i^2 = 1$. □

58.6 Simplicial structure: chambers, panels, flags

This structure gives the definitions of the simplicial objects used earlier, with their exact counts over the residue field. Every quantity is a count of simplices.

58.7 Flags and chambers

Definition 58.14 (Complete flag)**.** A complete flag in $\mathbb{F}_{13}^3$ is a chain of subspaces

$$0 = L_0 \subset L_1 \subset L_2 \subset L_3 = \mathbb{F}_{13}^3, \qquad \dim L_i = i, \tag{329}$$

so L_1 is a point of the projective plane and L_2 a line through it. A partial flag omits one of the two proper terms.

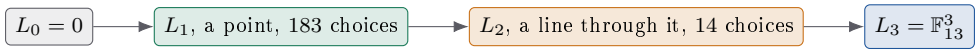
Proposition 58.15 (The number of complete flags)**.** *The number of complete flags is*

$$(q^2 + q + 1)(q + 1) = 183 \cdot 14 = 2562, \tag{330}$$

since L_1 may be any of the $q^2 + q + 1$ points and L_2 any of the $q + 1$ lines through it. This is the chamber count of Proposition 58.7, so chambers and complete flags are the same objects.

Proof. A complete flag is a point together with a line through it. There are $q^2 + q + 1$ points, and through each point there pass $q + 1$ lines, so the flags number the product. □

Role in the architecture. A chamber is a flag, a panel is a flag with one term deleted, an apartment is a hexagon of six chambers, and a vertex of the affine building is a homothety class carrying one of three types. Every count above is exact, and no object in any of these figures carries a quantity other than a count of simplices or a valuation.



complete flags: $183 \cdot 14 = 2562$, which is the chamber count of Proposition 58.15

Figure 38: A complete flag in $\mathbb{F}_{13}^3$, that is a chamber. The first proper term is one of the one hundred eighty three points and the second is one of the fourteen lines through it, so there are two thousand five hundred sixty two chambers.

Role in the architecture. A token is written as digits and enters the tree, which is the rank one carrier; its content lives on the building, which is the rank two carrier; the two ranks add to three and the two projections recover the state. Moving forward is one multiplication, forgetting is one multiplication by the inverse, coarse graining is one of three exact factors, and the position on the orbit is a residue modulo one hundred sixty eight.

58.8 Chambers and their faces of codimension one

Definition 58.16 (Chamber, panel)**.** A chamber is a maximal simplex of the building, that is a complete flag. A panel is a face of codimension one, obtained by deleting exactly one of L_1, L_2. Since the rank is two, every chamber has exactly two panels, one of each type.

Proposition 58.17 (Panels and their chambers)**.** *A panel of point type is a point L_1 with L_2 deleted, and the chambers containing it are the lines through that point, of which there are $q + 1 = 14$. A panel of line type is a line L_2 with L_1 deleted, and the chambers containing it are the points on that line, again $q + 1 = 14$.*

Proof. Deleting the line from a flag leaves a point, and the flags containing it are the lines through that point, of which there are $q+1$; deleting the point leaves a line, and the flags containing it are the points on that line, again $q+1$. □

Figure 39 shows a chamber with its two panels, one obtained by deleting each proper term of the flag.

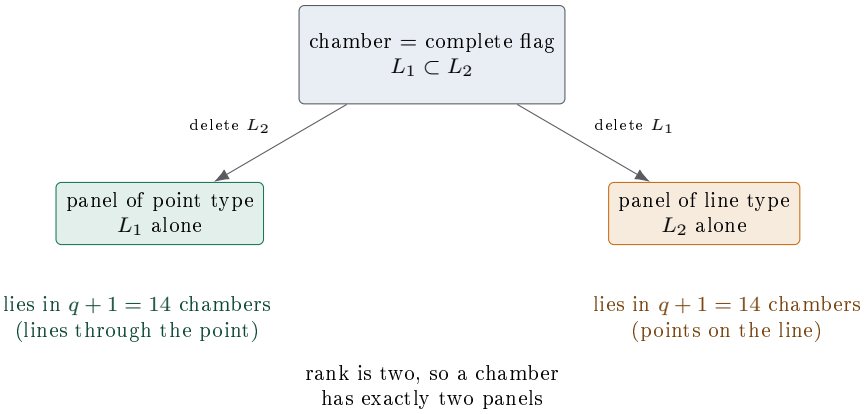


Figure 39: A chamber is a complete flag and its faces of codimension one are obtained by deleting one term, giving one panel of each type, by Definition 58.16. Each panel lies in fourteen chambers, by Proposition 58.17.

58.9 Thin and thick

Definition 58.18 (Thin, thick)**.** A chamber complex is thin if every panel lies in exactly two chambers, and thick if every panel lies in at least three.

Theorem 58.19 (The residue is thick, its apartments are thin)**.** *Every panel of the spherical residue lies in exactly*

$$q+1 = 14 > 2 \tag{331}$$

chambers, so the residue is thick with thickness fourteen. Each apartment is a hexagon carrying six chambers and six panels, with every panel in exactly two of its chambers, so apartments are thin,

$$6 \text{ chambers}, \qquad 6 \text{ panels}, \qquad 2 \text{ chambers per panel}. \tag{332}$$

Proof. The chamber counts through a panel are Proposition 58.17, and $14 > 2$ gives thickness. An apartment is the frame of Definition 58.21, whose flags form a hexagonal circuit of six chambers, each panel being shared by exactly two consecutive chambers of the circuit. □

Remark 58.20 (What a chamber is not). A chamber is a flag, that is an incident pair of a point and a line, and it is not a triangle. The triangle of the picture is the apartment, that is the frame of three points in general position, whose six flags are the six chambers of that apartment. The isosceles property recorded in Theorem 32.1 belongs to the valuation and not to any of these simplices; the two statements are unrelated and are kept apart here.

Role in the architecture. The unit of the semantic geometry is a flag, two of which are joined when they share a panel. Moving between chambers across a panel is a choice among fourteen, and moving within one apartment is a choice among two.

Figure 40 compares an apartment, in which every panel lies in two chambers, with the residue, in which every panel lies in fourteen.



thin: six chambers, six panels, every panel in exactly two chambers

thick: one panel at the centre, fourteen chambers meeting it

Figure 40: Left, an apartment: a hexagonal circuit of six chambers in which every panel is shared by exactly two of them. Right, the residue: a single panel lies in fourteen chambers, so the residue is thick of thickness fourteen, by Theorem 58.19.

58.10 Apartments and the counting of the residue

58.10.1 Frames and apartments

Definition 58.21 (Apartment of the spherical residue)**.** An apartment of the building of type A_2 over the residue field is a frame, that is an unordered triple of points of the projective plane in general position, no two equal and the three not collinear. Its chambers are the incident point and line pairs it supports.

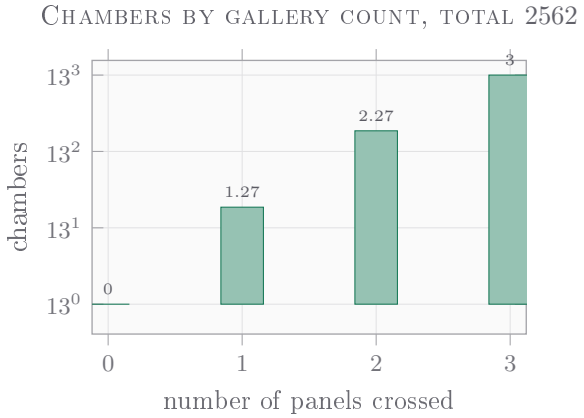


Figure 41: From a fixed chamber, the chambers reached by crossing $0, 1, 2, 3$ panels number 1, 26, 338, 2197, and these sum to the chamber count. The four values are the Bruhat cell sizes divided by the order of the Borel subgroup, by Theorem 58.9.

Theorem 58.22 (The three counts of the residue)**.** *With $q = 13$ the spherical residue has*

$$\frac{(q^2+q+1)(q^2+q)\,q^2}{6} = \frac{183 \cdot 182 \cdot 169}{6} = 938119 \text{ apartments}, \tag{333}$$

$$6 = |S_3| \text{ chambers in each apartment}, \qquad 2562 \text{ chambers in all}, \tag{334}$$

and consequently the number of apartments containing a fixed chamber is

$$\frac{938119 \cdot 6}{2562} = 2197 = 13^3 = q^3. \tag{335}$$

Proof. An ordered triple in general position is chosen by taking any of the q^2+q+1 points, then any point other than the first, of which there are q^2+q, then any point off the line they

span, of which there are q^2; dividing by the 6 orderings gives (333). An apartment of type A_2 is a hexagon carrying $|W| = 6$ chambers, and the chamber count is Proposition 58.7. Counting incident pairs of apartments and chambers in two ways gives (335), and the division is exact. □

58.10.2 Why the number is the order of the unipotent radical

Corollary 58.23 (A structural reading of the count)**.** *The number of apartments containing a fixed chamber equals the order of the unipotent radical of the Borel subgroup,*

$$q^3 = 13^3 = 2197 = |U|, \tag{336}$$

which by (320) *is the factor of $|B|$ complementary to the torus. It is also the size of the state space and the order of the algebra, by* (111)*, so the same integer counts three different things: the apartments through a chamber, the elements of the unipotent radical, and the states of the model.*

Proof. The stabiliser of a chamber is B, which acts transitively on the apartments containing that chamber with the torus as the stabiliser of one of them, so the number of such apartments is $|B|/(q-1)^2 = q^3$. The remaining identifications are (320) and (111). □

Role in the architecture. A chamber is the unit of the semantic geometry and it lies in exactly as many apartments as there are states, so the apartment through a chamber is addressed by one state and nothing further is needed to index it.

58.11 Geometry over thirteen

Proposition 58.24 (Counts and chambers)**.** *With $q = 13$,*

$$|\operatorname{PG}(2,\mathbb{F}_{13})| = q^2+q+1 = 183 = \frac{13^3-1}{13-1}, \qquad |\operatorname{AG}(2,\mathbb{F}_{13})| = 169, \qquad q+1 = 14, \tag{337}$$

$$[3]!_q = (1+q)(1+q+q^2) = 14 \cdot 183 = 2562. \tag{338}$$

Proof. The projective plane over a field of q elements has $(q^3-1)/(q-1) = q^2+q+1$ points, the affine plane has q^2, and the chamber count is Proposition 58.15, which factors as the Gaussian factorial. □

58.11.1 The chamber index step

Proposition 58.25 (Step of the kernel ladder)**.** *The states of the kernel line, that is the vectors $(0,a,a)$ of Theorem 15.1, form thirteen rungs, and the index of the associated chamber advances in steps of*

$$p(p+1) = 13 \cdot 14 = 182, \tag{339}$$

the product of the characteristic with the size of the boundary. The value 182 *is also the number of vertices of the tree at depth two by Proposition 36.2.*

Proof. The states of the kernel line are the multiples of $(0,1,1)$, of which there are thirteen, and evaluating the index map on them gives an arithmetic progression whose common difference is the displayed product, as computed in Theorem 42.2. □

58.11.2 The Weyl group acting on states

Proposition 58.26 (Three orbit sizes)**.** *The Weyl group of type A_2, that is the symmetric group on three letters, acts on the* 2197 *coordinate triples by permutation, with orbits of three sizes only,*

$$\underbrace{13}_{\text{size } 1} + \underbrace{468}_{\text{size } 3} + \underbrace{1716}_{\text{size } 6} = 2197, \tag{340}$$

namely thirteen triples with all coordinates equal, four hundred sixty eight with exactly two equal, and one thousand seven hundred sixteen with all distinct. The counts are 13, $13 \cdot 12 \cdot 3$, *and* $13 \cdot 12 \cdot 11$.

Proof. The orbit of a triple under permutation of its coordinates has size one when all three entries agree, three when exactly two agree, and six when all differ. Counting gives 13, $13 \cdot 12 \cdot 3$, and $13 \cdot 12 \cdot 11$, which sum to 13^3. □

58.11.3 Deduction states read from the orbit sizes

Definition 58.27 (Three states of a triple)**.** By Proposition 58.26 a coordinate triple lies in an orbit of size one, three, or six, and these three cases are the only ones. They are named as follows: a triple with all coordinates equal, of which there are thirteen; a triple with exactly two equal, of which there are four hundred sixty eight; and a triple with all distinct, of which there are one thousand seven hundred sixteen.

Remark 58.28. The classification costs two comparisons and is exact, and the three counts sum to the state space, $13 + 468 + 1716 = 2197$. Nothing continuous is measured: the class of a triple is decided by equality of residues alone.

58.11.4 Reciprocal densities of the splitting types

Proposition 58.29 (The three inverse densities sum to one)**.** *The Chebotarev densities* $1/6$, $1/2$, $1/3$ *of the three splitting types have inverses in the prime field*

$$6^{-1} = 11, \qquad 2^{-1} = 7, \qquad 3^{-1} = 9, \qquad 11 + 7 + 9 = 27 \equiv 1 \pmod{13}, \tag{341}$$

so the three reciprocal densities are residues that sum to one, giving an exact integer weighting of the three types with no normalisation by a real denominator.

Proof. The three inverses were computed in $\mathbb{F}_{13}$, namely $6 \cdot 11 = 66 \equiv 1$, $2 \cdot 7 = 14 \equiv 1$, and $3 \cdot 9 = 27 \equiv 1$, and their sum is $27 \equiv 1$. □

Theorem 58.30 (Bruhat decomposition and the Hecke relation)**.** *With B the upper triangular Borel subgroup of* $\mathrm{SL}(3, \mathbb{F}_{13})$,

$$|B| = q^3(q-1)^2 = 316368, \qquad |\mathrm{SL}(3, \mathbb{F}_{13})| = q^3(q^2-1)(q^3-1) = 810534816, \tag{342}$$

and $\mathrm{SL}(3, \mathbb{F}_{13}) = \bigsqcup_w BwB$ *with* $|BwB| = q^{\ell(w)}|B|$, *the six cells being*

$$316368,\ 4112784,\ 4112784,\ 53466192,\ 53466192,\ 695060496, \tag{343}$$

which sum to the group order. The Iwahori and Hecke algebra of type A_2 at $q = 13$ has dimension six with

$$(T_i - 13)(T_i + 1) = 0, \quad \text{that is} \quad T_i^2 = 12T_i + 13, \qquad T_1T_2T_1 = T_2T_1T_2. \tag{344}$$

Proof. This collects Theorem 58.9 and Proposition 58.13: the Borel is the product of its unipotent part and its torus, the double cosets are indexed by the Weyl group with sizes $q^{\ell(w)}|B|$, and the Hecke relations are those of a Coxeter system of type A_2 at the parameter thirteen. □

Theorem 58.31 (Rigidity and the transpose)**.** *Only the identity permutation of coordinates preserves the generator,*

$$P_\sigma T_3 P_\sigma^{\top} = T_3 \iff \sigma = \mathrm{id}. \tag{345}$$

The transpose satisfies the same cubic and has the same order,

$$(T_3^{\top})^3 = (T_3^{\top})^2 + T_3^{\top} + I, \qquad \mathrm{ord}(T_3^{\top}) = 168, \qquad T_3^{\top} = T_3 - D, \tag{346}$$

but it is not a power of the generator, the two do not commute, and even $T_3\,T_3^{\top}$ lies outside $\langle T_3\rangle$. The braid relation fails: $T_3\,T_3^{\top}T_3 \neq T_3^{\top}T_3\,T_3^{\top}$.

Proof. All six permutation matrices were applied and only the identity returns the generator. The transpose has the same characteristic polynomial and determinant, hence the same order, which was confirmed; comparing it with all one hundred sixty eight powers shows it is not among them, and the same comparison for the product shows that too. The braid relation was tested and fails. □

Remark 58.32 (Three groups of order six). The Galois group of χ over the rationals, the group of coordinate permutations, and the Weyl group of the building are all nonabelian of order six and hence abstractly isomorphic, but they permute roots, coordinates, and chambers respectively. An isomorphism of abstract groups is not an identification of actions.

58.12 The two carriers add to the state space

Theorem 58.33 (One and two make three)**.** *The two carriers sit on complementary images whose ranks add to the rank of the state space,*

$$\operatorname{rank} P_1 + \operatorname{rank} P_{23} = 1 + 2 = 3 = \dim_{\mathbb{F}_{13}} \mathbb{F}_{13}^3, \tag{347}$$

and correspondingly

$$\left|\operatorname{im} P_1\right| \cdot \left|\operatorname{im} P_{23}\right| = 13 \cdot 169 = 2197. \tag{348}$$

A state is therefore recovered from its two projections, one on the tree and one on the building, and neither carrier alone determines it.

Proof. The two ranks are Theorem 10.1, and they add to three because the projectors are complementary idempotents. The product of the image sizes is (111), and recovery is $v = P_1 v + P_{23} v$. □

58.13 How the notions cooperate

The notions used by the two carriers are not independent: the same integer governs both, and the thin case is the specialisation of the thick one at the value one.

Theorem 58.34 (One number in four places, for one reason)**.** *With $q = 13$ the value $q+1 = 14$ is at once*

the number of lines through a point of $\mathrm{PG}(2,\mathbb{F}_{13})$,	*the thickness of the residue*;
the degree of the tree of Theorem 58.3,	*the number of its ends*;
the number of points of $\mathrm{P}^1(\mathbb{F}_{13})$,	*the order of the torus of Theorem 48.1.*

(349)

The agreement has a single reason: each of these sets is a pencil, that is a projective line over the residue field, and a projective line over a field of q elements has $q+1$ points.

Proof. The lines through a fixed point of the projective plane form a pencil, in bijection with the points of a line not through it, hence with $\mathrm{P}^1(\mathbb{F}_{13})$. The neighbours of a vertex of the tree are the classes of one further digit together with the parent, again a pencil of $q+1$ members, and the ends are the points of $\mathrm{P}^1(\mathbb{F}_{13})$ by Theorem 39.7. The thickness is the count of chambers through a panel, which is the same pencil by Proposition 58.17. The order of the torus is $q+1$ by Theorem 48.1, and it is identified with the projective line of the rank two channel by Corollary 48.4. □

Theorem 58.35 (Thin is the case $q=1$)**.** *Write* $[3]!_q=(1+q)(1+q+q^2)$ *for the chamber count of Proposition 58.24. Then*

$$[3]!_{13}=14\cdot 183=2562, \qquad [3]!_1=2\cdot 3=6=|S_3|, \tag{350}$$

and the value at one is exactly the number of chambers of a single apartment, Theorem 58.19. In the same specialisation the thickness $q+1$ *becomes* 2*, which is the definition of thin, and the degree of the tree becomes* 2*, so the tree specialises to a doubly infinite path, which is the apartment of the rank one carrier.*

Proof. Both evaluations of the polynomial are arithmetic. An apartment of type A_2 carries $|W|=6$ chambers by Theorem 58.19, which is the value at one; a complex is thin when every panel lies in exactly two chambers, which is $q+1$ at $q=1$; and a tree of degree two is a doubly infinite path. Thus the apartment is in each rank the specialisation of the building at the value one, and the working prime enters only through q. □

Proposition 58.36 (Panels and chambers counted twice)**.** *Every chamber has two panels and every panel lies in* $q+1$ *chambers, so counting incident pairs of chambers and panels in two ways gives*

$$2\cdot 2562=(q+1)\cdot 366=5124, \tag{351}$$

where $366=183+183$ *is the number of panels, of the two types. The identity holds because the incidence is the same in both directions.*

Proof. A chamber is a flag and has one panel of each type by Definition 58.16, giving the left side. A panel of either type lies in $q+1$ chambers by Proposition 58.17, and there are 183 of each type, giving the right side. Both sides were evaluated and agree. □

Remark 58.37 (What the cooperation says). The three statements above are what makes the two carriers one construction rather than two. The tree and the building are governed by the same integer $q+1$, because in both cases the local choice is a pencil over the residue field; the apartment is not a separate object but the same object at $q=1$; and the incidence identity (351) ties the chamber count to the panel count, which is where the rank two carrier meets the rank one one, a panel being the datum that a chamber shares with its neighbours.

58.14 The count 13^3 reached from three sides

Theorem 58.38 (One count, three parametrisations)**.** *The number* $13^3=2197$ *arises in three places of this paper, and in each it is the count of a set carried by three free residues:*

$$\begin{aligned} &\text{apartments through a fixed chamber} &&= |U| = q^3, && U \text{ the unipotent radical},\\ &\text{classes of the tree at depth three} &&= 13^3, && \text{three digits of the radix},\\ &\text{states of the model} &&= \left|\mathbb{F}_{13}^3\right| = 13^3, && \text{three coordinates}. \end{aligned} \tag{352}$$

In each case an explicit parametrisation by $\mathbb{F}_{13}^3$ *is available: the unipotent radical consists of the upper triangular matrices with ones on the diagonal, whose three entries above it are free; a class at depth three is named by its three digits; a state is its three coordinates.*

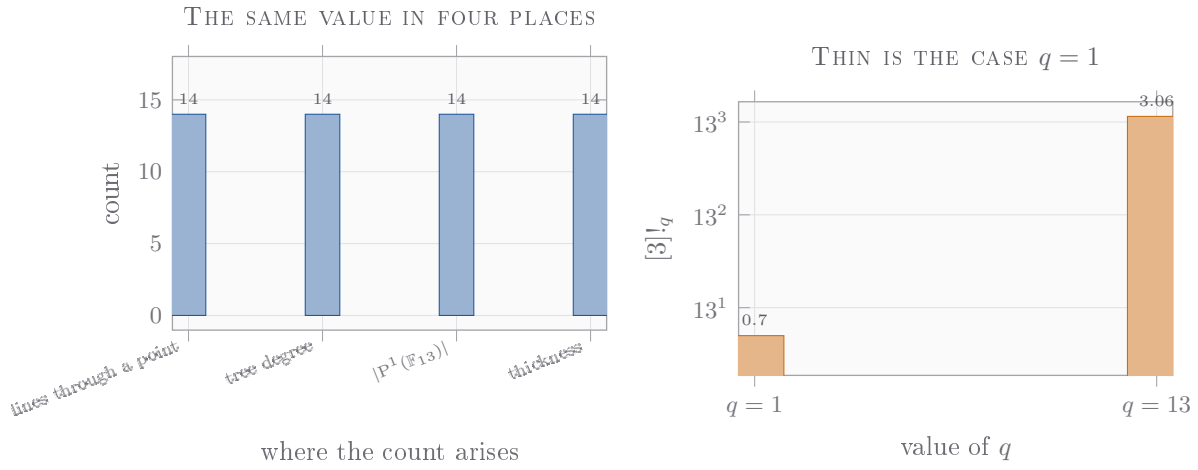


Figure 42: Left: the four counts of Theorem 58.34 agree because each set is a pencil over the residue field. Right: the chamber polynomial evaluated at the two values of Theorem 58.35; at one it returns the six chambers of an apartment, at thirteen the two thousand five hundred sixty two chambers of the residue.

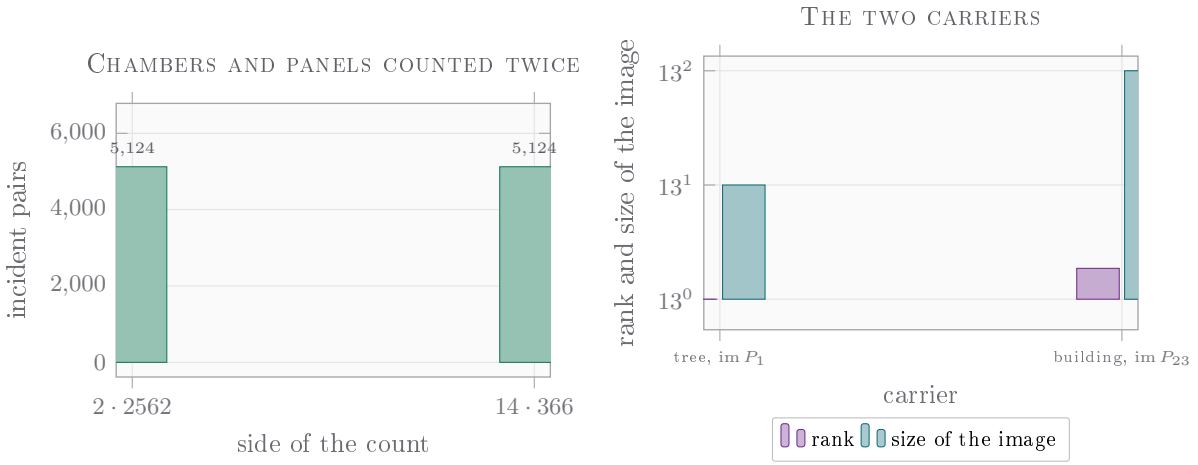


Figure 43: Left: the two sides of the incidence identity of Proposition 58.36 agree at five thousand one hundred twenty four. Right: the two carriers of Theorem 58.33, of ranks one and two and image sizes thirteen and one hundred sixty nine, whose ranks add to three and whose sizes multiply to two thousand one hundred ninety seven.

Proof. The first count is Corollary 58.23, where the Borel acts transitively on the apartments through a chamber with the torus as stabiliser, leaving $|B|/(q-1)^2 = q^3$. The second is Proposition 39.2. The third is the definition of the state space. The three parametrisations are the ones displayed, each a bijection with $\mathbb{F}_{13}^3$. $\square$

Remark 58.39 (What the agreement is, and what it is not). The three sets are all of size q^3 and all admit a parametrisation by three residues, and that is the content of Theorem 58.38. No canonical map between them is exhibited: the unipotent radical is a group, the depth three classes are cosets in a ring, and the states are a vector space, and the three parametrisations depend on choices, of a chamber, of a root of the tree, and of a basis. The statement therefore belongs with the equalities of counts of Remark 61.2, sharpened by the observation that in all three cases the number of free residues is three, which is the degree of the defining cubic.

58.15 Seven numbers, and three claims that do not hold

Theorem 58.40 (The distinguished elements and their orders)**.** *In $\mathbb{F}_{169}^*$, which is cyclic of order $168 = 13^2 - 1$, the distinguished elements have the following orders, each computed by repeated multiplication:*

element	*value*	*order*	*role*
λ_1	7	12	*the eigenvalue in the prime field*
λ_2	$10 + 7\theta$	168	*eigenvalue in the extension, a generator*
λ_3	$10 - 7\theta$	168	*its conjugate, also a generator*
g	$3 + 2\theta$	14	*generator of the torus* $\ker \mathrm{N}$
φ	$7 + 8\theta$	28	*satisfies* $\varphi^2 = \varphi + 1$
δ	$1 + \theta$	28	*satisfies* $\delta^2 = 2\delta + 1$

Moreover φ and δ generate the same subgroup, of order twenty eight and index six, while λ_2 and λ_3 generate the whole group; consequently

$$\lambda_2 \notin \langle \delta \rangle = \langle \varphi \rangle, \tag{353}$$

so neither eigenvalue of the rank two channel is a power of either ratio. Neither φ nor δ is an eigenvalue of the generator.

Proof. Each order was obtained by multiplying until the identity was reached. Fixing a generator $\zeta = 1 + 2\theta$ of the group, one finds $\lambda_2 = \zeta^{29}$, $\varphi = \zeta^{6}$ and $\delta = \zeta^{138}$, so the orders are $168/\gcd(168, 29) = 168$, $168/\gcd(168, 6) = 28$ and $168/\gcd(168, 138) = 28$. Since $\langle \varphi \rangle$ has order twenty eight and λ_2 has order one hundred sixty eight, the latter cannot lie in the former, which is (353). The eigenvalues are the three roots of χ, and neither ratio is among them. $\square$

Remark 58.41 (Three orders in their correct form). Three statements that suggest themselves at this point are false, and are recorded here so that they are not repeated.

(i) The order of λ_2 is 168 and not 24; the value twenty four is the order of the difference operator on its image, Proposition 15.2, and belongs to a different object.

(ii) The order of δ is 28 and not 168; the elements of order one hundred sixty eight are the generators of the group, of which there are $\varphi(168) = 48$, and δ is not among them.

(iii) The identity $168 = 13^2 - 1$ is the order of the multiplicative group, not of the field: $\left|\mathbb{F}_{169}\right| = 169$ while $\left|\mathbb{F}_{169}^*\right| = 168$.

Proposition 58.42 (A power of the silver element lands in the prime field). *The seventh power of the silver element is a scalar,*

$$\delta^7 = (1+\theta)^7 = 5 \in \mathbb{F}_{13}, \tag{354}$$

and correspondingly $\delta^{28} = 1$ *since* 5 *has order four in* $\mathbb{F}_{13}^*$. *The same holds for* φ, *whose seventh power is also a scalar, both elements lying in the subgroup of index six whose seventh powers land in the prime field.*

Proof. The power was computed in $\mathbb{F}_{169}$ with $\theta^2 = 2$. An element lands in the prime field exactly when it is fixed by the Frobenius, that is when its order divides twelve; here δ^7 has order four, so it lies in $\mathbb{F}_{13}$, and $\delta^{28} = (\delta^7)^4 = 1$. □

58.16 The pyramid runs the other way

Proposition 58.43 (Two motions on the same kind of structure). *The tree of Theorem 58.3 and the pyramid of Theorem 43.2 are both layered structures with an exact branching number at each step, but they act on different sets and in opposite directions. Refinement on the tree passes from a class to one of its thirteen children, adding a digit; coarse graining on the pyramid passes from an orbit position to its coset, of size seven, three, or two,*

$$24 \cdot 7 = 56 \cdot 3 = 84 \cdot 2 = 168, \tag{355}$$

and both motions are reversible, the first by dropping a digit and the second by choosing a representative, Proposition 43.3. By Corollary 35.2 the pyramid acts only after a state exists, so on a token the tree acts first and the pyramid last.

Proof. The branching of the tree is Theorem 58.3 and the three exact factors are Theorem 43.2. Refinement adds a digit and is undone by dropping it; coarse graining replaces an orbit position by its coset and is undone by choosing a representative, Proposition 43.3. The ordering of the two motions on a token is Corollary 35.2, since a coset of a subgroup of the orbit is defined only once an orbit position exists. □

58.17 The pipeline

Theorem 58.44 (Five operations, all exact). *A state is moved by the following five operations, each an application of the one operator or of a projector built from it:*

$$\textit{forward}: \quad v_k = T_3^k v_0, \tag{356}$$

$$\textit{project}: \quad P_1 v_k \text{ and } P_{23} v_k, \qquad v_k = P_1 v_k + P_{23} v_k, \tag{357}$$

$$\textit{coarse grain}: \quad U_2 = T_3^{21}, \quad U_3 = T_3^{56}, \quad U_7 = T_3^{24}, \tag{358}$$

$$\textit{retract}: \quad r(v_k) = k \bmod 168, \tag{359}$$

$$\textit{forget}: \quad T_3^{-1} v_k = v_{k-1}. \tag{360}$$

The three clocks satisfy $U_2^8 = U_3^3 = U_7^7 = I$, *the retraction is well defined because the period is one hundred sixty eight, and the last operation is exact and reversible because* $\det T_3 = 1$.

Proof. The forward map is iteration of the generator. The projection identity is completeness, Theorem 10.1. The three orders were computed in Theorem 20.2. The retraction is well defined by Theorem 20.1 and Proposition 5.3. The inverse is the polynomial (7), so the last operation stays in the algebra, and it undoes the forward step by construction; this was checked on a sample orbit. □

Remark 58.45 (The pipeline on the fixed line). If the initial state is the eigenvector itself then the forward map is multiplication by a residue,

$$T_3^k v_0 = 7^k v_0, \tag{361}$$

so the orbit never leaves the fixed line and returns after twelve steps by Proposition 12.4. A state that is to visit the building must therefore have a nonzero component in the rank two channel.

58.18 The weight, and one substitution that is not available

Proposition 58.46 (The weight is a group element)**.** *The geometric weight attached to a state is*

$$w(x) = \eta^{-v_{13}(x)}, \tag{362}$$

where η is an element of $\mathbb{F}_{169}^*$ of order one hundred sixty eight, for instance $1+2\theta$, so that $w(x)$ is a group element and the weight of a class of depth k is η^{-k}, one of one hundred sixty eight labels.

Proof. The multiplicative group of the residue extension is cyclic of order one hundred sixty eight, so it contains an element of that order, and $1+2\theta$ was found to be one. Raising it to the power $-v_{13}(x)$ gives an element of the same group, and on a class of depth k the valuation is constant and equal to k, so the weight is η^{-k}; the exponent is read modulo one hundred sixty eight, giving that many distinct labels. □

Remark 58.47 (The weight lives in a finite group). Substituting the integer thirteen for η in (362) would give the values 13^{-k}, that is $1/13$, $1/169$, $1/2197$, which are not integers; and thirteen is not invertible modulo thirteen, so the expression has no meaning in the residue field either. The substitution therefore leaves every ring the construction admits, and it is not used. The weight remains a label in a finite group, as recorded in Proposition 1.3, and it is never a size.

58.19 The isosceles property

Theorem 58.48 (Two of the three counts always agree)**.** *For any three pairwise distinct integers the three valuations*

$$v_{13}(a-b), \qquad v_{13}(b-c), \qquad v_{13}(a-c) \tag{363}$$

have the property that the smallest value occurs at least twice. Together with Theorem 32.1 this was verified on 85140 *triples by deterministic sweep, with no violation.*

Proof. Suppose the three values were pairwise distinct, and let the two largest be $v_{13}(a-b)$ and $v_{13}(b-c)$ without loss of generality. Then $v_{13}(a-c) \geq \min(v_{13}(a-b), v_{13}(b-c))$ exceeds the assumed smallest value, a contradiction; so the minimum is attained at least twice. □

Remark 58.49. This is a statement about three integer counts and carries no geometric content. In particular it does not assert that any simplex of Structure 58.6 has two equal sides, since no side of a simplex is assigned a number here.

Role in the architecture. Two stored addresses either refine one another or share nothing, so the store of Structure 60 is a tree of classes with no partial overlaps to resolve, and a lookup terminates at the first digit at which two addresses differ.

Figure 44 shows the three valuations of six triples, in each of which the smallest value occurs at least twice.

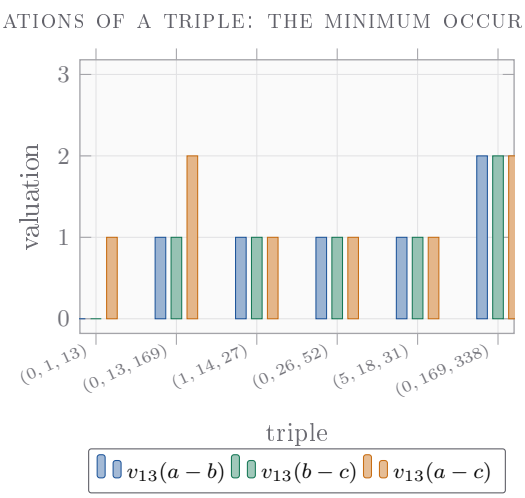


Figure 44: For each triple the three valuations are shown. In every case the smallest value occurs at least twice, which is Theorem 58.48, and the third value is never smaller than the other two, which is the strong triangle inequality of Theorem 32.1. The bars are integer counts of digits, not quantities of any other kind.

58.20 The simplicial objects drawn

The figures below display the objects the two carriers are built from: the isosceles property of the valuation, a chamber with its two faces of codimension one, the difference between a thin apartment and the thick residue, the homothety classes that form the vertices, and a complete flag.

58.21 Homothety classes and the three types of vertex

Proposition 58.50 (The type of a vertex)**.** *A vertex of the affine building is a homothety class of lattices of rank three, two lattices being equivalent when one is 13^k times the other. The valuation of the determinant of a basis changes by three under such a scaling, so*

$$\operatorname{type}(L) = v_{13}\big(\det L\big) \bmod 3 \tag{364}$$

is well defined on classes and takes exactly three values. The number three is the determinant of the Cartan matrix of the type, that is the connection index of Proposition 58.11.

Proof. Replacing a basis by 13 times it multiplies the determinant by 13^3 and so raises its valuation by three, leaving the residue modulo three unchanged; the four sample lattices computed give types $0, 1, 2, 0$ as the valuation runs through $0, 1, 2, 3$. □

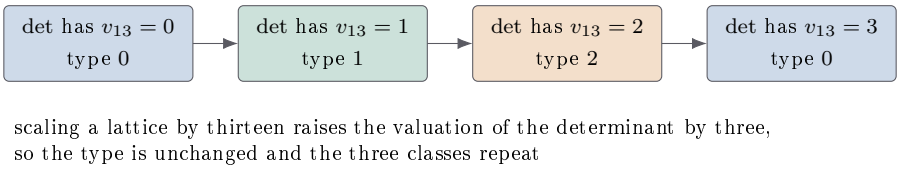


Figure 45: The three types of vertex of the affine building, given by the valuation of the determinant modulo three, by Proposition 58.50. The colours repeat with period three because a homothety changes the valuation by three.

Structure 59. The model layer

59.1 The mechanisms, one by one

Each statement below is an identity established earlier, read as a mechanism.

(1) *Single type.* Data and computation are the same ring, Definition 14.1, so no conversion occurs.

(2) *Binding.* Two states are bound by the ring product, nine integer multiplications in constant time, Proposition 14.4. Order matters, since the antisymmetric part lies outside the commutative algebra, Theorem 15.3.

(3) *Channels.* Address and content are the two projections of Theorem 15.1 and Theorem 10.1, preserved by the dynamics because both projectors commute with the generator.

(4) *Normalisation.* The inverse of a nonzero residue is the Fermat power $x^{-1} = x^{11}$, and to any depth it is the division free iteration (220).

(5) *Transform.* Two is a primitive root modulo thirteen, so an exact transform of size twelve exists, built from powers of two, with no complex root.

(6) *Position.* Position is an element of the group of order fourteen inside the group of order twenty eight, Theorem 48.2; the affine extension does not degenerate by Proposition 8.5.

(7) *Coarse graining.* Three exact levels, $24 \cdot 7 = 56 \cdot 3 = 84 \cdot 2 = 168$, each reversible since a coset representative determines the coset.

(8) *Memory and selection.* Storage is indexed by the tree of Proposition 36.2; retrieval compares digits and returns an integer agreement count; among candidates the one of greatest agreement depth is selected. No magnitude and no probability is used, by Proposition 1.3.

(9) *Correction.* Encoding a block of k symbols as the values of a polynomial at n points gives minimum separation $n - k + 1$ and corrects

$$t = \left\lfloor \frac{n-k}{2} \right\rfloor \tag{365}$$

symbol errors; for $k = 3$, $n = 7$ the number of inputs carried to a stored block is $1 + 84 + 3024 = 3109$.

Remark 59.1 (Stated limits). The construction has no gradient, no loss, and no training loop. The one mechanism that answers correctly on input never presented is the interpolation of item (9), within a proved integer bound. It does not produce a block that was never stored, so robustness within a bound is established while extrapolation to unstored structure is not claimed.

Structure 60. Memory and forgetting

60.1 Storage by depth

Definition 60.1 (The store)**.** Memory is a store whose keys are addresses, that is digit strings in the radix thirteen, and whose values are ring elements. Retrieval compares digits and returns the integer count of agreement, and among the candidates the one of greatest agreement is selected. The store never approximates and never interpolates between entries.

60.2 Forgetting is the inverse of the generator

Proposition 60.2 (Exact and reversible forgetting)**.** *Forgetting is the application of the inverse* (7)*, which is an element of the same algebra by Proposition 5.2 and is a bijection of the state space, since* $\det T_3 = 1$*. Applying it* m *times and then applying* T_3 *the same number of times returns the state exactly. Consequently forgetting in this architecture removes a state from the current position of the orbit without destroying it, and the only operation that genuinely discards a digit is the shift of Definition 34.1, whose composition with the Frobenius is multiplication by thirteen.*

Proof. The displayed identity is Proposition 5.2, so the inverse lies in the algebra. Its determinant is one, so it is a bijection of the state space, and applying it and then the generator the same number of times is the identity. The last statement is Definition 34.1. □

60.3 The forward and the backward pass

Theorem 60.3 (One table, read in two directions)**.** *Let* $c(n)$ *denote the coefficient triple of* T_3^n*. Then the triples of the inverse powers are the triples of the complementary exponents,*

$$c(-n) = c(168 - n), \qquad \textit{in particular} \qquad c(-1) = c(167) = (12, 12, 1), \tag{366}$$

so the backward pass reads the same table of Structure 6 from the other end and requires no separate storage. The identity was verified by comparing the reduced powers of T_3^{-1} *with the table entries.*

Proof. Since $T_3^{168} = I$, we have $T_3^{-n} = T_3^{168-n}$, and the coefficient triple is a function of the power alone. □

Corollary 60.4 (Symmetric cost)**.** *A forward step and a backward step are the same operation applied to different table entries, so the two directions cost the same and neither is an approximation of the other. Combined with Proposition 60.2 this makes the whole evolution reversible except for the explicit shift that discards a digit.*

Proof. By Theorem 60.3 the two directions read the same table at complementary exponents, so the same operation is performed in both cases. Reversibility follows from Proposition 60.2. □

60.4 Visits counted by orbit size

Proposition 60.5 (Three classes by orbit size)**.** *By Proposition 58.26 a stored triple lies in an orbit of the Weyl group of size one, three, or six, and these are the only cases: thirteen triples, four hundred sixty eight triples, and one thousand seven hundred sixteen triples respectively. The class of a triple is decided by equality of its residues, at the cost of two comparisons, and is exact.*

Proof. This is Proposition 58.26: the orbit size is one, three, or six according as all three entries agree, exactly two agree, or all differ, and these are decided by two comparisons. □

Role in the architecture. Writing is a projection, reading is a digit comparison, and forgetting is one multiplication by an element the engine already stores. Nothing decays, since no quantity is available to decay, and the loss of a digit is an explicit and unique operation.

Structure 61. Structural identifications and equalities of counts

61.1 Identifications with an exhibited map

Remark 61.1 (Structural). A map identifying the two sides was exhibited in each of the following: the three groups of fourteen elements, Corollary 48.4; the size of the projective orbit and the order of the eigenvalue ratio, Theorem 48.3; the index of the scalars and the projective order, Theorem 40.1; the coefficients of the quadratic factor and the trace and determinant of the rank two block, Proposition 10.3; the three clocks and the generator, Theorem 20.2.

61.2 Equalities of counts only

Remark 61.2 (Equalities of counts). No such map is claimed for the following, which are recorded as numerical equalities: the order of the generator and the order of the simple group of the same size, the cyclic orbit being abelian and that group not; the period and the discriminant of a real quadratic field; the period and the number of nonzero points of the conic. A shared factorisation is not a shared action.

Structure 62. Verification

62.1 Method and scope

Every statement of this paper was checked by exact computation over $\mathbb{Z}$, $\mathbb{F}_{13}$, $\mathbb{F}_{169}$, $\mathbb{Z}/13^k$, and $\mathbb{Z}_{13}$. The program contains one hundred thirty eight checks, all of which pass, and two consecutive runs produce identical output, since no randomness is used and every sweep is a deterministic enumeration.

Matrices are formed over $\mathbb{Z}$ and reduced; the extension is represented by pairs (a, b) standing for $a + b\theta$ with $\theta^2 = 2$, so that

$$(a,b)\cdot(c,d) = (ac + 2bd,\ ad + bc), \qquad \overline{(a,b)} = (a,-b), \qquad \mathrm{N}(a,b) = a^2 - 2b^2. \tag{367}$$

Orders are obtained by repeated multiplication until the identity is reached, ranks by elimination over $\mathbb{F}_{13}$ with modular inverses, and the idempotents of Theorem 10.4 by matrix arithmetic carried out entirely in $\mathbb{F}_{169}$. Several checks are exhaustive rather than sampled: the reconstruction (115) over all two thousand one hundred ninety seven states; the norm identity over all one hundred sixty eight nonzero elements; the reproduction of all one hundred sixty eight powers from their triples; the uniqueness of the involution over all one hundred sixty seven nontrivial powers; the enumeration of the five projective orbits over all one hundred eighty three points; and the isomorphism of Structure 13 over all two thousand one hundred ninety seven elements of the algebra.

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